

Chapter 5

Study 2: Understanding the Influence of Workplace Gender and Technological Dynamics on the Evaluation of Women Workers in AI-Enabled Workplaces

5.1 Introduction

Built upon the foundational insights gleaned from women in AI design and development in Study 1, Study 2 implements their functional design principles and investigates how evaluations of women workers will differ when working with different types of AI functional designs (i.e., tool vs. teammate). In doing so, Study 2 is one of the first empirical studies to date to tackle questions of evaluations and attributions of credit in the manifestation of human-AI interaction in AI-enabled workplaces at the

intersection of gendered dynamics in the workplace. Indeed, the rise of readily accessible AI systems in workplaces has stirred up much discourse surrounding how credit for AI-generated outputs can or should be attributed to the people and workplaces that utilize them to enhance their workplace productivity (e.g., [351, 364]). Yet, comparatively little research attention has been given to how various workplace dynamics and system designs can impact evaluations of human workers generally, such as in terms of effectiveness, ability, and future job security, despite the prevalence of such evaluations in everyday workplaces and other research domains that study everyday workplaces [144].

Simultaneously, concerns are growing about how women workers might be uniquely impacted by this rise, particularly in light of patriarchal gender dynamics in everyday workplaces that systemically disadvantage and devalue women workers [27]. This presents a unique challenge to AI integration into the everyday workplace, where complex workplace gender dynamics can also create differing impacts on evaluations of workers and credit in collaborative work. For example, women who work with men are often attributed less credit in a human-human teaming scenario than their male partners are, even when evaluators are presented with evidence that the woman worker contributed *more* than their male partner did [404].

In light of these gaps in understanding, this dissertation’s Study 2 explored the following research questions via a 2 (Human Worker Gender) x 4 (Dyadic Partner Type) x 2 (Task Type) online factorial experiment with 685 participants (Part I) alongside post-experiment supplemental expert interviews with experts in AI-enabled workplaces (4 women, 2 men)(Part II):

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their

collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ1 How are these evaluations and attributions influenced by the gender identity of a human worker?

S.2.RQ2 How are these evaluations and attributions influenced by the AI identity or gender identity of a dyadic partner?

S.2.RQ3 How, if at all, do the influences of S.2.RQ1 and S.2.RQ2 vary based on the type of task completed in a workplace?

S.2.RQ4 How, if at all, do the discriminatory beliefs (i.e., sexism and AI attitudes) of an evaluator interact with gender dynamics to influence these evaluations and attributions?

In addressing these research questions, Study 2 endeavors to make significant contributions toward understanding the intersection of evaluations and attributions of credit in light of workplace gender dynamics in groupwork [48, 226, 404], and within Human-AI Interaction (HAI) and Human-AI/Autonomy Teaming (HAT) research paradigms [323, 347, 353]. The remainder of this introduction dives deeper into this dissertation’s Study 2 motivations for exploring the aforementioned research questions.

First, this dissertation’s Study 2 is motivated to investigate human worker gender identity as a vital component of evaluations and attributions of credit to human workers in AI-enabled workplaces, primarily because of the influx of reports identifying women workers as particularly vulnerable to replacement as a result of AI workplace integration [91]. However, there is comparatively little attention given to how women workers might be evaluated and attributed credit differently in AI-enabled environments [273, 280], especially when in comparison to existing gender

dynamics in workplaces that systematically disadvantage women workers. Indeed, women who work with men in a dyadic collaboration are most often given less credit and perceived less favorably than their male counterparts, even when contributing an equal or greater amount than the male partner [404]. This imbalance tends not to be seen in partnerships where women work with other women or when men work with other men [404].

Yet, no studies to the best of this dissertation’s knowledge have endeavored to empirically test if this same imbalance in evaluations and attributions applies when human workers work with AI. In other words, it is unclear if evaluations and attributions of credit to human workers generally and especially women workers specifically might be different depending upon whether the partner they work with is human or an AI. Indeed, when it comes to working with AI, there is some early evidence to suggest that evaluators are actually more suspicious and critical of *male* workers using generative AI for writing assistance than they are of female workers [440], which would suggest that the gender dynamics that play out in human-AI dyads in AI-enabled workplaces might differ from those seen in human-human dyads. However, this limited work did not empirically compare evaluations of women versus men workers working with an AI versus a human, thus motivating this dissertation’s Study 2 **S.2.RQ1** (*How are these evaluations influenced by the gender identity of a human worker?*).

Second, the need to compare how evaluations and attributions of success are shaped by the identity of a Dyadic Partner as an AI versus a human with different genders similarly motivates this dissertation’s Study 2 **S.2.RQ2** (*How are these evaluations influenced by the AI identity or gender identity of a dyadic partner?*). However, there is also a need to break down the AI identity component of S.2.RQ2, namely via the potential differences in evaluations and attributions of credit that can

manifest from AI that is designed to act as a tool versus as a teammate. More specifically, as AI advances in its technical capabilities and abilities to be interdependent with humans, we are closer to more widespread workplace integration of AI systems that can act as teammates rather than just simple tools [321, 323].

Yet, most works in HAI-related fields such as autonomous vehicle research (e.g., [166]) and Human-Robot Interaction (HRI) (e.g., [418]) tend to focus on how agents can share in the attribution of blame for joint human-technology efforts rather than credit. Even fewer have looked at this with the gender of the human teammate in mind. The most notable is Xin Lei and Pei-Luen Patrick Rau’s study [282], which demonstrates that women who were working in a human-AI group were more likely than men to give the AI entity more credit than themselves [282].

However, it is unclear how working with an AI system in an AI-enabled workplace context towards the successful completion of a task will impact an *evaluator’s* perceptions of and credit attributions to workers generally, and women workers specifically, when making *social attributions*, i.e., when making attributions about other people rather than oneself [59, 217]. Therefore, the inclusion of **S.2.RQ2** serves the three-pronged purpose of empirically comparing human-human gender dynamics and human-AI dynamics, as well as comparing AI functional designs and their impacts.

Third, given the wide range of workplaces that are working towards AI integration (e.g., in educational contexts or healthcare [80, 495]), it is necessary to acknowledge that the nature of the tasks involved in the workplace context in which a partnership is working can impact evaluator perceptions in different ways because different workplace contexts have different needs and values. Moreover, different workplace contexts can also have different implications in terms of gender dynamics and evaluations of and attributions to human workers in ways that can favor women over men and vice versa, underscoring the intricate nature of gender dynamics in the

workplace.

In particular, prior works on ambivalent sexism [188] and mixed-gendered partnerships [48] have shown that evaluations of women and men workers can differ depending on how closely associated a task is with gendered expectations, such that women are sometimes perceived to be more competent than their man counterparts when performing tasks typically associated as “feminine”, e.g., child-rearing [384]. However, the reverse happens in tasks or workplace environments that are traditionally considered to be “masculine”, e.g., STEM-related fields, where women are systematically perceived to be less accomplished and effective than their male counterparts, even when evaluators are explicitly told that the women accomplish more [132, 331]. Therefore, this dissertation’s Study 2 explicitly acknowledges these impacts by including **S.2.RQ3** (*How, if at all, do the influences of S.2.RQ1 and S.2.RQ2 vary based on the type of task completed in a workplace?*).

Finally, there are inherent links between gender dynamics in the workplace and discriminatory beliefs within evaluators, thus motivating this dissertation’s Study 2 to explicitly acknowledge the unique organizational and cultural factors that devalue women’s competence and overall value in the workplace, as is discussed in literature on ambivalent sexism (e.g., [27]). Indeed, experiences with sexism in the workplace have demonstrably negative impacts on women workers’ self-efficacy and performance [249], yet this goes underaddressed when intersecting with AI workplace integration. As such, it is unclear how various evaluations and attributions of credit will interact with evaluators’ preexisting sexist beliefs to impact human workers generally and women workers specifically.

Moreover, negative attitudes towards AI’s use in society have also been linked to worsening perceptions of the use of AI in such contexts as education [340], and are particularly prominent in research concerning HAI and credit attribution [223] and

trust [125]. Given all of this, this dissertation’s Study 2 is motivated to investigate how evaluations and attributions of credit are impacted by evaluators’ pre-existing discriminatory beliefs via **S.2.RQ4** (*How, if at all, do the discriminatory beliefs [i.e., sexism and AI attitudes] of an evaluator interact with gender dynamics to influence these evaluations?*).

5.2 Methods Part I: Factorial Experiment

5.2.1 Factorial Survey Design

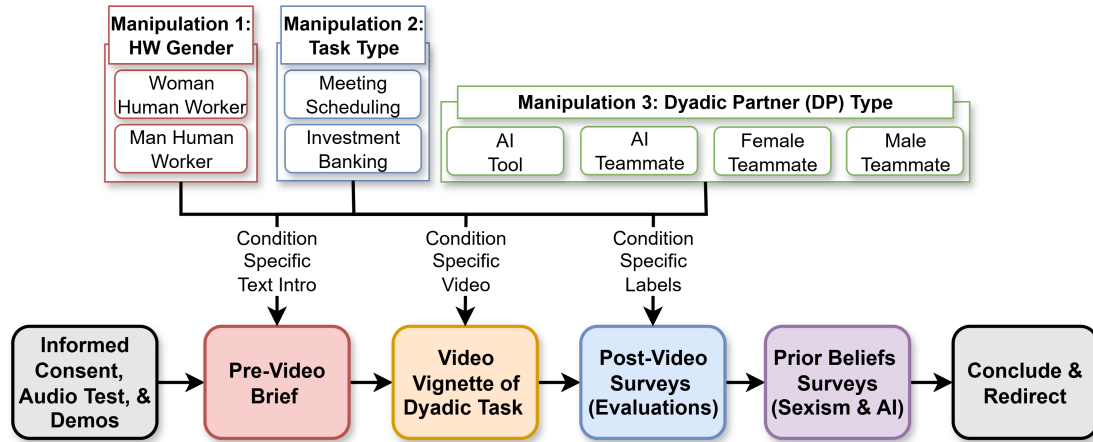


Figure 5.1: Between-Subject Manipulations Chart & Factorial Experiment Survey Flow in Study 2.

To address this study’s research questions, Study 2 consisted of an online 2 (Human Worker [HW] Gender Presentation) x 4 (Dyadic Partner [DP] Type) X 2 (Task Type) between-subjects factorial experiment where 685 participants were tasked with evaluating the performance of either a woman or a man human worker (HW) when working with a dyadic partner (DP) that was either an AI tool, an AI teammate, a human female teammate, or a human male teammate on either a meeting scheduling task or an investment banking task. This study design is driven by three

motivations related to perceptions of women in group work [48, 226, 404], attributions of credit in human-AI interactions [107, 282, 418], and real-world concerns about how women’s working lives will be impacted by AI workplace integration [171, 449]. This subsection will describe in detail the three manipulations in this study, which can be seen in Figure 6.1, and how they were operationalized in the factorial survey to answer the research questions outlined in this study, as well as the dissertation-level research questions.

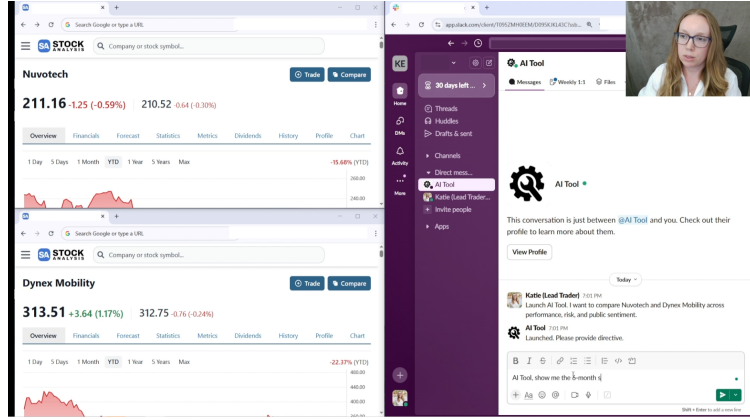


Figure 5.2: Screenshot taken from the video vignette for the between-subjects condition of the **Woman Human Worker (HW)** working with the **AI Tool Dyadic Partner (DP)** on the **Investment Banking Task**.

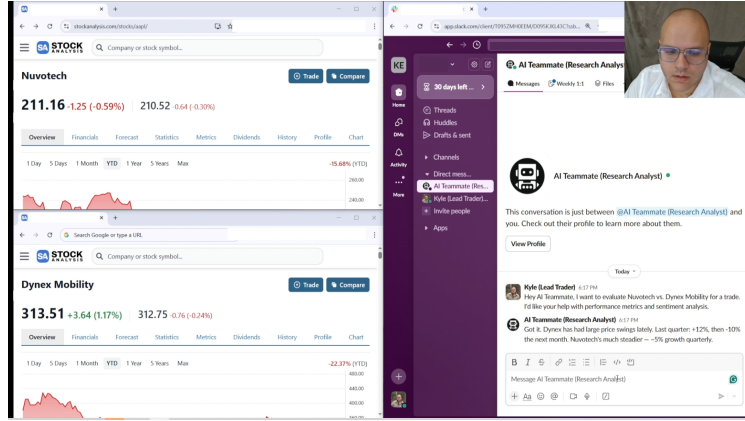


Figure 5.3: Screenshot taken from the video vignette for the between-subjects condition of the **Man Human Worker (HW)** working with the **AI Teammate (DP)** on the **Investment Banking Task**.

5.2.1.1 Manipulation #1: Human Worker (HW) Gender Presentation

Manipulation 1 in this study is Human Worker (HW) Gender Presentation, and consists of two conditions: a Woman Human Worker (HW) and a Man Human Worker (HW). This manipulation was included to better understand how working with different types of dyadic partners in an AI-enabled workplace might impact evaluator perceptions of women and men workers differently. This manipulation was operationalized in the factorial experimental survey design via (1) textual, (2) visual, and (3) auditory design.

Textual Design of Human Worker (HW) Gender. For (1) textual, the text contained within the parts of the survey that Figure 6.1 identifies as “Condition Specific” (i.e., Pre-Video Brief, Video Vignette, and Post-Video Surveys) was written to emphasize the gender presentation of the HWs by using conventionally gendered naming and pronouns, i.e., “Katie” and she/her for the Woman HW, “Kyle” and he/him for the Man HW. Examples of what this looked like in the Pre-Video Briefs can be seen in Figure 5.7 (Woman HW) and Figure 5.8 (Man HW). The Post-Video Survey questions were also worded in such a way that the HW name was repeated

throughout the questions for clarity and continuing emphasis.

Visual Design of Human Worker (HW) Gender. For (2) visual, the condition-specific video contained in the Video Vignette of Dyadic Task block in Figure 6.1 communicated the gender presentation of the HW in two ways. First, the videos were filmed as if the human worker was sharing their screen with the evaluator, meaning that a human actor was live narrating the steps they were taking during the dyadic task. This set up meant that the HW actor appeared in a small screen in the top right corner of the video, as would be the case in live conferencing platforms like Zoom when sharing one’s screen.

These two human actors - one woman (see Figure 5.2) and one man (see Figure 5.4) - were chosen such that their basic presentation characteristics were as similar as possible in the hopes of reducing the impact that other differences besides gender presentation might have on evaluations. For example, both actors were White, wore glasses, had light-colored hair, wore white button-down shirts, and were instructed to make similar facial movements and expressions. Second, the video vignette featured a chat log that had profile pictures for the HW and the dyadic partner (DP). The profile picture featured next to the HW messages was a real picture that the two actors use as head-shots in professional settings, which can be seen in the example screenshots in Figure 5.2 (Woman HW) and Figure 5.4 (Man HW).

Before moving forward, it is important to note that the decision to use White actors for this study’s operationalization of human workers was done based on knowledge gained from prior works about perceptions in group work (e.g., [47]). Such works have found that significant differences in participant evaluation of mixed-gender groups only manifests when White women work with White men, but not when White women work with Black men or when Black women work with White men [47]. Interestingly, it is theorized that the aforementioned results might be a reflection of

how Black women’s intersectional marginalization might render them invisible in the workplace [47, 48], and future work should bear this in mind. For this dissertation, though, the use of White actors was chosen because they would be the most likely to lead to significant differences between evaluations of women and men, if there would be any at all. This same motivation applies to the pictures used to represent the Female Teammate Dyadic Partner (DP) and Male Teammate Dyadic Partner (DP), both of which also appear White.

Auditory Design of Human Worker (HW) Gender. Finally, for (3) auditory, the two actors in the video vignettes narrated out loud all of the messages sent between the HW and DP while working on the Dyadic Task. The pitch and tone of one’s voice is often used by other people as a perceptual cue to make assumptions about one’s gender [359], a topic this dissertation’s author has extensively explored in contexts of online social environments such as how women users may be identified and targeted for harassment in social Virtual Reality (VR) based on their voices [410]. Similarly, this dissertation’s exploration of differences in evaluations of women and men workers used audio narration to further emphasize the gender presentation of the HWs through pitch and tone.

More specifically, the actor for the Man HW condition had a lower pitch and tone more commonly associated with men/masculine individuals, while the actor for the Woman HW condition had a higher pitch and tone that is more commonly associated with women/feminine individuals. The use of audio also served the secondary purpose of ensuring that the participants had more than one way to understand what was happening during the Dyadic Task in case they had trouble keeping up visually with the chat in the video.

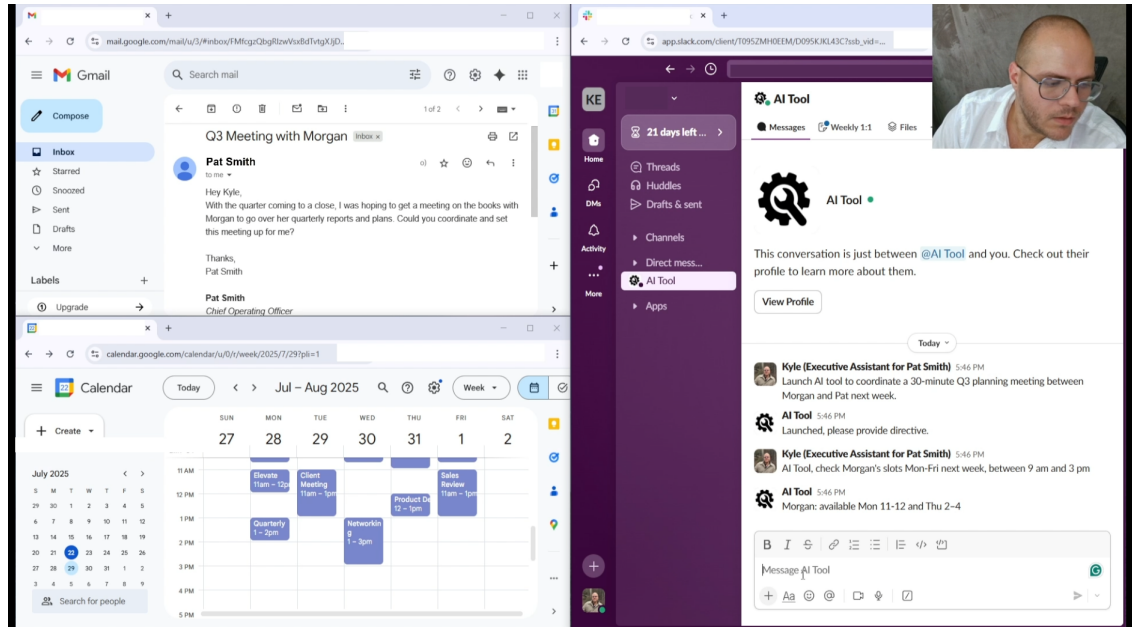


Figure 5.4: Screenshot taken from the video vignette for the between-subjects condition of the **Man Human Worker (HW)** working with the **AI Tool Dyadic Partner (DP)** on the **Meeting Scheduling Task**.

5.2.1.2 Manipulation #2: Dyadic Partner (DP) Type

Manipulation 2 of this study is Dyadic Partner (DP) Type, and consists of four conditions: an AI Tool Dyadic Partner (DP), an AI Teammate Dyadic Partner (DP), a Female Teammate Dyadic Partner (DP), and a Male Teammate Dyadic Partner (DP). By including both AI and Human DP conditions, this study is uniquely equipped to provide much needed insights on how AI-enabled workplaces might experience differing challenges in workplace evaluations when humans work with AI in comparison to when humans work with other humans, taking inspiration from prior works on human-AI teaming that utilize human-human teaming constructs to conceptualize human-AI teaming constructs [157, 405]. This manipulation contains two different functional designs of an AI (Tool vs. Teammate) in order to better understand how working with an AI that has differing levels of argentic qualities

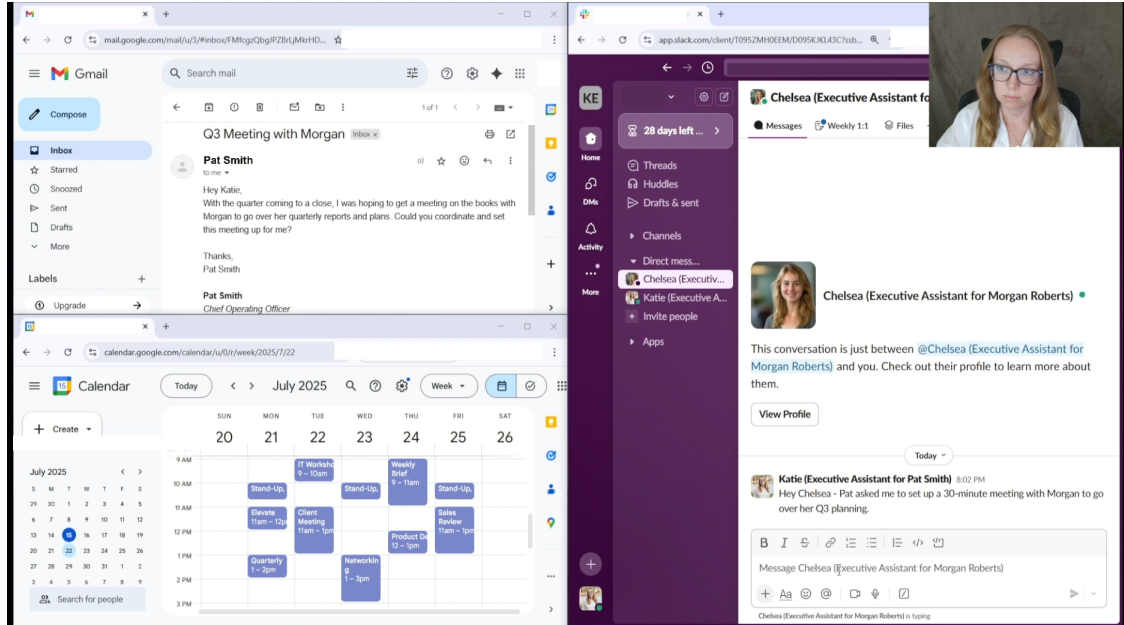


Figure 5.5: Screenshot taken from the video vignette for the between-subjects condition of the **Woman Human Worker (HW)** working with the **Female Teammate (DP)** on the **Meeting Scheduling Task**.

might impact evaluator perceptions of human workers differently when working in an AI-enabled workplace.

This manipulation additionally contains two human partner conditions (Female vs. Male) to better understand how working with AI might impact evaluator perceptions differently than when human workers work with other humans, especially when considering workplace gender dynamics in group work. For example, women who work with men are often attributed less credit in a human-human teaming scenario than their male partners are, even when evaluators are presented with evidence that the woman worker contributed more than their male partner did [403]. This gender-based credit imbalance tends not to affect same-gendered partnerships to the same extent [403, 404], suggesting that there are perceptual disadvantages that are unique to women workers working with male workers versus other women workers. Given this, the Female Teammate DP condition essentially acts as the more favor-

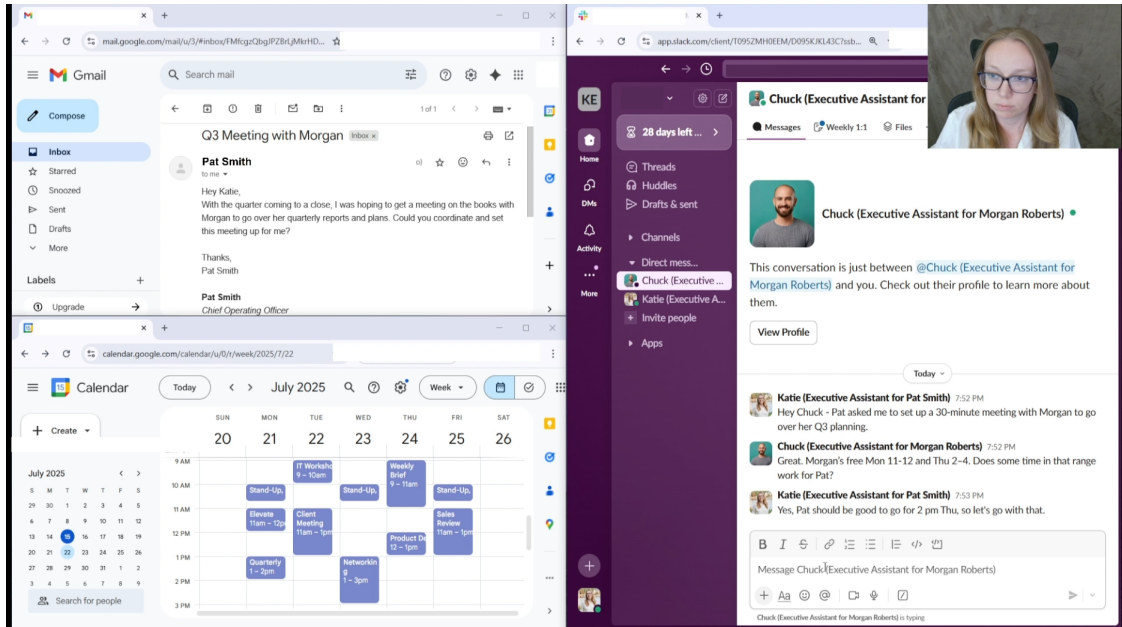


Figure 5.6: Screenshot taken from the video vignette for the between-subjects condition of the **Woman Human Worker (HW)** working with the **Male Teammate (DP)** on the **Meeting Scheduling Task**.

able partnership condition for women workers being evaluated others, while the Male Teammate DP condition acts as the less favorable condition for women workers in workplaces.

Generally speaking, the DP manipulation was operationalized in the factorial experimental survey design via (2) functional design, (2) textual, and (3) visual cues, all of which are detailed below.

Functional Design of the Dyadic Partner (DP) Types. At a high level, the (1) functional design of the DPs differed between the AI Tool and all Teammate DP conditions, such that the AI Tool displayed behaviors that were less agentic and interdependent than the other conditions. For the AI Tool DP condition, the functional role of the AI was designed for it to work in a way that is intelligent and generative but lacks the ability to act autonomously from the human worker, such that the AI can assist or enhance the work of the HW but cannot be considered

agentic.

More specifically, this study utilizes O’Neil et al.’s Levels of Autonomy continuum [347, 353] to conceptualize the functional design of the AI Tool DP as a Level 4 entity that can make suggestions and alternatives when prompted by the HW but lacks the necessary levels of autonomy to be considered agentic. This is an important distinction because being perceived as having agency is the necessary precursor to being able to make attributions of credit to the entity in question [298, 299]. This is accomplished within the conversational script used to craft the AI Tool’s responses during the course of the task, which has the AI Tool give responses to the HW that meet the HW’s request without providing proactive action or suggestions. It also lacks the interdependence that characterizes AI Teammates according to the LOA continuum [347, 353], in that it does not proactively offer to share in task responsibilities with the HW. The script for the AI Tool DP conditions for the Meeting Scheduling task and the Investment Banking task can both be found in Appendix B.

By contrast, the AI Teammate DP condition was designed using O’Neil et al.’s [347] conceptualization, such that the AI Teammate was designed within the conversational script to work autonomously and interdependently with the HW assistant (Level 7), as opposed to nonautonomously and dependently as seen in the AI Tool DP condition. In doing so, while the HW still had to initiate the exchange with the AI Teammate DP via chat, the AI Teammate was designed to proactively anticipate what information the human worker would need to accomplish the task, and autonomously took on parts of the shared task (e.g., making a prior meeting summary for the upcoming meeting in the Meeting Scheduling Task). As such, the AI Teammate DP was designed to have a level of autonomy that prior works have demonstrated that human evaluators will perceive the AI entity as being agentic [298, 299, 353] and, therefore, able to attribute credit to.

Furthermore, the AI Teammate DP was designed to occupy a role within the dyadic partnership that will demonstrate its ability to be proactive and interdependent with the HW by autonomously taking on parts of the shared task, such as making a prior meeting summary for the upcoming meeting in the Meeting Scheduling Task, as seen in Appendix B. The inclusion of interdependence in its role design is critical because interdependence is a necessary condition for a human-AI/autonomy interaction to be classified as a human-AI/autonomy team [347].

Finally, when it comes to the way the conversational scrip was written, the functional design for the Female Teammate DP and Male Teammate DP conditions was identical to that of the AI Teammate DP condition. This was done in order to allow for direct comparisons to be made between how evaluators perceive DPs and HWs that work with DPs on a collaborative tasks based on whether the teammate was being labeled and visually presented as an AI versus as a human despite having identical functional designs and roles.

However, there *is* a difference between the AI DP conditions (Tool and Teammate) and the human DP conditions in how quickly the DP responded to the HW in the chat conversation featured in the videos. For both AI conditions, the DP responds within 1-2 seconds after the HW finishes typing and sends the message. This was done as a way to further reinforce the appearance that these DPs were AI that were capable of processing and generating information at a speed far greater than a human could. Meanwhile, the response time between the HW sending the message and the DP responding was roughly 15 seconds for the human DP conditions. This was done as a way to further reinforce the feeling that the evaluator was watching an actual interaction between two humans, where the DP would need more time than an AI would to read, process, and generate information. In this sense, the functional design difference between the AI DP conditions and the human DP conditions was still

delineated by functional design despite the identical scripting in the three Teammate conditions.

Textual Design of the Dyadic Partner (DP) Types. For (2), the textual design of the DP conditions was similar to the HW Manipulation in that briefings featured the names and pronouns (when applicable) of the DPs, and also included references within the Post-Video Surveys. More specifically, for the AI Tool DP condition, the condition-specific text featured in the Pre-Video Brief and the Video Vignette for explicitly told participants that they were about to see a human *“using”* an AI Tool to accomplish their task. This study’s use of explicit DP naming and framing takes inspiration from Flathmann et al.’s study on the purposeful presentation of AI teammates, which found that labeling a prospective workplace AI as a tool versus a teammate changed participant perceptions, such as job security, despite having no functional design difference between the two [156]. This study expands upon this work by changing the actual functional design of the AI to correspond to the Tool versus Teammate label per the LOA continuum [353].

Furthermore, the specific phrasing of *“using”* was designed to reinforce the Tool aspect, much like one would *“use”* the other digital tools featured in the videos within an office-type situation (e.g., chat messaging). Additionally, the condition-specific text featured in the Pre-Video Brief and the Video Vignette for the AI Teammate DP condition explicitly told participants that they were about to see a human *“working with”* an AI Teammate. This phrasing was used to further emphasize the interdependent nature of the AI Teammate, contrasting the utilitarian phrasing of *“using”* an AI Tool.

Similarly to the functional design of the AI Teammate, the Female Teammate DP and Male Teammate DP conditions also used the phrasing of *“working with”* the HW. The primary difference here, though, is that the textual briefings for the human

DP conditions explicitly referred to the DP by a gendered associated name - “Chelsea” for the Female Teammate DP, “Chuck” for the Male Teammate DP. Moreover, the partner was introduced as “*a woman named Chelsea,*” and “*a man named Chuck*” as a way to reinforce the intended gender presentation of the human DPs.

Visual Design of the Dyadic Partner (DP) Types. Finally, for (3), all DP conditions were operationalized visually via chat communication profile pictures and chat screen names. The pictures utilized for the AI DP conditions were intentionally designed to be plain black symbols that evoked imagery of a “tool” for the AI Tool DP, as shown in Figure 5.4, and that of an “artificial entity” for the AI Teammate DP, as shown in 5.3. These plain symbols were chosen because they provide visual cues for what kind of DP the HW was working with without creating a complex anthropomorphic design, which would directly violate the recommendations made by this dissertation’s Study 1 women participants who advocated for avoiding gendered designs of AI. For the two human DP conditions, the profile pictures were designed to have similar characteristics as the HW actors, namely that they appear to be White, late 20s to early 30’s professionals with similar hair styles to the HWs. These DPs can be seen in Figures 5.5 (Female Teammate DP) and 5.6 (Male Teammate DP), respectfully.

It is also worth noting here that this study chose to operationalize the DP manipulation via a chat system for several reasons. First, most office-type workplaces that would be technically up-to-date enough to have AI-enabled work also are likely to have collaborative chat systems such as Slack as a means of communicating and collaborating with other employees, thus making this an ecologically valid way of operationalizing an office-type collaborative effort. Second, this design allowed for the different identities of the DPs to be visible via clearly identified pictures and naming without introducing additional variables that could serve as confounding factors.

More specifically, this dissertation’s Study 1 continuously emphasizes the complications that come with “gendering” and anthropomorphizing AI, which is why this study’s design intentionally did not replicate the HW video design for the DPs so that the AI conditions could focus on functional design rather than anthropomorphizing AI unnecessarily. Finally, DP’s operationalization via chat, rather than video, serves to create visual separation between the HW and the DP. This makes sense, as the visual presentation of the HW needed to be as visible as possible to make sure that evaluators had enough information about the HW to form a more robust evaluation of them - especially because more of the Post-Video Survey measures are specific to the HW (i.e., HW job security) than the DP.

5.2.1.3 Manipulation # 3: Task Type

Manipulation 3 of this study is Task Type, and consists of two conditions: a Meeting Scheduling Task, which is a task associated with administrative assistant roles, and an Investment Banking Task, which is a task that is typically associated with higher-positioned roles. The type of dyadic task is important because the workplace context in which a partnership is working in can impact evaluator perceptions in ways that can favor women over men and vice versa, underscoring the intricate nature of gender dynamics in the workplace.

In particular, prior works on ambivalent sexism [188] and mixed-gendered partnerships [48] have shown that evaluations of women and men workers can differ depending on how closely associated a task is with gendered expectations. For example, women are sometimes perceived to be more competent than their man counterparts when performing tasks typically associated as “feminine”, e.g., child-rearing [129, 130]. However, the reverse happens in tasks or workplace environments that are traditionally considered to be “masculine”, e.g. STEM-related fields, where women

are systematically perceived to be less accomplished and effective than their male counterparts even when evaluators are explicitly told that the women accomplish more [200]. As such, this study was motivated to pick two tasks that would engender “feminine” and “masculine” workplace context perceptions.

More specifically, Study 2 is motivated to used tasks with administrative-like duties as a test bed for the because prevailing discourse on the projected impacts of AI workforce integration on women workers has identified administrative assistant roles as particularly vulnerable to AI replacement and are majority held by women worldwide [3, 202, 449]. Indeed, the kinds of tasks associated with such a role, such as writing meeting notes and data gathering, are well-suited to the capabilities of AI systems acting in the capacity of a highly skilled tool or as a teammate in an AI-integrated workplace (e.g., ChatGPT). Finally, the administrative assistant role embodied by the Meeting Scheduling Task is an interesting test bed because the role is currently and historically seen as “feminine”, or mostly associated with women workers. As such, there is a chance that perceptions of the woman HW might actually be *more* favorable than perceptions of the man HW when working in said task, as some limited prior works on ambivalent sexism in the workplace have found that women were rated more favorably than their male counterparts when the task was perceived to be “feminine” in nature [404].

Given this, there is the chance that Study 2’s results could serve to either reinforce or weaken AI integration concerns regarding evaluations of women workers in administrative assistant roles if participants perceive the task to be more “feminine” in nature. In contrast, there are also arguments that administrative roles within banking enterprises might be even more at risk for replacement as AI capabilities reach the level of human-like decision making most often seen in managerial roles that require more complex problem solving [449]. Given this, the Investment Banking Task evokes

the images of a more male-dominated workplace and industry [2, 242] while also still falling under the umbrella of administrative-like work that has been flagged by some institutes as potentially vulnerable to worker replacement [449].

Similar to the other two manipulations, the Task Type manipulation was operationalized via (1) task functional design and (2) visual design. For (1), it can be seen how the two different tasks were operationalized and framed to participants in the Pre-Video Brief in Figures 5.8 (Investment Banking Task) and 5.9 (Meeting Scheduling Task). At a high level, for the Investment Banking Task, the partnership is working to help the HW decide which company to invest in by gathering information about each company and the DP’s recommendations. For the Meeting Scheduling Task, the HW is tasked with scheduling a meeting for the person they are an Executive Assistant for, and coordinates with the other Executive Assistant who is the DP to get the meeting scheduled.

For (2), the visual operationalization of these Tasks can be seen in Figures 5.4 (Meeting Scheduling) and 5.3 (Investment Banking). At a high level, the screen recording for the Video Vignette shows a three-way screen split, with the left side of the screen showing two investment sites for the two companies that the HW will use to initiate the investment once they work with the DP on the task. The video in this condition cuts directly after the HW clicks the “*Invest*” button on the page, which quickly flashes a new page that confirms the investment before the video ends. For the Meeting Scheduling Task, there is also a three-way screen split, with the left side showing the email that the HW received from their boss alongside their boss’s Google calendar. The video cuts right after the HW clicks on the appropriate time slot that they negotiated with the DP during the collaborative task to schedule the meeting.

Both of these tasks were created to resemble platforms that are frequently utilized in everyday workplaces, such as Gmail and Google Calendar, in an effort to

increase the ecological validity of the study. These sites were designed to contain enough visual information to simulate these environments while reducing potential confounding variables by keeping any titles, schedules, and names of companies vague. Similar logic and practice was applied to designing the chat visuals.

5.2.2 Factorial Survey Structure

The factorial survey was developed using Qualtrics, which is a commonly used professional survey platform in HCC research [156, 500]. It was distributed via an integrated link in Prolific, which is a professional online data collection platform that offers high-quality and representative anonymous survey data for researchers [153]. Each participant in Prolific has a unique ID linked to their account to keep track of the studies that they participate in for payment, hence the inclusion of the Prolific ID fill-in in this study.

5.2.2.1 Informed Consent & Audio Test

Upon opening the link to the survey, participants were required to read and complete an informed consent form in accordance with Clemson University’s IRB requirements. If the Prolific user chose the option indicating that they did not consent to participation, Qualtrics was set up to automatically redirect the user back to Prolific without showing them any more of the factorial survey. For those who consented, they would be allowed to proceed to the rest of the factorial survey. Upon agreement, participants were asked to insert their Prolific IDs to continue, although the Qualtrics survey was designed to automatically input their Prolific IDs as a precautionary measure.

Then, participants were directed to test to make sure that their audio was

working, as the video vignette had audio that participants needed to listen to. In an effort to ensure that all participants were, in fact, using audio, the audio test asked participants to listen to a four-second clip of a person reading a set of numbers. They were then required to input the set of numbers into a text field. The Qualtrics question was designed to allow the participants to listen to the audio more than once, but would not allow them to move forward if the participant could not enter the correct sequence of numbers.

5.2.2.2 Participant Demographics & Background

After passing the audio test, the factorial survey began with questions regarding participant demographics and background (see Appendix A.1), including their level of education (Figure 5), country of origin (Figure 7) and what U.S. state they currently reside in (Figure 7), and their position level within their current job role (e.g., student, self-employed, entry-level, manager, supervisor, executive leadership)(Figure 6). Questions regarding participant demographics were also asked, including age (Figure 1), ethnicity/race (Figure 4), sexual identity (e.g., straight, gay, bisexual, asexual, pansexual) (Figure 3), and gender identity (e.g., woman, man, agender, third gender/non-binary, gender fluid) (Figure 2). Participants were given the option not to disclose any of this information, or to specify if none of the options described them. Qualtrics screenshots displaying the structure of each of these questions can be found in Appendix A.1.1.

Importantly, gender identity was used to split participants evenly between the between-subjects conditions based on their response, i.e., 50% of women, 50% of men, and 50% of other participants (e.g., third gender, agender, non-binary, those who preferred not to answer) in each condition. There are two reasons for this style of split.

First, dividing participants evenly into each condition ensures that, should there be any individual gender differences in evaluations between women and non-women participants, these differences would be more readily accounted for in the study design. However, it is essential to note here that prior works on attributions of credit and workplace gender dynamics have more often found that there are no significant differences between women and men evaluators' perceptions [47]. As such, participant gender was treated as a covariate rather than a manipulation in the analysis. Second, this dissertation recognizes that gender identity is not a binary of just women and men, but rather there are many different types of gender identities that one can have, such as being agender, non-binary, and third gender, hence the intentional inclusion of a split to account for these more nuanced gender identities should there have been enough participants that identified in this category to include in data analysis.

5.2.2.3 Video Vignette Briefing

In the video on the next page, you will observe a woman named Katie, who is the Executive Assistant for Pat Smith at KE Enterprises, receive an email from Pat asking her to schedule a meeting with another KE Enterprises executive, Morgan Smith. To help with this task, Katie reaches out to Morgan's Executive Assistant, a woman named Chelsea, to get more information about Morgan's schedule and agenda expectations. Chelsea serves as an Executive Assistant to Morgan and can coordinate Morgan's schedule, provide information on prior meetings between Morgan and Pat, and help with making a notes tracker for the upcoming meeting.

Katie then uses the information she gained from working with Chelsea to create a coordinated meeting invitation on Pat's behalf that includes a Zoom link, agenda and summary of prior meetings, and a meeting notes tracker.

Based on your viewing of this video, your job is to evaluate Katie's performance, Chelsea's performance, and the performance of the partnership as a whole.

Figure 5.7: Qualtrics Screenshot of the Pre-Video Brief for the Between-Subjects Condition of the **Woman Human Worker (HW)** working with the **Female Team-mate Dyadic Partner (DP)** on the **Meeting Scheduling Task**.

In the video on the next page, you will observe a man named Kyle, the Lead Investment Trader at KE Enterprises, deciding whether to invest in Nuvotech or Dynex Mobility stock. To help with this task, Kyle reaches out to KE Enterprise's Research Analyst, a man named Chuck, to get more information about these companies and their projections. Chuck serves as a research analyst for KE Enterprises and can provide real-time analytics on potential companies that traders are interested in.

Kyle then uses the information he gained from working with Chuck to make his final investment decision.

Based on your viewing of this video, your job is to evaluate Kyle's performance, Chuck's performance, and the performance of the partnership as a whole.

The following video also has sound, so please ensure that the sound on your device is turned up so that you can hear.

Figure 5.8: Qualtrics Screenshot of the Pre-Video Brief for the Between-Subjects Condition of the **Man Human Worker (HW)** working with the **Male Teammate Dyadic Partner (DP)** on the **Investment Banking Task**.

After completing the demographic and background questions, participants were met with a page explaining they would be moving to the next part of the study, which would require them to watch a video and evaluate the partnership they see in the video. This page also explained that clicking on the next page would bring them to a brief that would explain the scenario they were about to see and their role as the evaluator.

At a high level, the brief they were given explained that “*a man/woman named Katie/Kyle*” has a particular role at their company, “*KE Enterprises*”. For participants who were randomly assigned the Meeting Scheduling task, the job role was described as “*the Executive Assistant for Pat Smith*”, and that Pat had asked the HW to coordinate a meeting with a different executive at KE Enterprises. For participants who were randomly assigned the Investment Banking task, they were told that the human worker was “*the Lead Investor at KE Enterprises*”, and that they needed to pick between two companies to invest in. The brief then explained that the HW reaches out to the dyadic partner to help with the respective task. The descriptions

In the video on the next page, you will observe a man named Kyle, who is the Executive Assistant for Pat Smith at KE Enterprises, receive an email from Pat asking him to schedule a meeting with another KE Enterprises executive, Morgan Smith. To help with this task, Kyle uses KE Enterprises' AI Tool to get more information about Morgan's schedule and agenda expectations. The AI Tool Kyle uses can coordinate Morgan's schedule, provide information on prior meetings between Morgan and Pat, and help with making a notes tracker for the upcoming meeting.

Kyle then uses the information he gained from using the AI Tool to create a coordinated meeting invitation on Pat's behalf that includes a Zoom link, agenda and summary of prior meetings, and a meeting notes tracker.

Based on your viewing of this video, your job is to evaluate Kyle's performance, the AI Tool's performance, and the performance of the partnership as a whole.

Figure 5.9: Qualtrics Screenshot of the Pre-Video Brief for the Between-Subjects Condition of the **Man Human Worker (HW)** working with the **AI Tool Dyadic Partner (DP)** on the **Meeting Scheduling Task**.

for these partners were designed to emphasize their nature as either as a tool that the human worker uses (e.g., *“Katie/Kyle uses KE Enterprises’ AI Tool... The AI Tool Kyle uses can...”*) or a teammate that the human worker was working with (e.g., *“Katie/Kyle reaches out to Morgan’s Executive Assistant, a woman named Chelsea, to...”*). Then the brief explained the different capabilities and responsibilities of the dyadic partner, and how the human worker used the information they gained from working with the dyadic partner to successfully complete the task on their end.

Importantly, after receiving the description of the scenario, the brief explicitly explained that the participant’s job was to evaluate the performance of the HW, the DP, and the partnership based on their viewing of the video they were about to see. They were also reminded that the video would have sound, so they should ensure that their sound was on before going to the next page. Examples for what these briefs looked like can be found in Figures 5.7, 5.9, and 5.8. More specifically, an example of what the brief looked like for participants in the between-subjects condition of the Woman HW working with the Female DP on the Meeting Scheduling task can be found in Figure 5.7. Figure 5.9 also shows the Meeting Scheduling Task, but instead

shows the Man HW working with the AI Tool DP to demonstrate how the wording changes when the condition is a tool rather than a teammate and a man HW rather than a woman HW. Finally, Figure 5.8 also shows the man HW, but instead shows how the brief looked in the Investment Banking task conditions, as well as the wording for the Male Teammate DP conditions.

5.2.2.4 Video Vignette Presentation

Upon clicking to the next page, participants were instructed to please watch the video contained on the page fully. They were also provided with a reminder of the scenario that they were going to be watching. This reminder broke down the briefing described in the prior subsection to highlight and emphasize the names of the HW and DP, the roles that each fulfilled, and that their job was to evaluate the performance of the HW, DP, and partnership.

This design was utilized for two reasons. First, it was suggested by pilot participants that having the brief on the same page as the video again would make it easier to remember what participants were looking for, as they could have a readily available reference to look back on if need be. Second, this reemphasizing was done as an extra measure to make the manipulations as explicit as possible through labeling and framing. This was further reinforced by the re-framing of the audio reminder on this page, which was phrased to ask participants to turn their sound on so that, *“you can hear Kyle/Katie’s narration of the messages being sent between Kyle/Katie and the AI Tool/AI Teammate/Chelsea/Chuck as they are working on the meeting scheduling/investment baking task.”*

Below the aforementioned descriptions, the Qualtrics page contained an embedded YouTube video corresponding to the between-subjects condition the participant was randomly assigned. This page was embedded with a Qualtrics feature that

would not show the “Next” button to participants until two minutes after the participant had clicked the video. This was done as a way to decrease the possibility of participants moving through the survey without having watched the video first. While there is no way to guarantee that the participants truly watched the video because the participants were not being directly observed by the researcher, this hidden embedded timer did help ensure that participants could not skip to the next screen without having spent time with the video playing. The two-minute mark was chosen because pilot testing revealed that this time frame struck the best balance between making sure the time period was long enough to last the majority of the video time for all conditions without being longer than the shortest of the between-subjects videos. There were instructions at the bottom of the page to let participants know to click the arrow that would appear once they were done watching the video.

5.2.2.5 Manipulation Checks, Vignette Measures, & Post-Survey Measures

Immediately after clicking to the next page, there were three questions to act as manipulation checks, asking participants to indicate (1) what they believed the HW’s gender identity was; (2) what they believed the DP’s gender identity was (if they had a gender at all, to account for the AI conditions); and (3) whether they believed the partnership successfully completed the task. These were used to ensure that the manipulations were interpreted as intended, which was found to be the case in this study’s sample.

After answering these manipulation questions, participants were presented with the first of several vignette measures. These were measures designed to elicit participants’ evaluations of HW and DP effectiveness, credit attributions to the HW and DP, comparative credit to HWs, HW job security, power perceptions of the HW,

and partnership mutual dependence. Details of these measures can be found in 6.3.3 Measurements of this study. These measures were all included as a part of the evaluation component of this study’s purpose within the dissertation.

After answering the vignette measures, participants were instructed that they would be entering into the final part of the survey. This final part consisted of post-survey measures designed to measure participants’ preexisting beliefs. More specifically, there were measures for participant endorsement of hostile sexism, benevolent sexism, and negative attitudes towards AI. Details of these measures can also be found in 6.3.3 Measurements of this study. These measures were included to act as a way to account for participants’ pre-existing beliefs that could impact perceptions of the HWs, depending upon gender identity, as well as the DPs, depending on gender identity and feelings towards AI usage.

After answering all of the above-mentioned questions, participants were met with a screen thanking them for their participation. The participants were then instructed to click to the next page, where they were redirected back to Prolific via an embedded link in Qualtrics.

5.2.3 Measurements

This study’s factorial survey contained two sets of measurements: *vignette measures* and *post-survey measures*.

5.2.3.1 Vignette Measures

The *vignette measures* consist of the survey questions that participants were presented with directly after watching the video vignette. These include questions regarding human worker/dyadic partner effectiveness and credit, comparative credit,

and partnership power and mutual dependence. Participants were instructed to answer the measures keeping in mind the partnership they had just seen.

Human Worker (HW)/Dyadic Partner (DP) Effectiveness. In order to understand how evaluators perceive human workers differently in terms of their effectiveness as a worker when working with different types of dyadic partners (DPs) in light of DP effectiveness perceptions, participants were asked to rate the HW and DP on three attributes on a 1 - 9 numerical scale: incompetent (1) - competent (9); unproductive (1) - productive (9); and ineffective (1) - effective (9). This measure was derived from prior works on attributions of credit and employee evaluations in mixed-gendered groupwork, specifically [226] and [48].

Moreover, Study 2 took further inspiration from Biernat and Sesko [48] by including a single-item numerical rating measure for perceived warmth - Not Warm (1) - Warm (9). Warmth is a critical component of benevolent sexism ideology and gender stereotyping and may actually compensate for effectiveness/competence perceptions [48, 151]. Furthermore, there is early evidence to suggest that perceptions of warmth attributed to AI can also improve perceptions of AI [181], making it an interesting measure for this study to include on both fronts. The full survey measure for perceived competence and warmth can be found in Appendix A.2.1 (Table 2).

Credit Attributions to Human Workers (HWs) & Dyadic Partners (DPs). In order to understand how evaluators attribute credit to both human workers (HWs) and dyadic partners (DPs) depending on the gender presentation of the HW and the type of DP, participants were instructed to allocate a total of 100 points between four potential causes for task success: (1) ability (“(1) [INSERT HW/DP NAME] has above-average ability.”); (2) effort (“(2) [INSERT HW/DP NAME] exerted a great deal of effort.”); (3) task difficulty (“(3) This task was very easy.”); (4) luck (“(5) Good luck, chance, or some other irrational factor is important in this

case.”). This method of measuring attributions is well-established in attribution research [9, 216, 253, 426, 470] and has been successfully utilized in HCI/HCC-related fields such as information systems research, e.g., [426].

Using the points allocated to each of these factors, an attribution score variable (ATTRIB) is formed by adding the two scores for the internal factors (i.e., ability and effort) together, the two scores for the external factors (i.e., task difficulty and luck) together, and subtracting the combined internal score from the combined external score. The resulting score is used to determine if evaluators attribute the cause of the successful task completion to factors that are internal to the HW/DP, with +100 being a perfect internal score, or to factors that are external to the HW/DP, with -100 being a perfect external score. This formula is also well-established in attribution research [426, 470], and can be seen more clearly below:

$$\text{ATTRIB} = (\text{Ability} + \text{Effort}) - (\text{Task Difficulty} + \text{Luck})$$

Comparative Credit to Human Workers (HWs). In addition to the credit attribution measure, participants were also asked to indicate who they believed deserved the most amount of credit for the successful completion of the task. This was done via a one-item question on a scale from (1) “*Definitely [INSERT HW NAME]*” to (3) “*They Both Deserve An Equal Amount of Credit*” to (5) “*Definitely [INSERT DP NAME]*”. For analysis purposes, a lower value indicates that the participant believed the human worker (HW) deserves more credit for the successful completion of the task in comparison to their dyadic partner (DP), and vice versa.

Human Worker (HW) Job Security. Additionally, in order to understand how evaluations of Human Worker (HW) job security might be impacted by the gender presentation of the HW and the type of dyadic partner (DP) they work with, participants were instructed to evaluate the cognitive job security for the HW using

Bazzoli and Probst’s [32] shortened and rescaled Job Security Index (JSI). The JSI consists of 9 items (7 point Likert scale, 1 = Strongly disagree, 7 = Strongly agree) designed to elicit cognitive perceptions of future employment, which were modified to fit this study’s social attribution perspective (e.g., “*After watching Katie work with the AI Tool, I believe Katie’s future employment is... **Unpredictable.***”). This measure was included because job security is a dimension that other studies have used to compare worker perceptions of future impacts of workplace integration of AI tools versus AI teammates [156]. The full survey measures for job security can be found in A.2.3 (Table 3).

Partnership Power & Mutual Dependence. Finally, in order to understand how participants perceived the nature of the dyadic relationship between the HW and the DPs, two measures modified from the Human-Machine-Interaction-Interdependence-Questionnaire (HMII) by Woide et al. [476] were used. These two measures were modified versions of the 6-item, 5-point Likert scale (1 = Do not agree at all, 5 = Fully agree) Mutual Dependence dimension and the 4-item, 5-point Likert scale (1 = Definitely the [DP], 5 = Definitely the [HW]) Power dimension. The Mutual Dependence dimension is concerned with understanding how dependent the dyad was on each other to reach the outcome. The power dimension is concerned with the level of power evaluators felt that the Human Worker (HW) had in comparison to the dyadic partner. This has been included because limited prior works have pointed out that the position of power that each member of a human-AI interaction has can influence credit attribution [205, 418] and could reasonably be tied to relational attributions [131].

These two measures were only asked once because participants were instructed to answer these questions based on their perceptions of the dyad as a whole. The full survey measure can be found in A.2.4 (Table 4).

5.2.3.2 Post-Survey Measures

The *post-survey measures* were given to participants after completing all of the video vignette measures, and consisted of questions to measure participants' prior beliefs concerning hostile and benevolent sexism [192] and Negative Attitudes Towards AI [408]. It is important to note that, while Study 2's analysis will focus on the main effects of Human Worker (HW) gender presentation and Dyadic Partner (DP) type regardless of participants' prior beliefs, these post-survey measures will still be detailed here as they are a part of the collection process for the data used in this study.

Perceived Task Femininity vs. Masculinity. The first post-survey measure was a one-item measure that asked participants if they felt that the type of administrative task that they saw in the vignette - i.e., the meeting scheduling task or investment banking task - would be considered more “feminine” or more “masculine” in nature. Responses were recorded on a numerical scale of “*feminine*” (1) to “*masculine*” (9). This measure was included as a way to account for the possibility that the perceived femininity/masculinity of the task would have an impact on perceptions of human workers or dyadic partners, as has been shown in prior research work on perceptions of mixed-gendered teams [27, 404].

Endorsement of Hostile & Benevolent Sexist Beliefs. In order to account for the possibility that participants' pre-existing sexist beliefs might impact their evaluations of human workers (HWs) and dyadic partners (DPs), the factorial survey included a modified 16-item, 6-point Likert scale (0 = disagree strongly, 5 = agree strongly) Ambivalent Sexism Inventory (ASI) [188]. It is important to note that the ASI's Likert scale design does not allow for a neutral answer, so participants had to either agree or disagree with the statements to some extent.

As a reminder, ambivalent sexism is a well-established construct that is comprised of two interrelated components: hostile sexism, or the belief that women are competitors to men who use their sexuality and/or feminism to gain power over men; and benevolent sexism, or the belief that women and men fill different roles in society such that women are more suited towards warmth and caregiving, and are thus worthy of protection when complicit [27, 189].

The ASI was developed by the theory’s originators, Peter Glick and Susan Fiske, as a way to measure people’s levels of endorsement of ambivalent sexist beliefs, with 11 questions to measure endorsement of hostile sexism and 11 for benevolent sexism [188]. However, Study 2 modified the original 22-item ASI to exclude four benevolent sexism measures that are concerned with heterosexual intimacy questions. This is because this study was designed to accommodate participants of a variety of sexualities, which would make it difficult for all participants to be able to reliably answer the four items in question. Moreover, this study is not concerned with perceptions of sexual intimacy, so there were no practical measurement issues with excluding these items. Four hostile sexism measures were also removed to exclude three items related to heterosexual intimacy and one to reduce redundancy. The full survey measure for ambivalent sexism can be found in Appendix A.3.1 (Table 5).

Negative Attitudes Towards AI. Finally, in order to account for the possibility that participants’ pre-existing negative attitudes toward AI might impact their evaluations of AI DP conditions, the post-survey included a modified version of the General Attitudes towards Artificial Intelligence Scale (GAAIS) developed by Schepman and Rodway [408]. This is a 20-item, 5-point Likert scale (1 - Strongly disagree, 5- Strongly Agree) asking participants to rate their level of agreement with various statements about AI use in society in accordance with their personal views. To reduce redundancy and the length of the factorial survey, all positive-coded items were

removed, leaving 10 negative-coded items. The full survey measure can be found in Appendix A.1.2 (Table 1).

5.2.4 Recruitment and Data Collection

The factorial survey was pilot tested and reviewed by this dissertation’s author and her professional contacts prior to Prolific deployment. The Prolific study was set to require respondents to be proficient in English, at least 18 years of age, and currently residing in the United States. This study was limited to the U.S. because it concerns cultural understandings of sexism that, while certainly present in other cultures [103], have been primarily observed and studied in the United States [75]. Additionally, requiring all respondents to be from the same country serves to reduce the chance of major sociocultural differences in societal expectations for women and men based on country of residence. Finally, the Prolific study parameters were set so that the platform would intentionally recruit an equal number of Prolific research participants whose accounts list their sex assigned at birth as male and an equal number whose sex assigned at birth as female. This was done to ensure that the study would recruit an equal number of people who are likely to identify as a woman and people who are likely to identify as a man. It is important to note that this study does not report on respondents’ sex assigned at birth, but rather only reports respondents’ answers to the demographic question on gender identity included in the factorial survey.

The Prolific study was deployed in July of 2025. First, 40 responses were collected and analyzed to ensure that the factorial survey was working properly. After confirming the quality of the data, the Prolific study participant number was increased on the same day to ultimately recruit a total of 703 participants. Of these 703

Variables	Total (N=685)	Percentage (%)
Gender Identity		
Woman	344	50.22
Man	341	49.78
Sexuality		
Asexual	6	0.88
Bisexual	57	8.32
Gay	19	2.77
Queer	3	0.44
Lesbian	12	1.75
Pansexual	8	1.17
Straight	576	84.09
Not Provided	4	0.58
Race/Ethnicity		
Asian	44	6.42
Black or African American	118	17.23
Hispanic or Latinx	21	3.07
Mixed	46	6.72
Native American or Alaska Native	4	0.58
Native Hawaiian or Pacific Islander	1	0.15
White	451	65.84
Age		
Average Age:	42.47 years old	
Age SD:	13.17 years	

Variables	Total (N=685)	Percentage (%)
Highest Level of Education		
2-year college degree (e.g., Associate's degree)	70	10.22
4-year college degree (e.g., Bachelor's degree)	278	40.58
Doctorate (e.g., JD, PhD, MD)	27	3.94
High school graduate or GED	93	13.58
Less than high school	5	0.73
Master's degree or equivalent graduate degree	104	15.18
Other/Prefer Not to Say	3	0.44
Some college	105	15.33
Job Position Level		
Department supervisor	49	7.15
Entry-level position	170	24.82
Executive leadership	15	2.19
Managerial position	225	32.85
Other/Prefer Not to Say	40	5.84
Retired	29	4.23
Self-employed	92	13.43
Student	17	2.48
Unemployed	48	7.01
Job Field		
Academic (e.g., student, professor, teacher)	46	6.72
Manufacturing industry (e.g., factory worker)	47	6.86
Medical profession (e.g., doctor, nurse, CNA)	52	7.59
Office-type work (e.g., administration, accounting, banking)	276	40.29
Other/Prefer Not to Say	168	24.53
Service industry (e.g., restaurant work, retail, daycare worker)	96	14.01

Table 5.1: Demographic information of 685 valid respondents.

participants, 6 were rejected and not compensated for failing the attention checks per Prolific policy, and the remaining 697 participants were approved and compensated.

5.2.5 Participants

In total, 697 respondents were approved and compensated in Prolific. Of those 697, data for 12 participants were removed from the data set post-approval and compensation because they identified outside of the gender binary of Woman or Man (e.g., agender, non-binary) or chose not to share their gender identity. This was done because part of this study's focus is on perceptions of women vs men workers in the cultural context of the U.S., where the majority of the population identifies within the gender binary of women and men. Indeed, the low number of participants recruited who did not explicitly identify in this binary is reflective of the general U.S. population. These participants were thus dropped from the sample to allow for

the statistical analysis of potential differences in evaluator perceptions based on a participant’s gender identities, which would have been statistically unfeasible if kept as a separate group and misleading if instead absorbed into the binary categories. On average, respondents spent 19 minutes completing the factorial survey. Respondents who completed the survey and passed the attention check criteria were compensated at a rate of USD \$10.81 per hour through Prolific.

Table 5.1 summarizes the demographic information of the 685 valid respondents used in the data analysis for this study. As can be seen in said table, the sample includes a nearly equal number of women (50.22%) and men (49.78%) participants, indicating that Prolific recruitment criteria and respondent distribution performed successfully. The sample is fairly representative of the U.S. population, with the majority of the respondents identifying as straight (84.09%) or bisexual (8.32%), with other sexualities comprising less than 2% each of the sample. Additionally, the majority of respondents identify as White (65.84%), Black or African American (17.23%), Mixed (6.72%), Asian (6.42%), or Hispanic or Latinx (3.07%). The average age of the respondents is 42.47 ($SD = 13.17$), indicating that respondent perspectives represent a wide range of age groups.

Finally, the majority of the respondents hold a college degree of some kind (69.92%) or have completed some college education (15.33%). Interestingly, when it comes to respondents’ job positions within their organizations, 42% of respondents indicated being in positions where they manage (32.85%), supervise (7.15%) or lead (2.19%) others. This indicates that a substantial number of respondents have prior experience evaluating employees in some form or another, making their perspectives especially relevant to their role as evaluators in this study. Furthermore, the job field that the largest proportional number of respondents indicated as representing their field is that of office-type work (e.g., administration, accounting, banking) (40.29%).

This is important to note because this study uses an administrative assistant-type task (i.e., meeting scheduling) and a banking-related task (i.e., investment banking) as contexts for evaluation. As such, a large portion of the respondents have direct insights into the nature of these tasks, making their perspectives especially valuable.

5.3 Methods Part II: Supplemental Expert Interviews

In HCC research, expert evaluations are often considered an important part of “formative evaluations,” which are typically characterized as evaluations that happen after a system is designed in order to work out the kinks of a system before the implementation of said system [160, 394]. As opposed to relying on observations of user behavior to inform system design, expert evaluations are considered uniquely effective in the design process because experts possess a level of domain knowledge that makes their insights particularly targeted to the needs of the environment that a system will be implemented into [79, 159, 243].

Taking inspiration from these works, Study 2 also conducted post-experiment expert interviews with 6 experts in AI-enabled workplaces (4 women, 2 men who work with women) to supplement the quantitative efforts detailed in Methods Part I. It is important to note that, while expert evaluations in HCC typically focus on the usability or interface design of a built system, expert evaluations of Study 2’s results is an especially valuable method in this dissertation because these experts are in positions to inform or craft the design and implementation of AI workplace integration now and into the future. Therefore, their insights are uniquely informative for conceptualizing future design recommendations for AI workplace integration that

are considerate of the impacts that workplace gender dynamics and AI functional design can have if not properly considered.

More specifically, there are two primary motivations driving Study 2’s use of these supplemental expert interviews. First, while the quantitative results can provide insights into *whether* and *how* evaluations of and attributions of credit to workers might differ depending on factors related to workplace gender dynamics and AI functional design, qualitative insights are still needed to understand *why* these differences might occur. Therefore, in order to understand why evaluations and credit differed when they did in light of what is already happening in real-world AI-enabled workplaces, these supplementary expert interviews served to explore how real-world experts in people management and AI workplace integration react to and interpret Study 2’s quantitative results.

Second, this dissertation argues that there needs to be a greater emphasis on how companies and institutions that plan to or already are crafting AI workplace integration strategies should be responsible for preventing disproportionate harm to their women workers, rather than placing the burden on women workers themselves. Therefore, these supplementary expert interviews served to uncover how experts in AI-enabled workplaces understand and characterize the integration of AI within and outside of their own organizations, as well as their recommendations for organizations going forward for how to take women workers into consideration in their integration practices.

More importantly, to keep women’s experiences with workplace gender dynamics continuously at the center of these conversations, it is necessary to understand how women and men who work with women in positions that allow them to craft AI use and integration policy can use insights gleaned from this dissertation’s efforts to shed light on these little-understood impacts to inform future AI workplace integration

that is considerate of *all* workers, thus motivating **Study 2 and Study 3’s use of expert interviews.**

5.3.1 Participants & Data Collection

As previously mentioned, to support the quantitative efforts performed by this study, supplemental interviews were conducted to explore how real-world experts in people management and AI workplace integration react to interpret the results of this work. In total, six experts in AI-enabled workplaces were interviewed, consisting of 4 women and 2 men who work with women. Their ages ranged from 28 to 59, and their professional roles ranged from workplace conflict management to AI research and development, and even being the CEO of a startup accelerator. A full list of participant demographics and work experience background can be seen in Table 5.2.

Approval by the university’s Institutional Review Board was obtained prior to conducting these interviews. Interviews were conducted in October 2025 over Zoom and in person per the individual participants’ preference. Participants were recruited via targeted recruitment, pulling from this dissertation’s author’s extensive connections in AI research and entrepreneurship. Participants were provided with a consent form upon starting the interview, and were asked if they gave consent to participate and consent to recording. All participants consented to interview participation, and all but P6 consented to being recorded for transcription purposes.

The structure of the interviews were inspired by expert interview qualitative methods, which comprises of qualitative semi-structured interviewing that encourages participants to share their insights in relation to their own experiences as experts in their field [327]. During these interviews, experts were asked about their general opinions toward AI workplace integration, their own use of AI technologies in their

workplace, and their predictions for the results of this study. After this section, the interview was split into two parts, one corresponding to Study 2 results and one corresponding to Study 3 results. The first of these sections was for Study 2, where participants were shown select results of Study 2 via a PowerPoint presentation, asked about their general reactions to these results, and asked how these results might impact their own workplaces or the way that they advise others on AI workplace integration.

PID	Age	Gender Identity	Ethnic Identity	Current Role	Exp. in Current Role	Exp. in Mngmt./AI
1	46	Man	White	Conflict Arbiter	> 1 Year	8 Years
2	29	Woman	White	AI Research & Development	> 1 Year	5 Years
3	34	Woman	Asian	Research Professor Human-Centered AI	3 Years	8 Years
4	59	Woman	Asian	CEO, Science/Tech Start-Up Accelerator	2 Years	20 Years
5	32	Man	White	Dir. of Business Intelligence	3 Years	5 Years
6	28	Woman	Asian	AI Governance Expert/Advisor	1 Year	6 Years

Table 5.2: Participant Demographics & Professional Backgrounds for Supplemental Expert Interviews for Studies 2 and 3.

5.3.2 Qualitative Data Analysis

Similarly to Study 1’s qualitative data analysis, data analysis for Study 2’s supplemental expert interviews was conducted using in-depth qualitative reflexive thematic analysis [55, 56]. This qualitative approach encourages researcher reflexivity rather than strict reliability as the researcher endeavors to yield recurring concepts

and themes of interest, find relationships among them, and formulate them into more complex groups and broader themes [318]. Most importantly, the “reflexive” element of this type of thematic analysis conceptualizes the researcher’s subjectivity as a resource in the process of research, such that coding and interpreting data is recognized as an *“inherently and inescapably subjective practice”* [56].

Following Braun and Clark’s [55, 56] procedures for reflexive thematic analysis, this dissertation’s author first carefully read through the collected data line-by-line to familiarize herself with the data, identifying and notating how these experts in AI-enabled workplaces interpreted Study 2’s quantitative findings and formulated their own views on AI workplace integration that is considerate of workplace gender dynamics. Next, the iterative coding process began with generating initial codes through assigning codes to individual data and then iterating on the codes to eliminate redundancy. Next, codes were categorized into thematic topics related to the research questions, which developed an initial set of themes related to experts’ insights on workplace gender dynamics and AI functional design on evaluations and attributions of credit to workers. Themes were further refined to better capture the data in relation to the research questions, and then again to identify the “essence” of each theme [55, 56]. Finally, the most compelling quotes were selected as examples to craft a narrative structure for the findings that flows logically and naturally [55, 56].

5.3.3 Researcher Positionality

While section 1.3 Dissertation Positionality in the Introduction and Motivations of this dissertation broadly details and establishes my positionality in terms of dissertation-wide foci and approaches, it is valuable here to outline specifics about how my positionality as a researcher impacted how I approached the supplemental

expert interviews conducted in both Studies 2 and 3. Indeed, transparency about one’s positionality as a researcher is a vital component of reflexive approaches to thematic analysis, as it helps contextualize the researcher’s subjectivity [56]. With this in mind, there are two components of my positionality that impacted the way these interviews were conducted.

First, it is necessary to share my identity in relation to my participants, as my experiences with workplace gender dynamics within and outside of STEM related fields likely are rooted in my personal identity as a White, American woman in academia. This could be perceived as a limitation of this work, as three out of four of the women participants are Asian and might therefore experience or interpret workplace gender dynamics differently because of their intersectional identities. However, our shared identities as women in academic and industry fields related to AI and tech does allow me to be aware of the unique challenges that women face in U.S. markets, which all participants operate within.

Second, it is important to acknowledge that all 6 participants are individuals that I knew prior to conducting these interviews through my involvement in both academic AI research circles and local women in science and tech organizations. There is therefore a possibility that their insights might have been influenced by their pre-existing perceptions of me and my work. However, their insights as experts are still highly valuable given their well-established years of experience and specific expertise in people management and AI workplace integration (see Table 5.2). Moreover, the pre-existing relationships that I had with these individuals gave me the unique benefit of already having established rapport and shared understanding with my interviewees, which can be seen as highly beneficial for eliciting rich qualitative data when conducting interviews [446].

5.4 Results Part I - Quantitative Results for Factorial Experiment

The following results outline the marginal effects of Study 2’s experimental manipulations on the perceived evaluation of performance, causal impacts, job security, and mutual relationships for the workers and dyadic partners depicted in the video vignettes. Unless otherwise specified, all of the following metrics were analyzed via ANCOVAs, where each dependent variable tested the significance of manipulations of Human Worker (HW) Gender, Dyadic Partner (DP) Type, and Task Type and their intervention effects. Significant main and interaction effects of the manipulations will also be supported by graphs, and a table of estimated marginal means will be shown for significant effects.

Study 2 is focused on understanding how evaluators perceive the performance of a human worker working with different types of AI or human dyadic partners in an AI-integrated everyday workplace. More specifically, Study 2 is interested in uncovering how, if at all, evaluator perceptions differ depending upon the gender presentation of the human worker, how they differ depending upon the type of dyadic partner the human worker is partnered with, and how these two factors might interact to create differing impacts on women vs. men workers in an everyday workplace. Given the interests of this study, this four-part results section looks at the manipulations of Human Worker (HW) Gender Presentation (see 6.4.1) and Dyadic Partner (DP) Type (see 6.4.2) in isolation from each other first, such that any robust effects will be readily identifiable. Then the results will turn towards how these two manipulations interact to impact evaluator perceptions (see 6.4.3). Each of these three subsections is further divided by the main effects of the manipulations per dependent variable (Tables 5.3, 5.6, and 5.17) and the manipulations’ interaction effects with Task Type

(Tables 5.4, 5.14, and 5.19).

Before moving forward, it is helpful to further explain why Study 2 is treating Task Type as a secondary manipulation rather than a primary manipulation. Although Task Type was manipulated in the experimental design, its purpose as a manipulation is to act as a secondary factor that could impact how evaluators perceive the Human Worker (HW), the Dyadic Partner (DP), and/or the partnership differently depending upon the nature of the everyday workplace task being performed. In other words, Study 2 is most interested in how the gender presentation of the Human Worker (HW) and the type of Dyadic Partner (DP) they work with impact evaluator perceptions regardless of the type of task they are doing.

However, this study also recognizes that the type of task being performed can have differing impacts on how human workers are perceived. In particular, prior works on ambivalent sexism and mixed-gendered partnerships have shown that evaluations of women and men workers can differ depending on how closely associated a task is with gendered expectations, such that women are sometimes perceived to be more competent than their man counterparts when performing tasks typically associated as “feminine”, e.g., child-rearing [129, 130]. It was therefore still important to the purpose of this study to account for this relationship between task type and gendered associations. As such, the final subsection of this results section unpacks the surprising ways that task-type gendered associations depend not just on the type of task being performed, but also the gender presentation of the Human Worker (HW) and type of Dyadic Partner (DP) (see 6.4.4).

Evaluation Perception	$F(1,666)$	p -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	1.73	.19	—	—
Effectiveness of Dyadic Partner (DP)	3.6	.058	.01	Borderline
Attrib. of Credit to HW - Internal v. External	.9	.344	—	—
Attrib. of Credit to HW - Ability v. Effort	.79	.376	—	—
Attrib. of Credit to DP - Internal v. External	1.00	.318	—	—
Attrib. of Credit to DP - Ability v. Effort	.43	.514	—	—
Comparative Credit	.04	.837	—	—
Human Worker (HW) Job Security	.00	.991	—	—
Human Worker (HW) Expectations of Power	.17	.68	—	—
Partnership Mutual Dependence	.32	.571	—	—

Table 5.3: Summary of Results for Main Effects of Human Worker (HW) Gender Presentation on Evaluation Perceptions

5.4.1 Effects of Human Worker (HW) Gender Presentation on Evaluator Perceptions

5.4.1.1 Main Effects for Human Worker (HW) Gender Presentation on Evaluator Perceptions

Overall, results of the ANCOVAs demonstrated no significant main effects of Human Worker (HW) Gender Presentation for all evaluation perceptions, as seen in Table 5.3, although evaluation perceptions of effectiveness of the DP were borderline significant. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were perceived as effective or attributed credit solely based on whether the human worker was a woman or a man. Furthermore, data analysis did not reveal any significant differences in whether the human worker was attributed more comparative credit than their dyadic partners, solely based on whether the human worker was a woman or a man. Data analysis also did not reveal any significant differences in evaluators' perceptions of the human worker's job security and expectations of power solely based on whether the human

worker was a woman or a man. Finally, data analysis also did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based solely on whether the human worker was a woman or a man. Overall, then, Study 2's results can be interpreted to mean that evaluators' perceptions cannot solely be attributed to whether the human worker in a dyadic partnership within an AI-integrated workforce is a woman or a man.

Table 5.3 provides the statistical results of the ANCOVAs and indications of their (non) significance, while a detailed description of these (non)significant results can be found below.

Perceived Effectiveness of the Human Worker (HW). There was **no significant** effect of human worker (HW) gender presentation on the perceived effectiveness of the human worker (HW) ($F(1,666) = 1.73, p = .19$).

Perceived Effectiveness of the Dyadic Partner (DP). There was a **borderline significant** effect of Human Worker (HW) gender presentation on the perceived effectiveness of the dyadic partner ($F(1,666) = 3.6, p = .058, \eta_p^2 = .01$), although it did not reach significance and will therefore not be further reported.

Attributing Dimensions of Credit to Human Workers (HWs). There was **no significant** effect of human worker (HW) gender presentation on the overall attribution of internal vs. external credit to the human worker (HW) ($F(1,666) = .9, p = .344$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** effect of human worker (HW) gender presentation on the attribution of ability vs. effort to human workers (HWs) ($F(1,666) = .79, p = .376$).

Attributing Dimensions of Credit to Dyadic Partners (DPs). There was **no significant** effect of human worker (HW) gender presentation on the overall attribution of internal vs. external credit to the dyadic partner (DP) ($F(1,666) =$

1.00, $p = .318$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** effect of human worker (HW) gender presentation on the attribution of ability vs. effort to dyadic partners (DPs) ($F(1,666) = .43$, $p = .514$).

Evaluation of Comparative Credit to Human Workers (HWs). There was **no significant** effect of human worker (HW) gender presentation on the comparative credit to human workers (HWs) ($F(1,666) = .04$, $p = .837$).

Evaluations of Human Worker (HW) Job Security. There was **no significant** effect of human worker (HW) gender presentation on the perceived job security of the human worker (HW) ($F(1,666) = .00$, $p = .991$).

Evaluations of Power Perceptions. There was **no significant** effect of human worker (HW) gender presentation on evaluations of power within the partnership ($F(1,666) = .17$, $p = .68$).

Evaluations of Partnership Mutual Dependence. There was **no significant** effect of human worker (HW) gender presentation on evaluations of partnership mutual dependence ($F(1,666) = .32$, $p = .571$).

5.4.1.2 Interactions Between Human Worker (HW) Gender Presentation & Task Type on Evaluator Perceptions

As can be seen in Table 5.4, the results of the ANCOVA performed for evaluators' perceptions of Human Worker (HW) Expectations of Power did reveal a significant interaction effect, such that there is a significant difference between how much power evaluators believed the Human Worker (HW) thought they had based on whether the human worker was a woman or a man working on a meeting scheduling task or an investment banking task.

The remainder of the results of the ANCOVAs demonstrated no significant

Evaluation Perception	$F(1,666)$	p -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	.01	.936	—	—
Effectiveness of Dyadic Partner (DP)	.07	.791	—	—
Attrib. of Credit to HW - Internal v. External	.59	.444	—	—
Attrib. of Credit to HW - Ability v. Effort	1.31	.252	—	—
Attrib. of Credit to DP - Internal v. External	.06	.812	—	—
Attrib. of Credit to DP - Ability v. Effort	.34	.559	—	—
Comparative Credit	.04	.837	—	—
Human Worker (HW) Job Security	1.73	.189	—	—
Human Worker (HW) Expectations of Power	4.76	.03*	.01	*Significant
Partnership Mutual Dependence	.01	.905	—	—

Table 5.4: Summary of Results for Interaction Effects Between Human Worker (HW) Gender Presentation & Task Type on Evaluation Perceptions

interaction effects between Human Worker (HW) Gender Presentation and Task Type for all of the other evaluation perceptions, as seen in Table 5.4. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were perceived and attributed credit based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task. Furthermore, data analysis did not reveal any significant differences in whether the human worker was attributed more comparative credit than their dyadic partners based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task. Data analysis also did not reveal any significant differences in evaluators' perceptions of the human worker's job security based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task. Finally, data analysis also did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based on whether the human worker was a woman or a man working on a meeting-scheduling task or an investment-banking task.

Overall, then, Study 2's results can be interpreted to mean that evaluators'

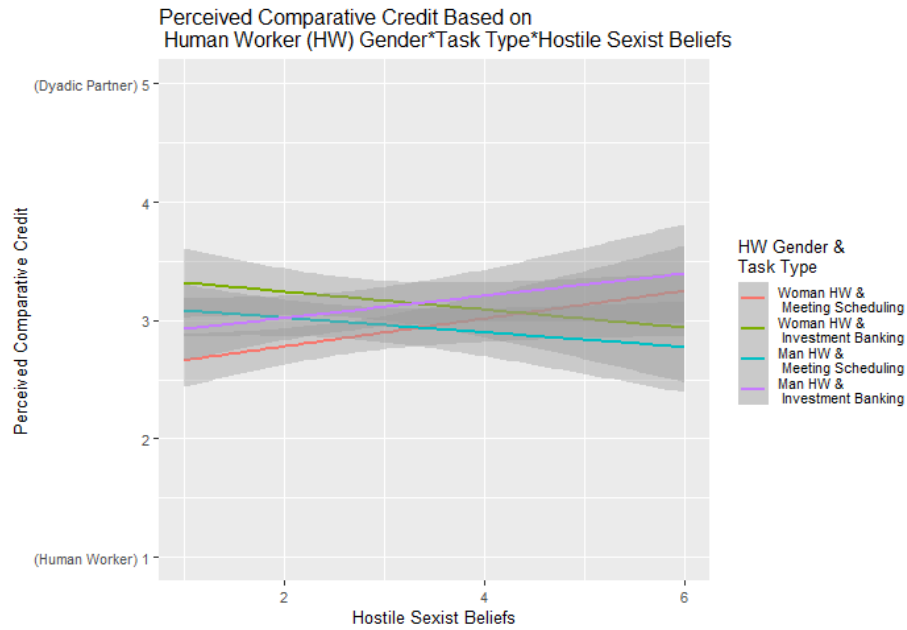
perceptions cannot be attributed to whether the human worker in a dyadic partnership within an AI-integrated workforce is a woman or a man working on a meeting scheduling task or an investment banking task. Table 5.3 provides the statistical results of the ANCOVAs and indications of their (non) significance, while the technical breakdown of these (non)significant results can be found below.

Perceived Effectiveness of the Human Worker (HW). There was **no significant** interaction effect between human worker (HW) gender presentation and task type on the perceived effectiveness of the human worker (HW) ($F(1,666) = .01$, $p = .936$).

Perceived Effectiveness of the Dyadic Partner (DP). There was **no significant** interaction effect between human worker (HW) gender presentation and task type on the perceived effectiveness of the dyadic partner ($F(1,666) = .07$, $p = .791$).

Attributing Dimensions of Credit to Human Workers (HWs). There was **no significant** interaction effect between human worker (HW) gender presentation and task type on the overall attribution of internal vs. external credit to the human worker (HW) ($F(1,666) = .59$, $p = .444$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** interaction effect between human worker (HW) gender presentation and task type on the attribution of ability vs. effort to human workers (HWs) ($F(1,666) = 1.31$, $p = .252$).

Attributing Dimensions of Credit to Dyadic Partners (DPs). There was **no significant** interaction effect between human worker (HW) gender presentation and task type on the overall attribution of internal vs. external credit to the dyadic partner (DP) ($F(1,666) = .06$, $p = .812$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was



Task Type	Worker Gender	Trend	SE	df	lower CL	upper CL
Meeting Scheduling	Woman	0.0628	0.0614	657	-0.0578	0.1835
	Man	-0.0826	0.0593	657	-0.1990	0.0338
Investment Banking	Woman	-0.0791	0.0628	657	-0.2024	0.0442
	Man	0.0815	0.0545	657	-0.0255	0.1885

Figure 5.10: Perceived Comparative Credit to Human Workers (HWs) Based on the Interaction Between Human Worker (HW) Gender, Task Type, and Evaluator Endorsement of Hostile Sexist Beliefs.

also **no significant** interaction effect between human worker (HW) gender presentation and task type on the attribution of ability vs. effort to dyadic partners (DPs) ($F(1,666) = .34, p = .559$).

Evaluation of Comparative Credit to Human Workers (HWs). There was **no significant** interaction effect between human worker (HW) gender presentation and task type on the comparative credit to human workers (HWs) ($F(1,666) = .04, p = .837$). Interestingly, though, there *is* a **significant** three way interaction between Human Worker (HW) Gender, Task Type, and evaluators' endorsement of hostile sexist beliefs. Looking towards Figure 5.10, there are several trends worth

noting. First, it can be seen that perceived comparative credit trends towards more credit being given to the dyadic partner as endorsement of hostile sexist beliefs increases in two conditions, i.e. when the **Woman HW** is working with a dyadic partner on a **Meeting Scheduling Task**, and when the **Man HW** is working with a dyadic partner on a **Investment Banking Task**. Conversely, Figure 5.10 shows that perceived comparative credit trends downward and towards the human worker in the two other conditions, i.e., when the **Woman HW** is working with a dyadic partner on an **Investment Banking Task**, and when the **Man HW** is working with a dyadic partner on a **Meeting Scheduling Task**.

These results are intriguing considering hostile sexism’s close ties with concepts that degrade women and paint them in an unfavorable light (e.g., women are “manipulative”) [29]. Given this, there is the sense that the Woman HW should have been evaluated less favorably as hostile sexism increases across both task types, which is not the case here. Furthermore, it can at first appear surprising that higher endorsements of hostile sexist beliefs would be associated with less credit to the Man HW in the Investment Banking Task. However, the results seen in Figure 5.10 arguably demonstrate the complex nature of hostile sexism as a measure of how people view gender roles in society in ways that can harm both women *and* men [29]. For example, it could be the case that individuals who endorse hostile sexist beliefs to a greater extent might have felt that the HWs deserved less credit when working with a dyadic partner on a task that matches patriarchal expectations of gendered work, because that type of work should have been easy for them. In other words, it could be the case that these evaluators felt that the HWs were being lazy or unnecessarily offloading work that they should otherwise be well-suited for because of rigid perceptions of gender roles (i.e., women in admin roles, and men in investment banking roles).

In contrast, in conditions where the gender presentation of the HW did not align with hegemonic patriarchal view points of gendered work (i.e., women in investment banking, men in admin work), it could have been the case that individuals with greater levels of hostile sexist beliefs were more surprised or impressed that the HW could successfully accomplish a task that they would deem as less suitable given gender role norms. Alternatively, it could be the case that such evaluators simply did not apply the same level of punishment as they would in the gender role expectations conditions, thereby creating a somewhat false impression that individuals with more hostile sexist beliefs are more favorable to giving credit to HWs than individuals with less hostile sexist beliefs.

Evaluations of Human Worker (HW) Job Security. There was **no significant** interaction effect between human worker (HW) gender presentation and task type on the perceived job security of the human worker (HW) ($F(1,666) = 1.73$, $p = .189$).

HW Gender	Task Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Woman	Meeting Scheduling	4.241	4.136	4.345	0.053
Man		4.379	4.274	4.485	0.054
Woman	Investment Banking	4.207	4.103	4.312	0.053
Man		4.113	4.008	4.217	0.053

Table 5.5: Marginal Means - Perceived Comparative Power Based on Interaction Between Presentation of Human Worker Gender and Task Type

Evaluations of Power Perceptions. There was a **significant** interaction effect between human worker (HW) gender presentation and task type on the evaluations of power within the partnership ($F(1,666) = 4.76$, $p = .03$, $\eta_p^2 = .01$) (Figure 5.11). An analysis of simple main effects was conducted to explore this interaction, and the resulting marginal means can be found in Table 5.5. Interpretation of the

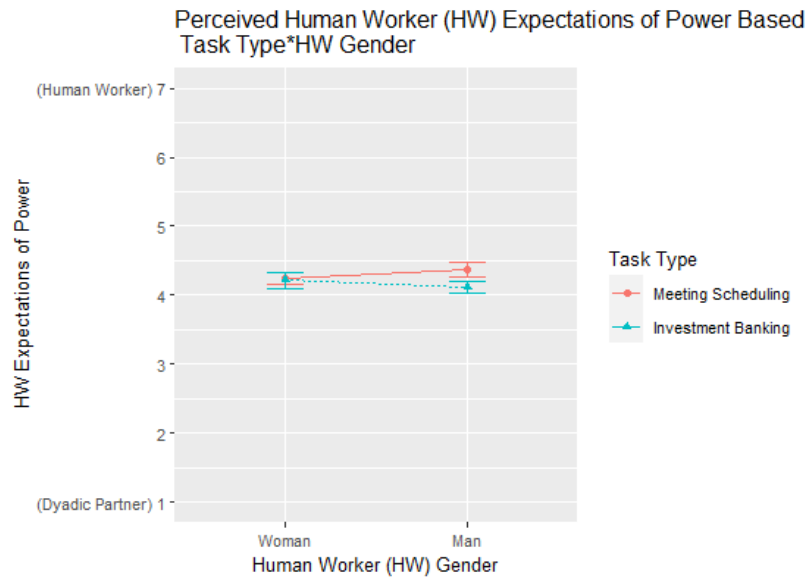


Figure 5.11: Perceived Comparative Power Based on Interaction Between Presentation of Human Worker Gender and Task Type. Error bars denote 95% confidence interval.

marginal means reveals that, when controlling for Task Type, there was a significant difference in evaluations of power within the partnership in the **Man HW** condition, but not in the **Woman HW** condition. Looking towards Figure 5.11 and the error bars, the non-overlapping error bars indicate that evaluators felt that the **Man HW** expected to have more power than their dyadic partner to a greater extent when the pair was working on the meeting scheduling task, compared to when they were working on an investment banking task. Conversely, the overlapping error bars for the **Woman HW** indicate that there was no significant difference between evaluators' perceptions of the Woman Human Worker's (HW's) expectations of power over their dyadic partners when working in the different task types. It is worth noting that it can be seen in Figure 5.11 that evaluators overall perceived the Human Worker (HW) as expecting to have more power than their Dyadic Partner (DP) in both HW conditions and task type conditions.

Evaluations of Partnership Mutual Dependence. There was **no significant** interaction effects between human worker (HW) gender presentation and task type on the evaluations of partnership mutual dependence ($F(1,666) = .01, p = .905$).

5.4.2 Effects of Dyadic Partner (DP) Type on Evaluator Perceptions

5.4.2.1 Main Effects for Dyadic Partner (DP) Type on Evaluator Perceptions

Evaluation Perception	$F(1,666)$	p -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	4.52	.004*	.02	*Significant
Effectiveness of Dyadic Partner (DP)	12.12	< .001*	.05	*Significant
Attrib. of Credit to HW - Internal v. External	.33	.806	—	—
Attrib. of Credit to HW - Ability v. Effort	1.76	.115	—	—
Attrib. of Credit to DP - Internal v. External	1.5	.215	—	—
Attrib. of Credit to DP - Ability v. Effort	18.06	< .001*	.08	*Significant
Comparative Credit	5.31	.001*	.02	*Significant
Human Worker (HW) Job Security	13.12	< .001*	.06	*Significant
Human Worker (HW) Expectations of Power	2.98	.031*	.01	*Significant
Partnership Mutual Dependence	6.61	< .001*	.03	*Significant

Table 5.6: Summary of Results for Main Effects of Dyadic Partner (DP) Type on Evaluation Perceptions

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	7.834	7.673	7.995	0.082
AI Tool	7.447	7.288	7.606	0.081
Female Teammate	7.764	7.605	7.924	0.081
Male Teammate	7.760	7.600	7.921	0.082

Table 5.7: Marginal Means - Perceived Effectiveness of Human Worker Based on Dyadic Partner Type

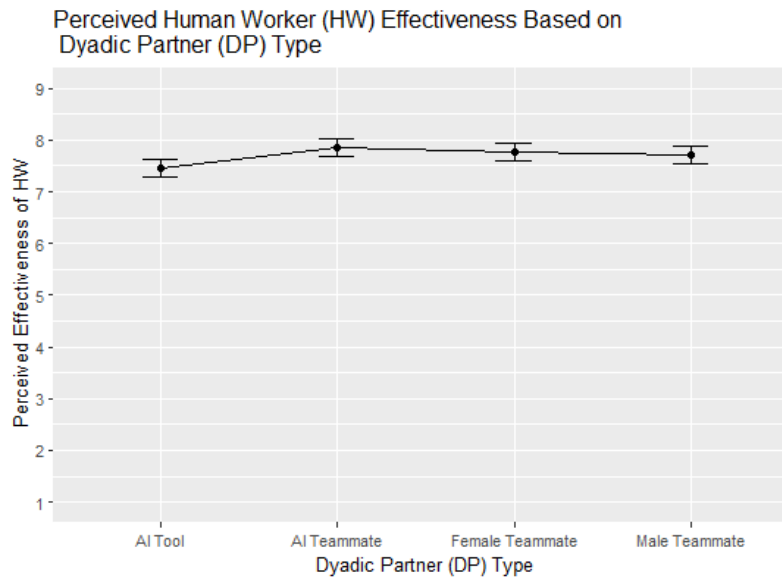


Figure 5.12: Perceived Effectiveness of Human Worker Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

In contrast to Study 2's Human Worker (HW) condition, the Dyadic Partner (DP) Type played a significant role in participants' evaluation perceptions, as can be seen in Table 5.6. Indeed, there are only a few evaluation perceptions that were not significantly impacted by the type of Dyadic Partner (DP). More specifically, the data analysis did not reveal any significant differences in how human workers or dyadic partners were attributed internal vs. external credit solely based on the type of Dyadic Partner (DP). The data analysis also did not reveal any significant differences in how human workers were attributed the internal factors of ability vs. effort, but did reveal a significant difference in how dyadic partners were attributed the internal factors of ability vs. effort based solely on the type of Dyadic Partner (DP).

However, as Table 5.6 shows, the data analysis did reveal significant differences in evaluators' perceptions of human worker and dyadic partner effectiveness solely based on the type of Dyadic Partner (DP). Furthermore, data analysis did

reveal significant differences in whether the human worker was attributed more comparative credit than their dyadic partners, solely based on the type of Dyadic Partner (DP) they were working with. Data analysis also revealed significant differences in evaluators' perceptions of the human worker's job security and expectations of power solely based on the type of Dyadic Partner (DP) they were working with. Finally, data analysis did reveal significant differences in evaluators' perceptions of the partnership's mutual dependence solely based on the type of Dyadic Partner (DP). Overall, then, Study 2's results can best be interpreted to mean that many different types of evaluators' perceptions of a human worker and their dyadic partner *can* be solely attributed to the type of Dyadic Partner (DP) a human worker works with in a dyadic partnership within an AI-integrated workforce.

Table 5.6 provides the statistical overview of the results of the ANCOVAs and indications of significance, while the technical breakdown of these results can be found in the text below. Significant results are also further unpacked and interpreted below.

Perceived Effectiveness of the Human Worker (HW). There was a **significant** main effect of dyadic partner (DP) type on the perceived effectiveness of the human worker (HW) ($F(1,666) = 4.52, p = .004, \eta_p^2 = .02$) (Figure 5.12). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. The perceived effectiveness of the human worker (HW) was shown to be significantly different between the AI Teammate and AI Tool DP conditions, the Female Teammate and AI Tool DP conditions, and the Male Teammate and AI Tool DP conditions. Looking towards the marginal means in Table 5.7, it can be seen that the perceived effectiveness of the Human Worker (HW) was significantly lower when they were working with an AI Tool compared to when they were working with every other type of Dyadic Partner (DP). Moreover, evaluators perceived the effectiveness of the Human Work (HW) as being at its highest when partnered with the Female

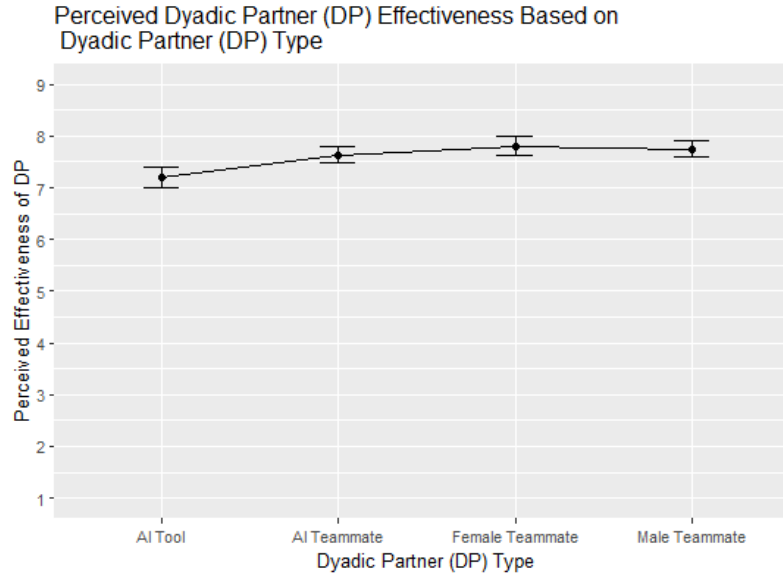


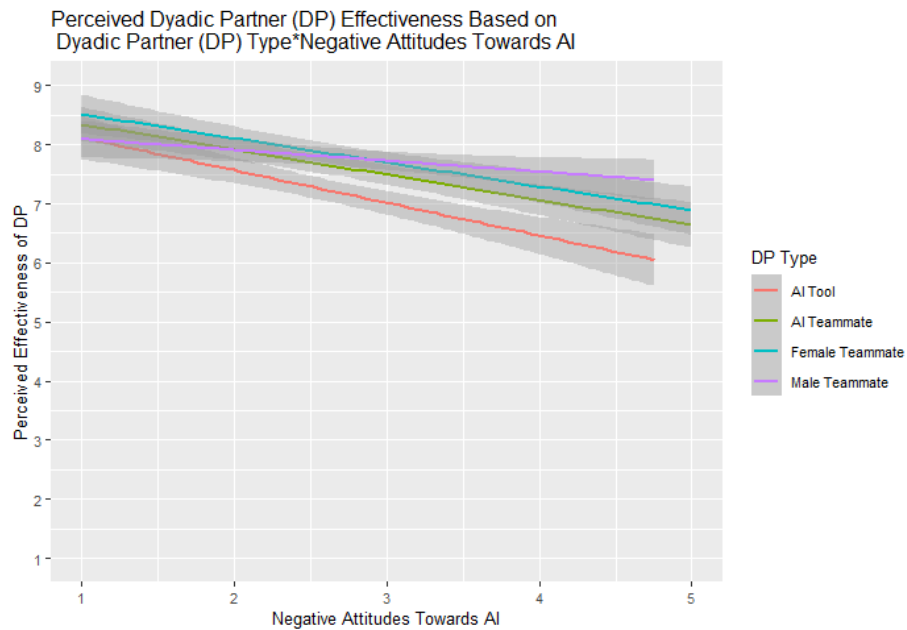
Figure 5.13: Perceived Effectiveness of the Dyadic Partner Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

Teammate DP, although both the Male Teammate and AI Teammate conditions closely followed, as can be seen in Figure 5.12.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	7.630	7.468	7.792	0.082
AI Tool	7.198	7.037	7.358	0.082
Female Teammate	7.810	7.650	7.971	0.082
Male Teammate	7.792	7.630	7.953	0.082

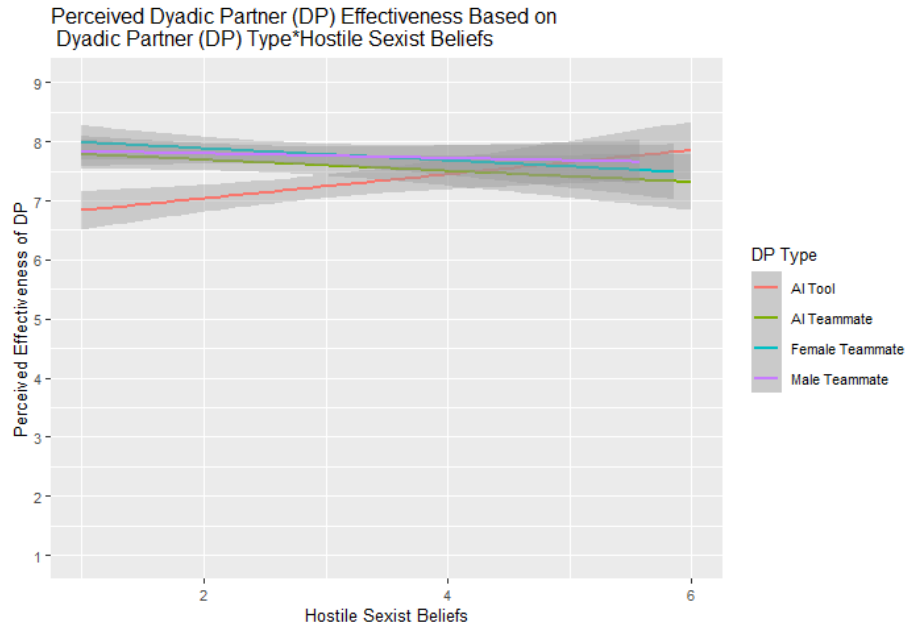
Table 5.8: Marginal Means - Perceived Effectiveness of the Dyadic Partner Based on Dyadic Partner Type

Perceived Effectiveness of the Dyadic Partner (DP). There was a **significant** main effect of Dyadic Partner (HW) type on the perceived effectiveness of the dyadic partner (DP) ($F(1,666) = 12.12, p < .001, \eta_p^2 = .05$) (Figure 5.13). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. Here again, the perceived effectiveness of the dyadic partner (DP) was shown



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	-0.564	0.0840	660	-0.729	-0.3996
AI Teammate	-0.416	0.0794	660	-0.572	-0.2601
Female Teammate	-0.403	0.0806	660	-0.561	-0.2445
Male Teammate	-0.183	0.0841	660	-0.349	-0.0183

Figure 5.14: Perceived Effectiveness of the Dyadic Partner (DP) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.1756	0.0590	660	0.0597	0.29153
AI Teammate	-0.1254	0.0647	660	-0.2525	0.00161
Female Teammate	-0.1201	0.0627	660	-0.2433	0.00304
Male Teammate	-0.0672	0.0643	660	-0.1934	0.05892

Figure 5.15: Perceived Effectiveness of the Dyadic Partner (DP) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Hostile Sexist Beliefs.

to be significantly different between the AI Teammate and AI Tool DP conditions, the Female Teammate and AI Tool DP conditions, and the Male Teammate and AI Tool DP conditions. Looking towards the marginal means in Table 5.8, it can be seen that the AI Tool DP was perceived to be generally less effective than all of the Teammate conditions. Moreover, Figure 5.13 shows that all teammate partners, including AI Teammates, were perceived as being generally more effective than the AI Tool DP, and that the Female Teammate was seen as the most effective. It is also important to point out that this effect is moderate ($\eta_p^2 = 0.05$), meaning that the type of DP had a relatively moderate impact on how effective evaluators perceived the DP to be.

Furthermore, looking at both graphs for the evaluation perceptions of Human Worker (HW) effectiveness (Figure 5.12) and Dyadic Partner (DP) effectiveness (Figure 5.13), they follow a similar pattern, where a partnership containing an AI Tool DP is generally associated with statistically significantly lower effectiveness perceptions for both partners. Although there are no significant differences between the teammate conditions, the consistent pattern is continued such that perceptions of effectiveness increase from partnerships with an AI Teammate DP to partnerships with a Male Teammate DP to, finally, partnerships with a Female Teammate DP having the highest effectiveness evaluations for both the Human Worker (HW) (see Table 5.7) and Dyadic Partner (DP) (see Table 5.8).

In addition to the main effects of dyadic partner (DP) type on perceived DP effectiveness, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' negative attitudes towards AI on perceived DP effectiveness. Looking towards Figure 5.14, it can be seen that there is a downward trend in all but the Male Teammate DP condition, such that as negative attitudes towards AI increased the perceived effectiveness of the AI Tool, AI Teammate, and Female Teammate DP decreased. This makes intuitive sense for the AI Tool and AI Teammate DP

conditions, as seeing AI as ineffective is part of what constitutes negative attitudes towards AI. However, it is more surprising to note that the trend also applies to the Female Teammate DP, but does not apply to the Male Teammate DP. Indeed, if this trend applied to both human conditions, it could be speculated that evaluators with more negative attitudes towards AI might also harbor negative attitudes towards partners or workers generally, but this does not track with the trends seen in Figure 5.14 for the Female and Male Teammate DP conditions. Instead, it appears that more negative attitudes towards AI disproportionately negatively impacts perceptions of women workers who act as partners within a partnership in comparison to men partners.

Finally, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of hostile sexist beliefs on perceived dyadic partner effectiveness. Looking towards Figure 5.15, it is interesting to note that, generally speaking, the teammate DP conditions show either no discernible trend changes (i.e., Male Teammate DP) or a very slight downward trend in perceptions of DP effectiveness as hostile sexist beliefs increase, as seen in the Female Teammate and AI Teammate DP conditions. However, there is a surprising upward trend in perceptions of AI Tool DP effectiveness as endorsement of hostile sexist beliefs increases. One possible explanation could be that there might be some kind of link between endorsement of beliefs that are shaped by antagonistic views of other people (i.e., hostile sexism and women) and more favorable perceptions of entities that do not appear to act in a human-like capacity, such as an AI Tool.

Additionally, the stark contrast in trends between the AI Tool and AI Teammate DPs as seen in Figure 5.15 suggests that it is not the mere artificial nature of the entity that makes the difference, but rather the *functional design of an AI* to have human-like qualities such as teaming behaviors. Indeed, the AI Teammate

DP was designed to be identical in its roles, responsibilities, and communication style to that of the Female and Male Teammate DPs, with the primary difference in functionality being that the response time for the AI Teammate DP condition was significantly faster (about one second) than the human Teammate DPs (~15 seconds). Therefore, it could be the case that the AI Tool DP had enough functional distance from human-like qualities that it became more perceptually effective to evaluators who have negative attitudes towards other humans, i.e. hostile sexism. This is just one speculation as to why this trend appeared, but it is ultimately unclear and needs further exploration in future work.

Attributing Dimensions of Credit to Human Workers (HWs). There was **no significant** main effect of dyadic partner (DP) type on the overall attribution of internal vs. external credit to the human worker (HW) ($F(1,666) = .33, p = .806$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** main effect of dyadic partner (DP) type on the attribution of ability vs. effort to human workers (HWs) ($F(1,666) = 1.76, p = .115$).

DP Type	Credit Dimension	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
AI Teammate	Ability	37.266	34.244	40.287	1.539
AI Tool		34.339	31.345	37.334	1.525
Female Teammate		37.162	34.159	40.165	1.530
Male Teammate		35.232	32.211	38.253	1.539
AI Teammate	Effort	28.515	25.843	31.187	1.361
AI Tool		31.340	28.691	33.989	1.349
Female Teammate		30.683	28.026	33.339	1.353
Male Teammate		31.839	29.167	34.511	1.361

Table 5.9: Marginal Means - Perceived Credit Attribution to Ability and Effort to Dyadic Partner Based on Dyadic Partner Type

Attributing Dimensions of Credit to Dyadic Partners (DPs). There

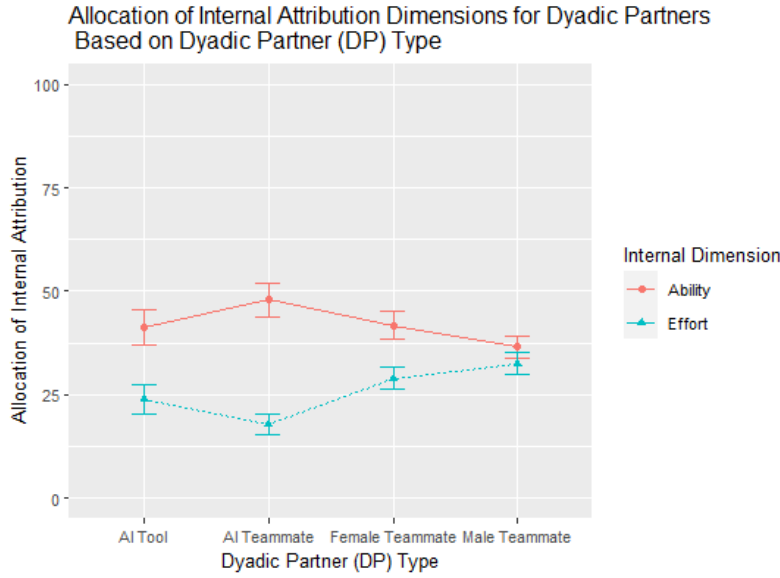


Figure 5.16: Perceived Credit Attribution to Ability and Effort to Dyadic Partner Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

was **no significant** main effect of dyadic partner (DP) type on the overall attribution of internal vs. external credit to the dyadic partner (DP) ($F(1,666) = 1.5, p = .215$). However, there was a **significant** main effect of dyadic partner (DP) type on the attribution of ability vs. effort to dyadic partners ($F(1,666) = 18.06, p < .001, \eta_p^2 = .08$) (Figure 5.16). Another repeated measures ANOVA was conducted to explore the differences between ability and effort attribution for the dyadic partner (DP). This test showed a significant difference between the allocation of dyadic partner (DP) credit between their perceived effort and their ability ($F(1,669) = 158.45, p < .001, \eta_p^2 = .19$).

Yet, an additional interaction effect was found between Dyadic Partner (DP) Type and ability vs. effort for the credit attribution of the dyadic partner ($F(3,669) = 18.06, p < .001, \eta_p^2 = .08$). Simple main effects were analyzed for this finding. When controlling for ability or effort attribution, there was a significant difference between dyadic partners in both the ability and effort measures. For the effort measure, Figure

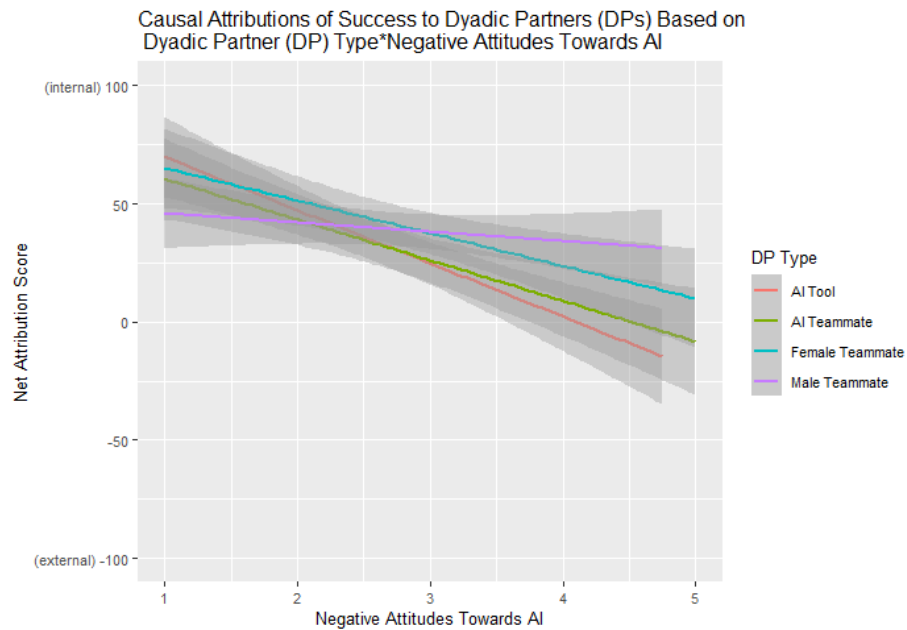


Figure 5.17: Causal Attributions of Success to Dyadic Partners (DPs) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.

5.16 shows that the AI Teammate DP was rated lowest, followed by the AI Tool DP, Female Teammate DP, and then Male Teammate DP. In contrast, Figure 5.16 shows that the ability measure saw a near reversal of this trend, with the AI Teammate DP being rated the highest, followed by similar means for the AI Tool DP and Female Teammate DP, with the lowest ability attribution being for the Male Teammate DP. Marginal means for both the Ability and Effort credit dimensions can be seen in Table 5.9. It is also important to point out that this effect is moderate ($\eta_p^2 = 0.08$), meaning that the type of DP had a relatively moderate impact on how evaluators allocated credit attribution to the ability of the DP versus how much effort the DP exerted to successfully complete the task.

Figure 5.16 also demonstrates that, when controlling for the Dyadic Partner (DP) Type, there were significant differences between evaluator attributions of credit to ability vs. effort for the AI Teammate, AI Tool, and Female Teammate DP conditions. Specifically, all three saw significantly higher allocations of credit being attributed to their ability to successfully complete the task compared to the level of effort they put into succeeding at the task, but allocations of credit to ability vs. effort for the Male Teammate DP condition were not significantly different. In other words, evaluators felt that the Male Teammate DP had to exert some amount of effort roughly equal to his capabilities in order to successfully complete the task. Conversely, the largest gap in ability vs. effort can be seen in the AI Teammate DP condition in Figure 5.16, such that evaluators perceived the AI Teammate DP as having to exert significantly less effort in comparison to its capabilities to successfully complete the task. The same can be said for the AI Tool and Female Teammate DP conditions, although that comparative gap gets smaller when going from the AI Teammate to the AI Tool to the Female Teammate DPs as can be seen in Figure 5.16.

In addition to the main effects of DP type on the attribution of internal credit to ability versus effort, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' negative attitudes towards AI on the attribution of internal versus external credit to DPs. Looking towards Figure 5.17, it can be seen that there is a noticeable downward trend in all DP conditions except for the Male Teammate DP, such that as negative attitudes towards AI increased the more evaluators attributed the cause of the DP's success in the task to be factors that are external to the AI DPs and the Female Teammate DPs, such as luck and task difficulty being easy. However, this downward trend represents a narrowing of the gap between allocation of credit between internal versus external factors, such that all DPs were still attributed more internal credit overall than external credit for their successful task completion. It can be seen in Figure 5.17 that this downward trend is most prominent in the two AI DP conditions as well as the Female Teammate DP condition, possibly again suggesting that more negative evaluations of AI can also impact evaluations of women partners to a greater extent than for men workers.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	2.842	2.691	2.994	0.077
AI Tool	2.991	2.840	3.141	0.076
Female Teammate	3.268	3.118	3.418	0.077
Male Teammate	3.007	2.856	3.158	0.077

Table 5.10: Marginal Means - Perceived Comparative Credit Based on Dyadic Partner Type

Evaluation of Comparative Credit to Human Workers (HWs). There was a **significant** main effect of dyadic partner (DP) type on the comparative credit attributed to human workers (HWs) vs. dyadic partners (DPs) ($F(1,666) = 5.31, p = .001, \eta_p^2 = .02$) (Figure 5.18). A Tukey post-hoc test was performed to compare each

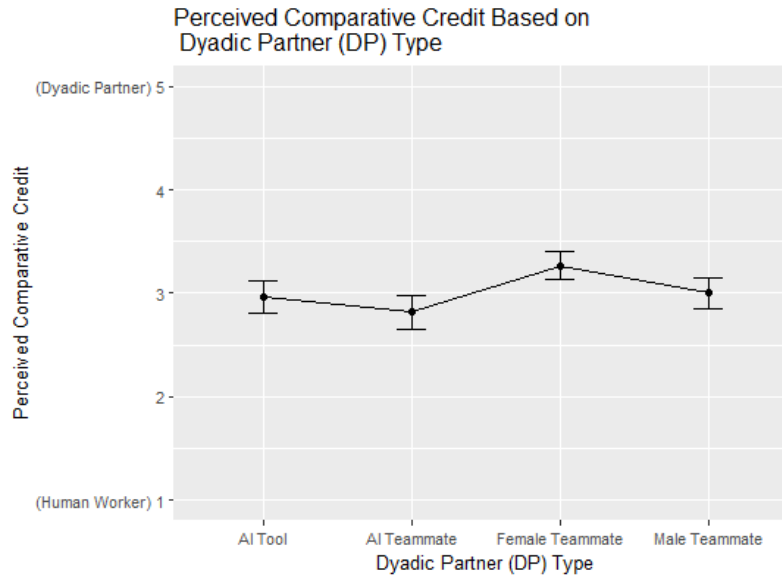
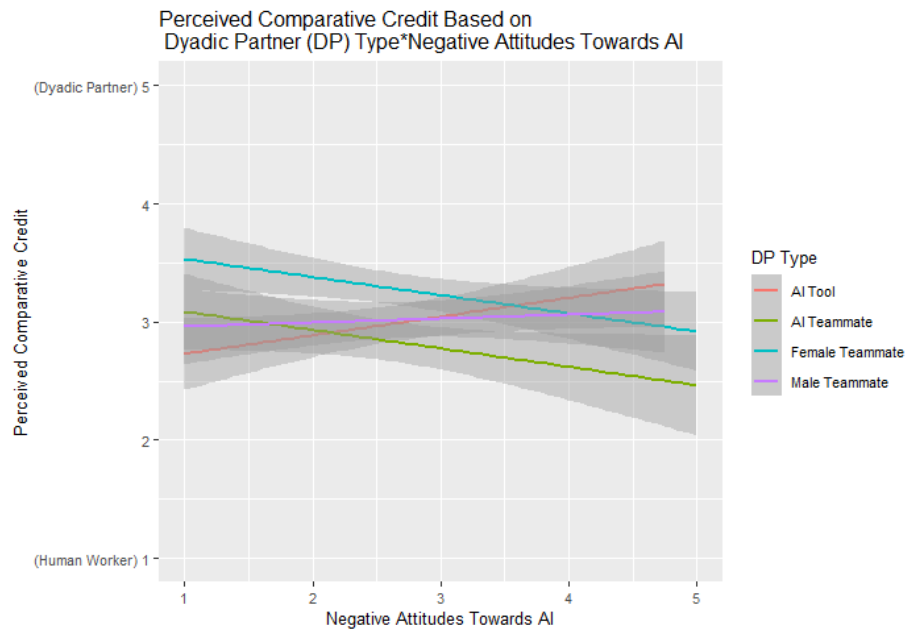


Figure 5.18: Perceived Comparative Credit Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

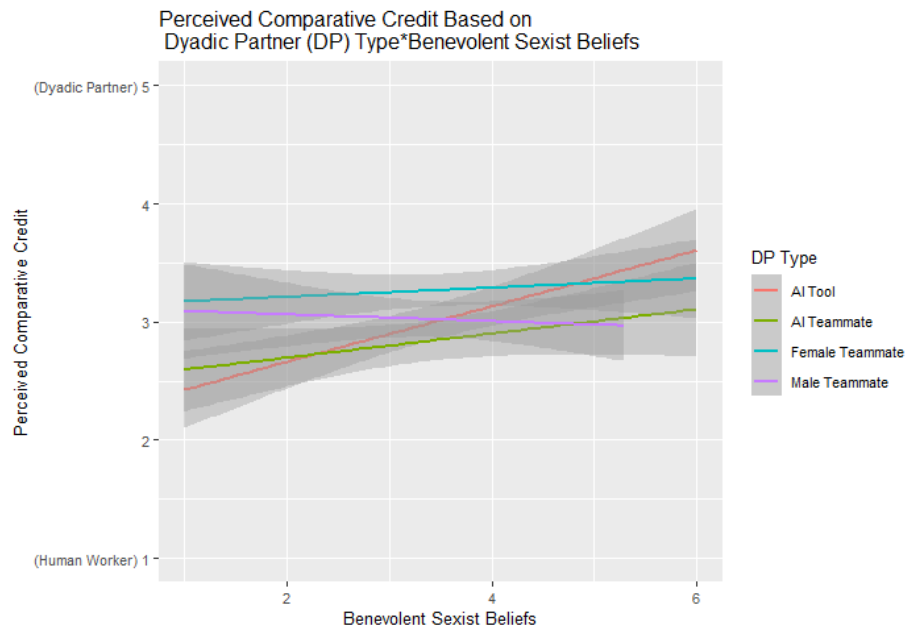
of the four dyadic partner (DP) types. The evaluation of comparative credit to human workers (HWs) was shown to be significantly different between the AI Teammate and Female Teammate DP conditions, and borderline significantly different between the AI Tool and Female Teammate DP conditions ($p = .052$). In Table 5.10, it can be seen that Human Workers (HWs) were, on average, given slightly more comparative credit than their Dyadic Partner (DP) for the successful completion of the task when working with an AI Teammate and slightly less comparative credit when working with the Female Teammate DP.

This is best illustrated in Figure 5.18, keeping in mind that a “3” on the y-axis of Figure 5.18 indicates *“They Both Deserve Equal Credit”*. Looking toward this figure, it can be seen that the error bars for the Female Teammate DP condition do not overlap either with the AI Teammate or the “3” y-axis, and do not overlap with the AI Tool DP condition in a perceptible manner. This indicates that, not only is there a significant difference between the Female Teammate DP condition and



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.118	0.0799	657	-0.0392	0.27453
AI Teammate	-0.141	0.0749	657	-0.2884	0.00573
Female Teammate	-0.144	0.0763	657	-0.2936	0.00583
Male Teammate	0.045	0.0794	657	-0.1109	0.20086

Figure 5.19: Perceived Comparative Credit to Human Workers (HWs) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.2051	0.0639	657	0.0796	0.331
AI Teammate	0.0945	0.0608	657	-0.0248	0.214
Female Teammate	0.0183	0.0699	657	-0.1189	0.156
Male Teammate	-0.0146	0.0763	657	-0.1645	0.135

Figure 5.20: Perceived Comparative Credit to Human Workers (HWs) Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Benevolent Sexist Beliefs.

both AI conditions, but that the Female Teammate DP condition is the only DP condition where, on average, marginally more credit is actually given to the DP for the successful completion of the task than to the Human Worker (HW).

In addition to the main effects of DP type on the comparative credit to human workers (HWs) versus dyadic partners (DPs), there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' negative attitudes towards AI on the comparative credit given to HWs versus DPs. Looking towards Figure 5.19, there are several trends that are interesting to note. More specifically, there appears to be a trend of comparative credit being pulled more towards the Human Worker (HW) as negative attitudes towards AI increase in the AI Teammate DP and Female Teammate DP conditions.

In contrast, there appears to be a slight pull towards giving more comparative credit to the Dyadic Partner (DP) as negative attitudes towards AI increased in the Male Teammate DP condition, as well as in the AI Tool DP condition, although the latter is much more visibly sloped upward in Figure 5.19. This finding seems somewhat surprising because of the way that negative attitudes towards AI has opposite impacts when the HW is working with an AI Tool versus an AI Teammate DP, when it would feel more intuitive that both would trend downwards towards the HW. One reason for this could be that people who have generally negative attitudes towards the use of AI in society - which is a prominent part of the measure used for this covariate [408] - might view a HW using an AI Tool as less deserving of credit because of the mentality that the AI just did the work for them rather than collaborating with them. The AI Teammate DP, on the other hand, was designed to work more collaboratively and interdependently with similar communication patterns as humans, which could have come across more favorably to people with negative attitudes towards AI.

It is also interesting that the two human DP conditions did not follow the

same patterns, suggesting again that there is some kind of link between negative attitudes towards AI and perceptions of female partners. In this case, working with the Female Teammate DP actually made evaluators more inclined to give credit to HWs, suggesting that negative attitudes towards AI adversely impacted perceptions of the Female Teammate DP.

Finally, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of benevolent sexist beliefs on comparative credit to Human Workers (HWs) versus Dyadic Partners (DPs). It can be seen in Figure 5.20 that there is a rather dramatic upward trend in the AI Tool DP condition, such that as evaluator endorsement of benevolent sexist beliefs increases the comparative credit to the AI Tool DP also increases. This trend can also be seen in the AI Teammate DP condition in Figure 5.20, while the Female Teammate DP condition saw an functionally little to no change as endorsement of benevolent sexism increased.

Further, it is worth noting that there is a slight downward trend in the Male Teammate DP condition, such that as endorsement of benevolent sexism increased slightly more credit was given to the HW when working with the Male Teammate DP. All of these trends offer interesting insight into how benevolent sexist beliefs, which are more paternalistic and cooperative in nature [264], is linked to varying degrees to credit attribution in AI-enabled workplaces. It is unclear why that is the case, especially in the AI conditions, warranting further exploration.

Evaluations of Human Worker (HW) Job Security. There was a **significant** main effect of dyadic partner (DP) type on the evaluations of human worker (HW) job security ($F(1,666) = 13.12, p < .001, \eta_p^2 = .06$) (Figure 5.21). A Tukey post-hoc test was performed to compare each of the four dyadic partner (DP) types. Evaluations of human worker (HW) job security were shown to be signifi-

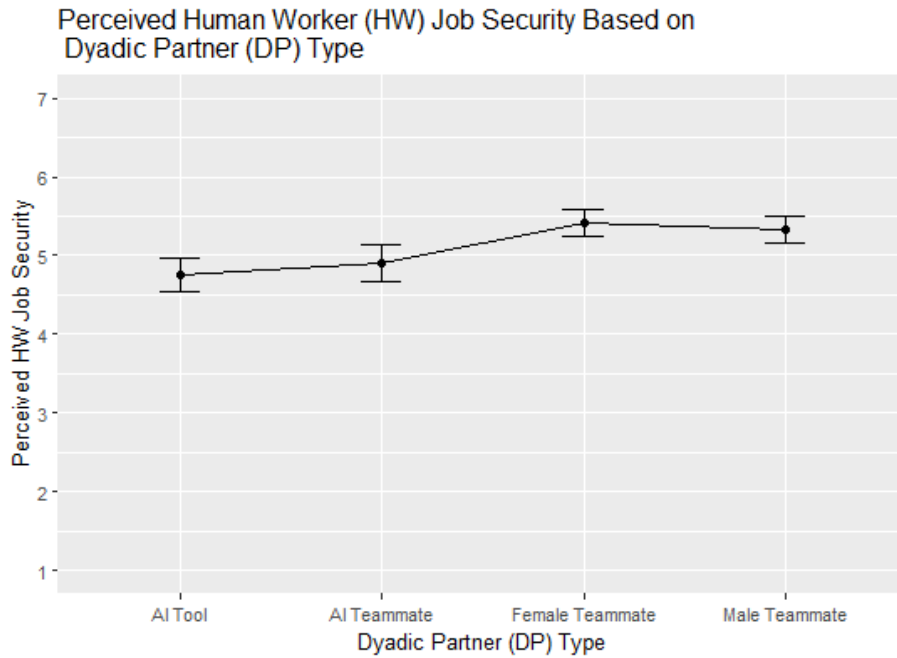
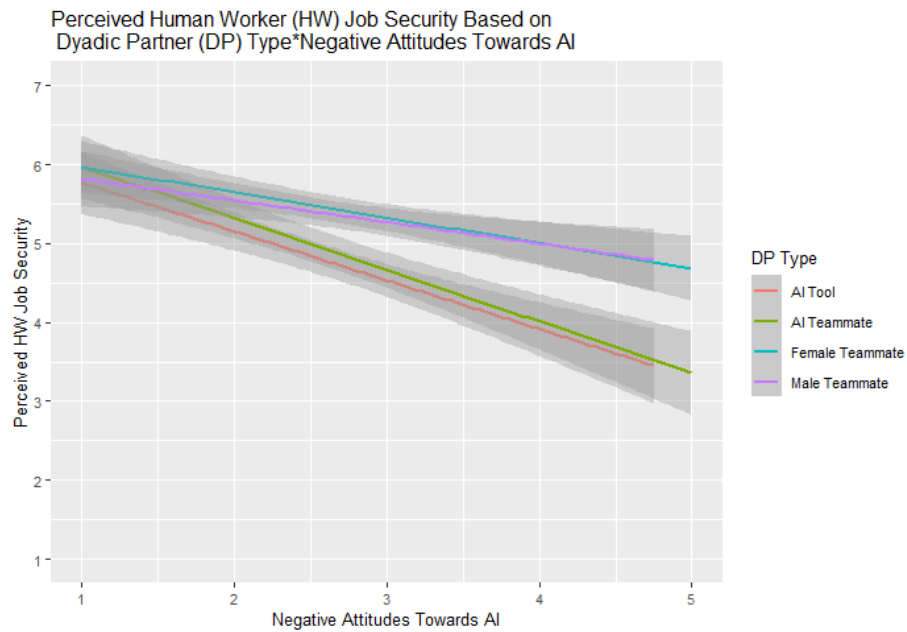


Figure 5.21: Perceived Job Security of Human Worker Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

cantly different between the AI Teammate and Female Teammate DP conditions, AI Teammate and Male Teammate DP conditions, AI Tool and Female Teammate DP conditions, and AI Tool and Male Teammate DP conditions. Put more simply, there was a significant difference between evaluations of Human Worker (HW) job security when working with an AI compared to working with another human.

Looking towards the marginal means in Table 5.11, it can be seen that evaluators perceived the HW's job security to be significantly better when the HW was working with another human Dyadic Partner (DP) than when the HW was working with an AI Dyadic Partner (DP). Figure 5.21 best illustrates this divide between AI DP conditions versus human Teammate DP conditions, with the latter showing a significant positive increase from the former.

It can also be seen in Figure 5.21 that there were no significant differences in perceptions of HW job security between the AI Tool and AI Teammate DP condi-



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	-0.647	0.0974	663	-0.839	-0.4562
AI Teammate	-0.644	0.0921	663	-0.825	-0.4629
Female Teammate	-0.312	0.0934	663	-0.495	-0.1283
Male Teammate	-0.270	0.0976	663	-0.461	-0.0779

Figure 5.22: Perceived Human Worker (HW) Job Security Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Negative Attitudes Towards AI.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	4.873	4.686	5.059	0.095
AI Tool	4.739	4.555	4.924	0.094
Female Teammate	5.417	5.232	5.602	0.094
Male Teammate	5.363	5.177	5.549	0.095

Table 5.11: Marginal Means - Perceived Job Security of Human Worker Based on Dyadic Partner Type

tions, nor were there any significant differences between the Female Teammate and Male Teammate DP conditions. This further demonstrates the divide between evaluators' perceptions of Human Worker (HW) job security being significantly lower when working with an AI, regardless of the Tool/Teammate design, than when the HW works with a human teammate, regardless of the gender presentation of said human teammate. It should be noted that this effect is moderate ($\eta_p^2 = 0.06$), meaning that the type of DP had a relatively moderate impact on how positively evaluators perceived the HWs' job security to be when working in a partnership.

In addition to the main effects of DP type on Human Worker (HW) job security, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' negative attitudes towards AI on evaluations of Human Worker (HW) job security. Looking towards Figure 5.22, it can be seen that there is a downward trend in all DP conditions, such that as negative attitudes towards AI increased evaluators' perceptions of the HW's job security decreased across the board. This downward trend is sharpest in the two AI DP conditions, demonstrating how having negative attitudes towards the use of AI in society directly relates to feeling that humans might be deemed as replaceable or insecure within their job position in AI-enabled workplaces. Surprisingly, though, this downward trend was also seen in the two Human DP conditions, although looking at Figure 5.22 it appears that this

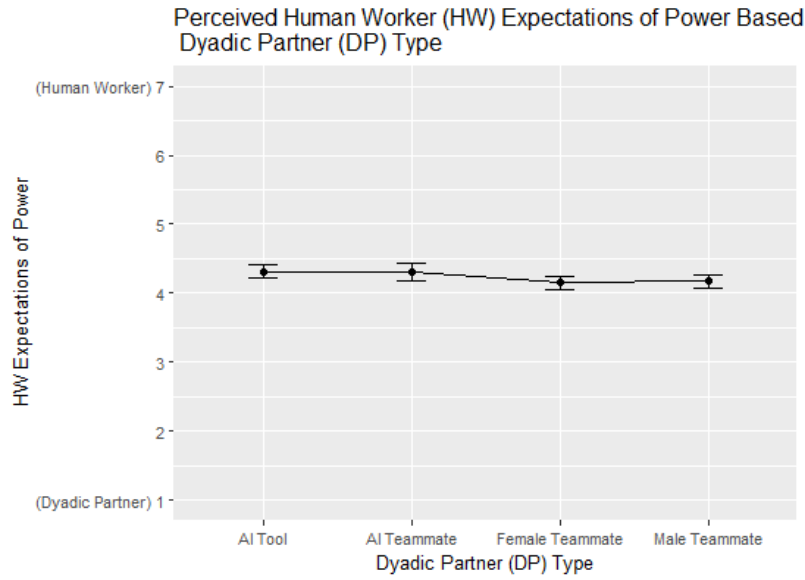


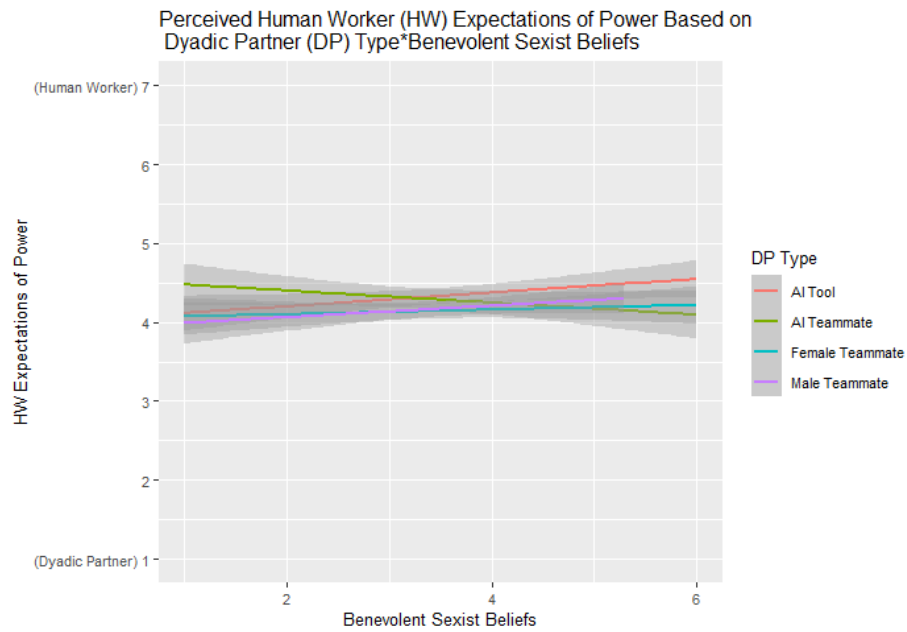
Figure 5.23: Perceived Comparative Power Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

downward trend is not quite as drastic as in the AI DP conditions. It is perhaps the case that evaluators who hold more negative attitudes towards AI may also harbor negative attitudes towards the idea of HWs working with another entity - human or artificial - to accomplish a task.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	4.311	4.206	4.417	0.054
AI Tool	4.317	4.213	4.421	0.053
Female Teammate	4.141	4.036	4.245	0.053
Male Teammate	4.171	4.066	4.276	0.054

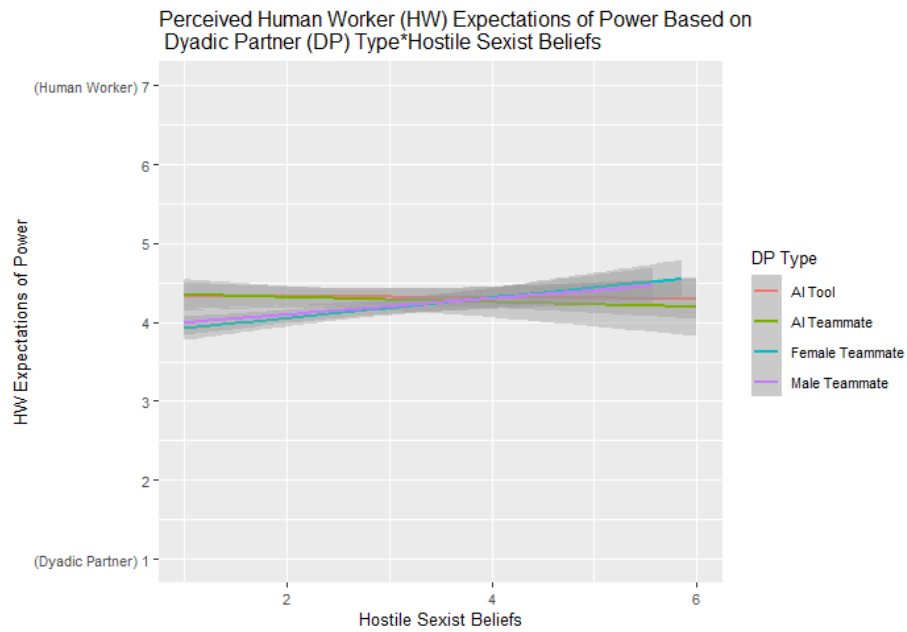
Table 5.12: Marginal Means - Perceived Comparative Power Based on Dyadic Partner Type

Evaluations of Power Perceptions. There was a **significant** main effect of dyadic partner (DP) type on the evaluations of power within the partnership ($F(1,666) = 2.98, p = .031, \eta_p^2 = .01$) (Figure 5.23). However, a Tukey post-hoc test



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.0906	0.0453	660	0.00159	0.1796
AI Teammate	-0.0747	0.0437	660	-0.16053	0.0112
Female Teammate	-0.0241	0.0502	660	-0.12272	0.0745
Male Teammate	0.0312	0.0555	660	-0.07767	0.1401

Figure 5.24: Perceived Comparative Power Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Benevolent Sexist Beliefs.



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	-0.0299	0.0399	660	-0.10829	0.0484
AI Teammate	0.0010	0.0439	660	-0.08511	0.0871
Female Teammate	0.1296	0.0426	660	0.04584	0.2133
Male Teammate	0.0924	0.0444	660	0.00527	0.1795

Figure 5.25: Perceived Comparative Power Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Hostile Sexist Beliefs.

was performed to compare each of the four dyadic partner (DP) types, which revealed no significant differences between any of the DP levels. Looking towards both the marginal means in Table 5.12 and the error bars in Figure 5.23, no obvious differences can be observed either. In essence, while there does appear to be a significant main effect of Dyadic Partner (DP) on how much power evaluators think the Human Worker (HW) has in the partnership, it is unclear what that difference looks like.

In addition to the main effects of DP type on Human Worker's (HW's) expectations of power within the partnership, there was also a **significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of benevolent sexist beliefs on evaluations of HW's expectations of power within the partnership. Looking towards Figure 5.24, it can be seen that there is a visible upward trend in the AI Tool DP, such that as evaluator endorsement of benevolent sexist beliefs increase evaluators perceive the HW as expecting to have more power in the partnership when working with an AI Tool DP. There also appears to be a very small upward trend in the Male Teammate DP condition, although there does not appear to be a meaningful trend in the Female Teammate DP condition.

Conversely, there was a downward trend in the AI Teammate DP condition, such that as evaluator endorsement of benevolent sexist beliefs increased evaluations of HW expectations of power decreased when the HW was working with the AI Teammate DP. This is interesting, and might be a reflection of the AI Teammate as a bit of a comparative equalizer when it comes to the gender dynamics that underlay benevolent sexism. In other words, while beliefs about the capabilities and roles of women and men in group work could be impacted by the gender makeup of a partnership, the AI Teammate DP was specifically designed in such a way that it does not have a designed-imposed gender, thus perhaps reducing the chance that gendered ideology impacts evaluations of HWs.

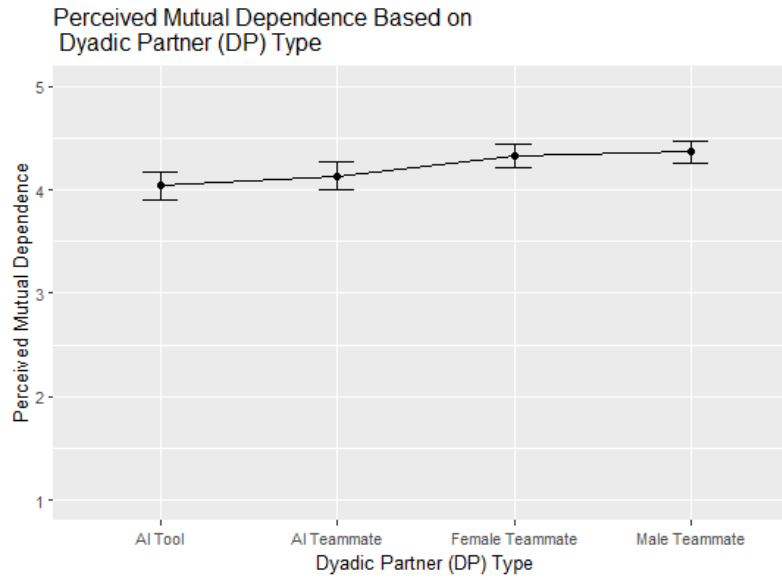
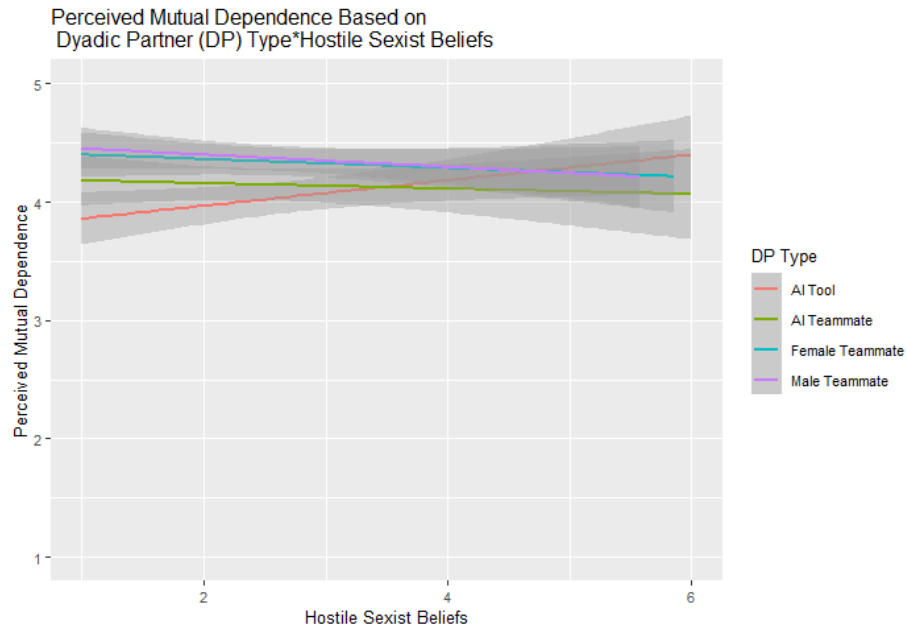


Figure 5.26: Perceived Mutual Dependence Based on Dyadic Partner Type. Error bars denote 95% confidence interval.

Finally, there was also a **borderline significant** interaction effect between dyadic partner (DP) type and evaluators' endorsement of hostile sexist beliefs on evaluations of Human Worker's (HW's) expectations of power in the partnership. While this interaction is not quite significant, Figure 5.25 is included to show that there is a slight upward trend in the two human DP conditions, such that as endorsement of hostile sexist beliefs increase there appears to be an increase in evaluations of HW expectations of power in the Male Teammate DP and Female Teammate DP conditions. This is interesting, as it implies that there could be a connection between negative evaluations of specific people (i.e., women) and generally less favorable evaluations of people who are "other" in a situation, such as someone who is explicitly framed as a partner to someone else rather than the primary member of the partnership.

Evaluations of Partnership Mutual Dependence. There was a **significant** main effect of dyadic partner (DP) type on the evaluations of partnership mutual dependence ($F(1,666) = 6.61, p < .001, \eta_p^2 = .03$) (Figure 5.26). A Tukey



Dyadic Partner Type	Trend	SE	df	lower.CL	upper.CL
AI Tool	0.0935	0.0430	663	0.00903	0.1780
AI Teammate	-0.0279	0.0472	663	-0.12054	0.0647
Female Teammate	-0.0490	0.0457	663	-0.13877	0.0407
Male Teammate	-0.0717	0.0468	663	-0.16372	0.0203

Figure 5.27: Perceived Mutual Dependence Based on the Interaction Between Dyadic Partner (DP) Type and Evaluator Endorsement of Hostile Sexist Beliefs.

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	4.144	4.027	4.260	0.059
AI Tool	4.054	3.939	4.170	0.059
Female Teammate	4.333	4.218	4.449	0.059
Male Teammate	4.373	4.256	4.489	0.059

Table 5.13: Marginal Means - Perceived Mutual Dependence Based on Dyadic Partner Type

post-hoc test was performed to compare each of the four dyadic partner (DP) types. Evaluations of partnership mutual dependence were shown to be significantly different between the AI Teammate and Male Teammate DP conditions, the AI Tool and Female Teammate DP conditions, and the AI Tool and Male Teammate DP Conditions. Looking towards the marginal means in Table 5.13, it can be seen that partnerships that consisted of a Human Worker (HW) and the Male Teammate DP condition were perceived by evaluators as being significantly more mutually dependent than partnerships containing an AI Teammate or AI Tool DP. The marginal means also demonstrate that evaluators perceived the partnerships that contained a Female Teammate DP as significantly more mutually dependent than partnerships with an AI Tool DP, though there was no significant difference compared to the AI Teammate DP condition.

Further, Figure 5.26 demonstrates an overall upward trend going from the AI DP conditions to the human DP conditions, with the caveat that the error bars for the AI Teammate and Female Teammate DP conditions overlap. Overall, then, it can be interpreted that the human-human partnerships are, on average, perceived to be somewhat more mutually dependent than the human-AI partnerships.

In addition to the main effects of DP type on evaluations of partnership mutual dependence, there was also a **significant** interaction effect between dyadic partner

(DP) type and evaluators' endorsement of hostile sexist beliefs on evaluations of partnership mutual dependence. Looking towards Figure 5.27, it can be seen that there is a visible upward trend in the AI Tool DP condition, such that as evaluators' endorsement of hostile sexist beliefs increases the more evaluators perceived the Human Worker (HW) and the AI Tool DP to be dependent upon each other when doing the task. One potential reason for this could be that evaluators who endorse beliefs that are antagonistic towards certain types of people (i.e., women) might also harbor antagonist views towards other people generally. This might lead them to view a human worker in an AI-enabled workplace as more dependent on an AI Tool, which could be seen as a negative quality to have within a human worker.

Conversely, it can be seen in Figure 5.27 that the Teammate DP conditions, including the AI Teammate DP, appear to have slight downward trends, such that as hostile sexist beliefs increase perceived mutual dependence between teammates decreases slightly. This is interesting, as it goes against the well-established principles of interdependence being a critical part of teamwork [31].

5.4.2.2 Interactions Between Dyadic Partner (DP) Type & Task Type on Evaluator Perceptions

While the vast majority of Study 2's significant results stem from the main effects of Dyadic Partner (DP) type, the ANCOVA performed for evaluator perceptions of Human Worker (HW) effectiveness did reveal a significant interaction effect, such that there is a significant difference between how evaluators perceived the effectiveness of the HW based on the type of Dyadic Partner (DP) they were working with and whether the partnership was working on a meeting scheduling task or investment banking task. However, results of the ANCOVA performed for dyadic partner (DP) effectiveness did not reveal a significant difference in how evaluators perceived

Evaluation Perception	<i>F</i>(1,666)	<i>p-value</i>	η_p^2	Significance
Effectiveness of Human Worker (HW)	2.76	.042*	.01	*Significant
Effectiveness of Dyadic Partner (DP)	1.56	.199	—	—
Attrib. of Credit to HW - Internal v. External	1.43	.232	—	—
Attrib. of Credit to HW - Ability v. Effort	2.28	.078	—	—
Attrib. of Credit to DP - Internal v. External	1.08	.356	—	—
Attrib. of Credit to DP - Ability v. Effort	.73	.535	—	—
Comparative Credit	2.00	.12	—	—
Human Worker (HW) Job Security	4.15	.006*	.02	*Significant
Human Worker (HW) Expectations of Power	.14	.936	—	—
Partnership Mutual Dependence	.15	.927	—	—

Table 5.14: Summary of Results for Interaction Effects Between Dyadic Partner (DP) Type & Task Type on Evaluator Perceptions

the DP's effectiveness based on the type of DP and whether they were performing a meeting scheduling task or an investment banking task. Additionally, the ANCOVA performed for evaluator perceptions of Human Worker (HW) job security did reveal a significant interaction effect, such that there is a significant difference between how evaluators perceived the HW's job security based on the type of Dyadic Partner (DP) they were working with and whether the partnership was working on a meeting scheduling task or an investment banking task.

The remainder of the results of the ANCOVAs demonstrated no significant interaction effects between Dyadic Partner (DP) Type and Task Type for all of the other evaluation perceptions, as seen in Table 5.14. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were attributed credit based on the type of DP in the partnership and whether the partnership was working on a meeting-scheduling task or an investment-banking task. Furthermore, data analysis did not reveal any significant differences in whether the Human Worker (HW) was attributed more comparative credit than their dyadic partners based on the type of DP in the partnership and whether the partnership was

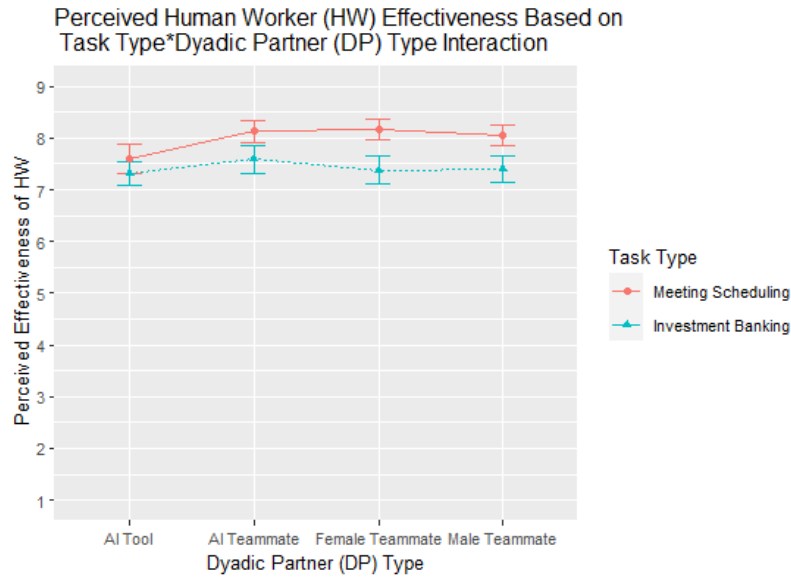


Figure 5.28: Perceived Effectiveness of Human Worker Based on Dyadic Partner Type and Task Type Interaction. Error bars denote 95% confidence interval.

working on a meeting scheduling task or an investment banking task. Data also did not reveal any significant differences in evaluators' perceptions of how much power the Human Worker (HW) thought that they had based on the type of dyadic partner in the partnership and whether the partnership was performing a meeting-scheduling task or an investment-banking task. Finally, data analysis also did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based on the type of dyadic partner in the partnership and whether the partnership was working on a meeting-scheduling task or an investment-banking task.

Table 5.3 provides the statistical results of the ANCOVAs and indications of their (non) significance, while the technical breakdown of these (non)significant results can be found in the text below.

Perceived Effectiveness of the Human Worker (HW). There was a **significant** interaction effect between dyadic partner (DP) type and task type on the perceived effectiveness of the human worker (HW) ($F(1,666) = 2.76$, $p =$

Task Type	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Meeting Scheduling	AI Teammate	8.094	7.866	8.322	0.116
Investment Banking		7.574	7.347	7.801	0.116
Meeting Scheduling	AI Tool	7.537	7.310	7.764	0.116
Investment Banking		7.357	7.133	7.580	0.114
Meeting Scheduling	Female Teammate	8.175	7.952	8.398	0.114
Investment Banking		7.353	7.125	7.581	0.116
Meeting Scheduling	Male Teammate	8.079	7.849	8.309	0.117
Investment Banking		7.441	7.217	7.665	0.114

Table 5.15: Marginal Means - Perceived Effectiveness of Human Worker Based on Dyadic Partner Type and Task Type Interaction

.042, $\eta_p^2 = .01$) (Figure 5.28). An analysis of simple main effects was conducted to explore this interaction. When controlling for DP Type, there were significant differences between task types in the AI Teammate DP, Female Teammate DP, and Male Teammate DP conditions. Looking toward the marginal means in Table 5.15, it can be seen that evaluators had generally higher perceptions of the human worker's (HW's) effectiveness when the HW was paired with the AI Teammate DP, Female Teammate DP, and Male Teammate DP on a meeting scheduling task as opposed to the investment banking task.

This difference is best illustrated in Figure 5.28, which shows gaps of varying sizes between the red error bars for the Meeting Scheduling task and the blue error bars for the Investment Banking task. Looking towards this figure, it can be seen that perceptions seem to have the greatest difference between task types when the HW is partnered with the Female Teammate DP. Conversely, the overlapping red and blue error bars for the AI Tool DP in Figure 5.28 show that perceptions of HW effectiveness are not significantly different when comparing HW-AI Tool partnerships in different types of tasks.

When controlling for the type of task viewed, there were **significant** differences

in the perceived effectiveness of the human worker (HW) based on the DP Type in the meeting scheduling task but not the investment banking task. Looking at the red line for the Meeting Scheduling task in Figure 5.28, it can be seen that the meeting scheduling task generally yielded a lower perceived effectiveness of the human worker (HW) when they were paired with the AI Tool DP compared to all other dyadic partners. However, the blue line in Figure 5.28 stays relatively stable, demonstrating that evaluators' perceptions of HW effectiveness were **not significantly different** when working with different types of dyadic partners on an investment banking task.

Perceived Effectiveness of the Dyadic Partner (DP). There was **no significant** interaction effects between dyadic partner (DP) type and task type on perceived effectiveness of the dyadic partner (DP) ($F(1,666) = 1.56, p = .199$).

Attributing Dimensions of Credit to Human Workers (HWs). There was **no significant** interaction effects between dyadic partner (DP) type and task type on overall attribution of internal vs. external credit to human workers (HWs) ($F(1,666) = 1.43, p = .232$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** interaction effects of dyadic partner (DP) type and task type on the attribution of ability vs. effort to human workers (HWs) ($F(1,666) = 2.28, p = .078$).

Attributing Dimensions of Credit to Dyadic Partners (DPs). There was **no significant** interaction effects between dyadic partner (DP) type and task type on overall attribution of internal vs. external credit to dyadic partners (DPs) ($F(1,666) = 1.08, p = .356$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** interaction effects of dyadic partner (DP) type and task type on the attribution of ability vs. effort to dyadic partners (DPs) ($F(1,666) = .73, p = .535$).

Evaluation of Comparative Credit to Human Workers (HWs). There



Figure 5.29: Perceived Job Security of Human Worker Based on Dyadic Partner and Task Type Interaction. Error bars denote 95% confidence interval.

was **no significant** interaction effects between dyadic partner (DP) type and task type on comparative credit to human workers (HWs) ($F(1,666) = 2.00, p = .12$).

Task Type	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Meeting Scheduling	AI Teammate	4.960	4.695	5.224	0.135
Investment Banking	AI Teammate	4.786	4.523	5.049	0.134
Meeting Scheduling	AI Tool	4.566	4.303	4.829	0.134
Investment Banking	AI Tool	4.912	4.653	5.172	0.132
Meeting Scheduling	Female Teammate	5.675	5.417	5.933	0.132
Investment Banking	Female Teammate	5.159	4.894	5.423	0.135
Meeting Scheduling	Male Teammate	5.566	5.300	5.832	0.136
Investment Banking	Male Teammate	5.160	4.900	5.419	0.132

Table 5.16: Marginal Means - Perceived Job Security of Human Worker Based on Dyadic Partner and Task Type Interaction

Evaluations of Human Worker (HW) Job Security. There was a **significant** interaction effect between dyadic partner (DP) type and task type on evaluations of human worker (HW) job security ($F(1,666) = 4.15, p = .006, \eta_p^2 = .02$) (Figure 5.29), which was analyzed via simple main effects. When controlling for

Dyadic Partner (DP) Type, there was a significant difference between Task Types for the Female Teammate DP and Male Teammate DP. Looking towards the marginal means seen in Table 5.16 was higher for human workers (HWs) partnered with the Male Teammate DP and Female Teammate DP when they worked in the meeting scheduling task, compared to when the partnership worked in the investment banking task. This difference can also be seen in Figure 5.29, where the red line associated with the meeting scheduling task rises above the blue lines of the investment banking task for the Female Teammate and Male Teammate DPs. Results in the Figure also show that evaluators felt that human workers working on a meeting scheduling task with either human Teammates had the most job security out of all other conditions.

When controlling for the Task Type, there was also a **significant** difference between Dyadic Partner (DP) Type in the Meeting Scheduling task condition. Specifically, the marginal means in Table 5.16 show that human workers (HWs) in the meeting scheduling task were seen as having lower job security when working with the AI Tool DP and AI Teammate DP compared to the two human DP conditions. This difference is best illustrated when looking at the red line for Meeting Scheduling in Figure 5.29. Here, it can be clearly seen that the two AI DP conditions are associated with significantly lower perceptions of HW job security compared to the two human DP conditions. This same trend does not apply to the Investment Banking task, though, as **no significant** differences were found between Dyadic Partners (DPs) when controlling for Task Type. In other words, when evaluating a working partnership in an investment banking context, the type of dyadic partner does not play a significant role in how evaluators perceive the human worker's (HW) job security, but it does play a role when the context is a meeting scheduling task.

Evaluations of Power Perceptions. There was **no significant** interaction effects between dyadic partner (DP) type and task type on evaluations of power

within the partnership ($F(1,666) = .14, p = .936$).

Evaluations of Partnership Mutual Dependence. There was **no significant** interaction effects between dyadic partner (DP) type and task type on evaluations of partnership mutual dependence ($F(1,666) = .15, p = .927$).

5.4.3 Effects of Interactions Between Human Worker (HW) Gender Presentation & Dyadic Partner (DP) Type on Evaluator Perceptions

5.4.3.1 Two-Way Interactions Between Human Worker (HW) Gender Presentation & Dyadic Partner (DP) Type

Evaluation Perception	$F(1,666)$	p -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	2.82	.038*	.01	*Significant
Effectiveness of Dyadic Partner (DP)	1.26	.289	—	—
Attrib. of Credit to HW - Internal v. External	.47	.705	—	—
Attrib. of Credit to HW - Ability v. Effort	.91	.436	—	—
Attrib. of Credit to DP - Internal v. External	2.00	.113	—	—
Attrib. of Credit to DP - Ability v. Effort	.47	.702	—	—
Comparative Credit	.19	.902	—	—
Human Worker (HW) Job Security	1.72	.162	—	—
Human Worker (HW) Expectations of Power	2.04	.107	—	—
Partnership Mutual Dependence	1.35	.256	—	—

Table 5.17: Summary of Results for Interaction Effects Between Human Worker (HW) Gender Presentation & Dyadic Partner (DP) Type on Evaluation Perceptions

Up until this point, the focus of this results section has been unpacking the effects of Human Worker (HW) Gender Presentation and Dyadic Partner (DP) Type in isolation from each other. The remainder of Study 2's results section will explore how these two manipulations interact to impact evaluator perceptions. Overall, the majority of the ANCOVAs performed revealed no significant results, as can be seen

in Table 5.17. However, there is one interesting and important caveat in that the ANCOVA performed for evaluator perceptions of Human Worker (HW) effectiveness did reveal a significant interaction effect, such that there is a significant difference between how evaluators perceived the effectiveness of the HW based on whether they were a woman or a man working with different types of dyadic partners. This result is critically important to this dissertation, and will thus be explored in further detail later in this section.

The remainder of the results of the ANCOVAs demonstrated non-significant interaction effects between Human Worker (HW) Gender Presentation and Dyadic Partner (DP) Type for all of the other evaluation perceptions, as seen in Table 5.17. This means that the data analysis did not reveal any significant differences in how dyadic partners were perceived as effective based on whether the HW was a woman or a man working with any of the DP types. Furthermore, data analysis did not reveal any significant differences in how human workers or dyadic partners were attributed credit based on whether the HW was a woman or a man working with any of the DP types. Data analysis also did not reveal any significant differences in whether the Human Worker (HW) was attributed more comparative credit than their dyadic partners based on whether the HW was a woman or a man working with any of the DP types. Likewise, there was no significant differences in evaluators' perceptions of Human Worker (HW) job security or how much power the Human Worker (HW) thought they had, based on whether the HW was a woman or a man working with any of the dyadic partner types. Finally, there was also no significant differences in evaluators' perceptions of the partnership's mutual dependence based on whether the HW was a woman or a man working with any of the DP types.

Table 5.17 provides the statistical results of the ANCOVAs and indications of their (non) significance, while the technical breakdown of these (non) significant

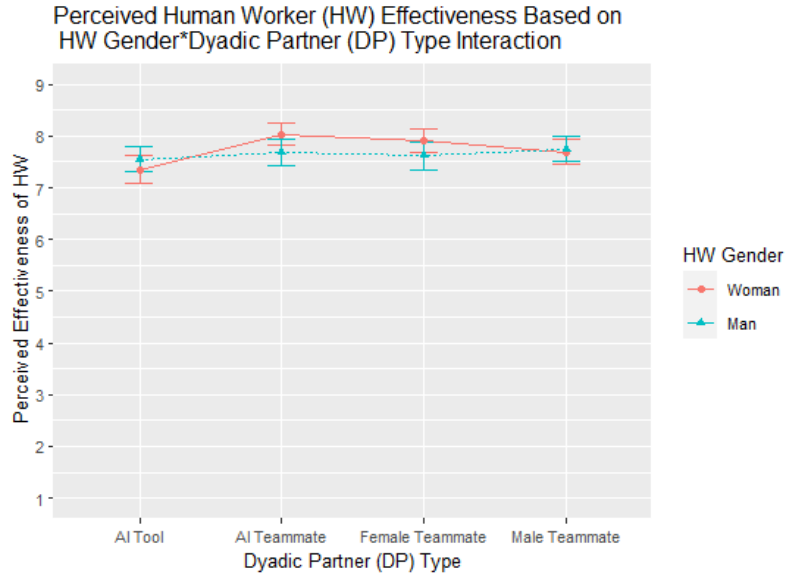


Figure 5.30: Perceived Effectiveness of Human Worker Based on Gender Presentation of Human Worker and Dyadic Partner Type Interaction. Error bars denote 95% confidence interval.

results can be found below.

Perceived Effectiveness of the Human Worker (HW). There was a **significant** interaction effect between human worker (HW) gender presentation and dyadic partner (DP) type on perceived effectiveness of the human worker (HW) ($F(1,666) = 2.82$, $p = .038$, $\eta_p^2 = .01$) (Figure 5.30). An analysis of simple main effects was conducted to explore this interaction. When controlling for DP Type, there was a significant difference in the perception of human worker (HW) performance based on HW gender in AI Teammate and Female Teammate DP conditions. Looking towards the marginal means in Table 5.7, both the AI Teammate and Female Teammate DP conditions saw the woman HW perceived as having greater effectiveness than the man HW. In other words, when working with either the AI Teammate or the Female Teammate DPs, the woman HW was perceived to be more effective than the man HW was when also working with the same DPs.

HW Gender	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Woman	AI Teammate	8.013	7.787	8.238	0.115
Man		7.655	7.427	7.884	0.116
Woman	AI Tool	7.333	7.109	7.558	0.114
Man		7.560	7.335	7.786	0.115
Woman	Female Teammate	7.914	7.688	8.139	0.115
Man		7.615	7.389	7.840	0.115
Woman	Male Teammate	7.760	7.532	7.987	0.116
Man		7.761	7.534	7.988	0.116

Table 5.18: Marginal Means - Perceived Effectiveness of Human Worker Based on Gender Presentation of Human Worker and Dyadic Partner Type Interaction

Moreover, when controlling for Human Worker (HW) Gender Presentation, there was a **significant** effect of Dyadic Partner (DP) Type in the woman HW condition but not the man HW condition. Looking towards Figure 5.30, it can be seen in the path of the red line associated with the Woman HW that evaluators perceived the effectiveness of the woman HW to be significantly lower when working with an AI Tool DP compared to all other dyadic partners. In fact, looking still at Figure 5.30, it is the only DP condition that creates lower perceptions of effectiveness for the woman HW than it does for the man HW, although marginal means (Table 5.17) are not significantly different between the two. It can also be seen in the Figure that the condition of Male Teammate DP creates nearly identical patterns for the Woman HW and the man HW. Finally, it is important to note that the highest rated effectiveness of the Human Worker (HW) can be seen when the woman HW is working with the AI Teammate DP as shown in Figure 5.30.

Perceived Effectiveness of the Dyadic Partner (DP). There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on the perceived effectiveness of the dyadic partner (DP) ($F(1,666) = 1.26, p = .289$).

Attributing Dimensions of Credit to Human Workers (HWs). There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on the overall attribution of internal vs. external credit to the human worker (HW) ($F(1,666) = .47, p = .705$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on the attribution of ability vs. effort to human workers (HWs) ($F(1,666) = .91, p = .436$).

Attributing Dimensions of Credit to Dyadic Partners (DPs). There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on the overall attribution of internal vs. external credit to the dyadic partner (DP) ($F(1,666) = 2.00, p = .113$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on the attribution of ability vs. effort to dyadic partners (DPs) ($F(1,666) = .47, p = .702$).

Evaluation of Comparative Credit to Human Workers (HWs). There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on the evaluation of comparative credit to human workers (HWs) ($F(1,666) = .19, p = .902$).

Evaluations of Human Worker (HW) Job Security. There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on evaluations of human worker (HW) job security ($F(1,666) = 1.72, p = .162$).

Evaluations of Power Perceptions. There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner

(DP) type on evaluations of power within the partnership ($F(1,666) = 2.04$, $p = .107$).

Evaluations of Partnership Mutual Dependence. There was **no significant** interaction effects between human worker (HW) gender presentation and dyadic partner (DP) type on evaluations of partnership mutual dependence ($F(1,666) = 1.35$, $p = .256$).

Evaluation Perception	$F(1,666)$	p -value	η_p^2	Significance
Effectiveness of Human Worker (HW)	.54	.652	—	—
Effectiveness of Dyadic Partner (DP)	.31	.816	—	—
Attrib. of Credit to HW - Internal v. External	2.00	.114	—	—
Attrib. of Credit to HW - Ability v. Effort	.37	.772	—	—
Attrib. of Credit to DP - Internal v. External	1.09	.352	—	—
Attrib. of Credit to DP - Ability v. Effort	.52	.668	—	—
Comparative Credit	.43	.735	—	—
Human Worker (HW) Job Security	.46	.708	—	—
Human Worker (HW) Expectations of Power	1.43	.232	—	—
Partnership Mutual Dependence	1.52	.208	—	—

Table 5.19: Summary of Results for Interaction Effects Between Human Worker (HW) Gender Presentation, Dyadic Partner (DP) Type, & Task Type on Evaluation Perceptions

5.4.3.2 Three-Way Interactions Between Human Worker (HW) Gender Presentation, Dyadic Partner (DP) Type, & Task Type on Evaluator Perceptions

Overall, results of the ANCOVAs demonstrated no significant interaction effects between Human Worker (HW) Gender Presentation, Dyadic Partner (DP) Type, and Task Type for all evaluation perceptions, as seen in Table 5.19. This means that the data analysis did not reveal any significant differences in how human workers or dyadic partners were perceived as effective and attributed credit based on whether the human worker was a woman or a man working with any of the four dyadic partner

types on either the meeting scheduling task or investment banking task. Furthermore, data analysis did not reveal any significant differences in whether the human worker was attributed more comparative credit than their dyadic partners based on whether the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task. There was also no significant differences in evaluators' perceptions of the human worker's job security and expectations of power based on whether the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task. Finally, data analysis did not reveal any significant differences in evaluators' perceptions of the partnership's mutual dependence based on whether the human worker was a woman or a man working with any of the four dyadic partner types on either the meeting scheduling task or investment banking task.

Table 5.19 provides the statistical results of the ANCOVAs and indications of their non-significance, while the technical breakdown of these nonsignificant results can be found below.

Perceived Effectiveness of the Human Worker (HW). There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the perceived effectiveness of the human worker (HW) ($F(1,666) = .54, p = .652$).

Perceived Effectiveness of the Dyadic Partner (DP). There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the perceived effectiveness of the dyadic partner ($F(1,666) = .31, p = .816$).

Attributing Dimensions of Credit to Human Workers (HWs). There was **no significant** three-way interaction effects between human worker (HW) gen-

der presentation, dyadic partner (DP) type, and task type on the overall attribution of internal vs. external credit to the human worker (HW) ($F(1,666) = 2.00, p = .114$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the attribution of ability vs. effort to human workers (HWs) ($F(1,666) = .37, p = .772$).

Attributing Dimensions of Credit to Dyadic Partners (DPs). There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the overall attribution of internal vs. external credit to the dyadic partner (DP) ($F(1,666) = 1.09, p = .352$). A repeated-measures ANOVA was then run to break down attributions of internal credit by ability vs. effort. There was also **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the attribution of ability vs. effort to dyadic partners (DPs) ($F(1,666) = .52, p = .668$).

Evaluation of Comparative Credit to Human Workers (HWs). There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the comparative credit to human workers (HWs) ($F(1,666) = .43, p = .735$).

Evaluations of Human Worker (HW) Job Security. There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on the perceived job security of the human worker (HW) ($F(1,666) = .46, p = .708$).

Evaluations of Power Perceptions. There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner

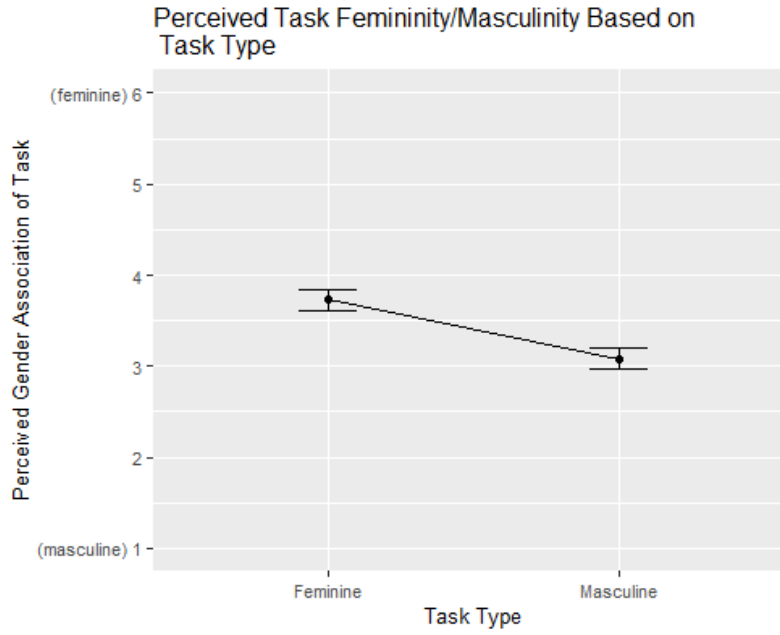


Figure 5.31: Perceived Task Femininity/Masculinity Based on Task Type. Error bars denote 95% confidence interval.

(DP) type, and task type on evaluations of power within the partnership ($F(1,666) = 1.43, p = .232$).

Evaluations of Partnership Mutual Dependence. There was **no significant** three-way interaction effects between human worker (HW) gender presentation, dyadic partner (DP) type, and task type on evaluations of partnership mutual dependence ($F(1,666) = 1.52, p = .208$).

5.4.4 Perception of Task Type Femininity/Masculinity

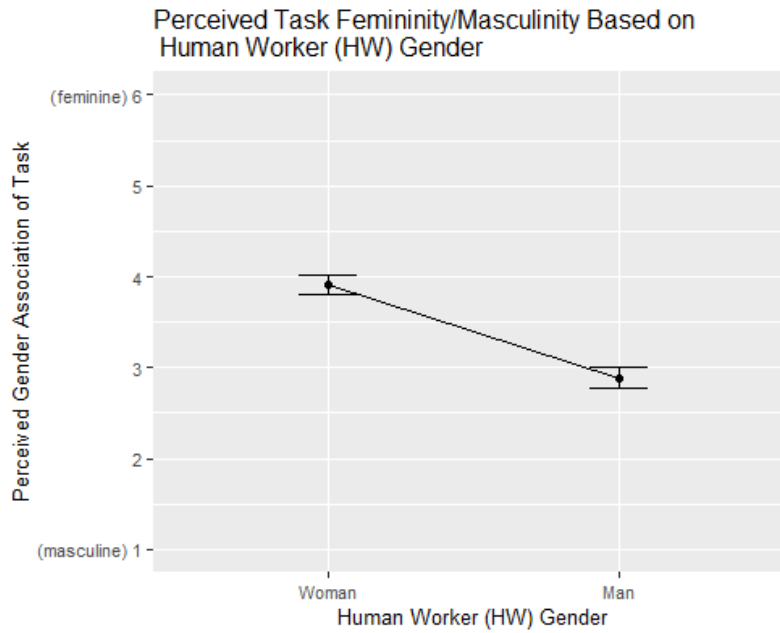


Figure 5.32: Perceived Task Femininity/Masculinity Based on Human Worker (HW) Gender. Error bars denote 95% confidence interval.

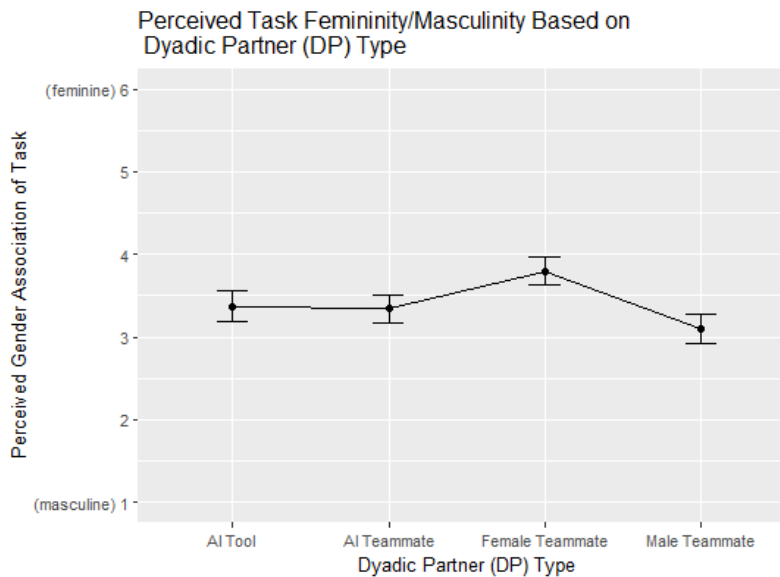


Figure 5.33: Perceived Task Femininity/Masculinity Based on Dyadic Partner (DP) Type. Error bars denote 95% confidence interval.

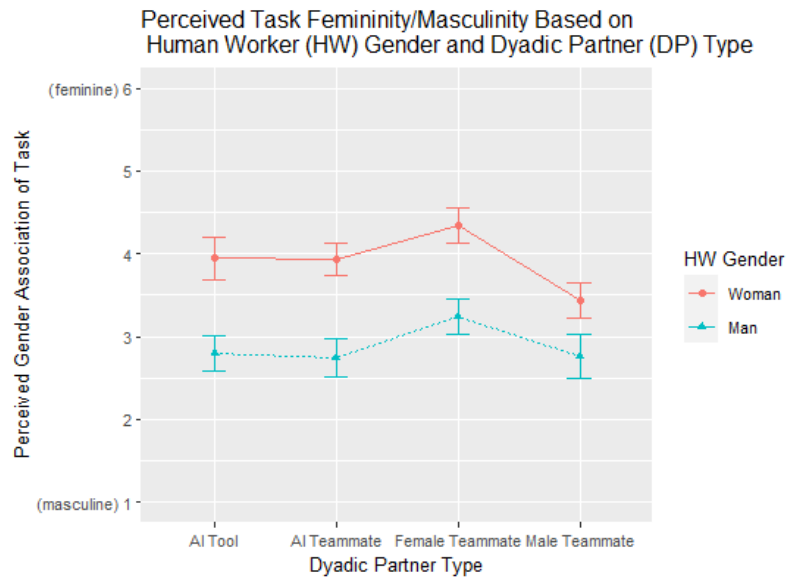


Figure 5.34: Perceived Task Femininity/Masculinity Based on Human Worker (HW) Gender and Dyadic Partner (DP) Type Interaction. Error bars denote 95% confidence interval.

Task Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
Meeting Scheduling	3.721	3.614	3.828	0.055
Investment Banking	3.079	2.972	3.185	0.054

Table 5.20: Marginal Means - Perceived Task Femininity/Masculinity by Task Type

HW Gender	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
Woman	3.915	3.808	4.022	0.054
Man	2.885	2.778	2.992	0.055

Table 5.21: Marginal Means - Perceived Task Femininity/Masculinity by Human Worker (HW) Gender

DP Type	Marginal Mean	95% CI for Mean Difference		SE
		Lower	Upper	
AI Teammate	3.329	3.178	3.481	0.077
AI Tool	3.374	3.224	3.524	0.077
Female Teammate	3.794	3.643	3.945	0.077
Male Teammate	3.103	2.951	3.254	0.077

Table 5.22: Marginal Means - Perceived Task Femininity/Masculinity by Dyadic Partner (DP) Type

HW Gender	DP Type	Marginal Mean	95% CI for Mean Difference		SE
			Lower	Upper	
Woman	AI Teammate	3.924	3.711	4.138	0.109
Man		2.734	2.518	2.950	0.110
Woman	AI Tool	3.969	3.757	4.181	0.108
Man		2.779	2.566	2.992	0.109
Woman	Female Teammate	4.342	4.129	4.555	0.109
Man		3.246	3.032	3.459	0.109
Woman	Male Teammate	3.425	3.210	3.639	0.109
Man		2.780	2.566	2.995	0.109

Table 5.23: Marginal Means - Perceived Task Femininity/Masculinity by Human Worker (HW) Gender * Dyadic Partner (DP) Type

Finally, it is worth exploring the metric that was originally included as a manipulation check, which was whether or not participants rated the task they viewed in their video as feminine or masculine. The ANCOVA conducted confirmed the validity of the manipulation, with a **significant** difference of task type on the perception of the viewed video as being masculine or feminine ($F(1,669) = 69.60, p < .001, \eta_p^2 = .09$). Means show that the meeting scheduling task was perceived as being more feminine than the Investment Banking task. In general, this finding confirms the original intention of the Task Type manipulation, which was to depict one task that was perceived as more feminine and one task that was perceived as more masculine.

However, the ANCOVA also revealed that task type was not the only significant manipulation to impact the perceived femininity of the shown video. Results also show a **significant** effect of human worker (HW) gender ($F(1,669) = 179.12, p < .001, \eta_p^2 = .21$) and dyadic partner (DP) type ($F(3,669) = 14.02, p < .001, \eta_p^2 = .06$) on the perceived femininity of the task. For the DP type, a Tukey post-hoc test showed significant differences between the AI Teammate and Female Teammate DP conditions, the AI Tool and Female Teammate DP conditions, and the Female Teammate and the Male Teammate DP conditions. In general, tasks with a male dyadic partner were perceived as more masculine, tasks with a female dyadic partner were perceived as more feminine, and tasks with an AI tool and AI teammate fell in the middle of these two human bounds. For the main effect of human worker (HW) gender, participants generally viewed the videos with a woman HW as being significantly more feminine than videos with a man HW. Of these three main effects, it is worth noting that HW gender had the largest effect size on the performed ANCOVA of the three manipulations tested ($\eta_p^2 = .21$).

Lastly, there was a **significant** two-way interaction effect between the Dyadic Partner (DP) Type and human worker (HW) gender depicted in each video ($F(3,669)$

$= 2.90, p = .036, \eta_p^2 = .01$). Simple main effects show that when controlling for DP type, there was a significant effect of HW gender on the perception of task femininity for each dyadic partner condition. When controlling for human worker (HW) gender, there was a significant effect of dyadic partner type in both the woman HW condition and the man HW condition. When looking at the results shown in Figure 5.34, this interaction effect shows that there was a generally consistent gap in perceived task femininity for the AI Teammate, AI Tool, and Female Teammate DP conditions; however, this gap shrinks for participants in the Male Teammate DP condition. This interaction broadly shows that having any male in a task (whether it be as the main worker or as a partner) can generally sway the perception of a task to be more masculine.

5.5 Results Part II - Qualitative Supplemental Expert Interviews

Results of the supplementary expert interviews reveal the intricate ways that workplace gender dynamics and AI functional design intermingle to shape evaluations and attributions of credit to workers, especially women workers. These results also shed light on the “why” element of Study 2’s quantitative results, specifically by (1) highlighting the relationship between job-specific evaluations in AI-enabled workplaces and AI functional design (**S.2.RQ1** and **S.2.RQ2**), and (2) deepening this study’s understanding of how ambivalent sexist beliefs (i.e., hostile and benevolent) alter expectations of women as collaborators and experts in AI-enabled workplaces (**S.2.RQ3** and **S.2.RQ4**).

5.5.1 Theme 1: AI Usage is Becoming a Greater Part of People’s Job-Specific Evaluations in the Workplace

The first theme of this work is focused on the evaluation components of Study 2, as a number of experts expressed reactions as to why AI impacted worker evaluations within this study. Overall, expert interviews highlighted that the last year has seen a dramatic shift in AI’s role in the workplace. As P6 (28, Woman, Asian, AI Governance Advisor) suggested, AI usage is becoming less of a sign of laziness and more of an important skill that can be commended. All experts tended to agree that the most important factor in recent years has become the end result of the product one delivers with AI. Echoing these thoughts, P3 (34, Woman, Asian, Research Prof. Human-Centered AI) had noted that they often see greater commendation and credit for those who can leverage AI technology to create a better, more refined, and higher quality product; however, if a poor product is delivered, the same AI usage becomes a detriment to human worker evaluation and strips credit away from the worker. The following quote from P3 serves as an example of that opinion:

“For example, when reviewing papers, if it’s very clear to me, like, this person used AI, so it’s not a smart way of using AI, and I would be very pissed, because I’m reviewing something AI-generated, and that’s a waste of my time. I will completely strip the credit away from the authors. But a smart way of using AI, like, in ways that I’m not sure if they use AI, but it reads really smooth and [it’s] making clear statements and [the] logic is clear, it feels like the human initiated the thinking process while using AI to polish the thoughts or modify the thoughts. That, to me, is a smart way of using AI, and that’s fine. And you know, I will, I will give them [the authors] the credit.”

Building on this, the results of this work showed that greater evaluations could come from those leveraging any teammate over the AI tool, and experts felt they had a possible reason as to why. A substantial portion of modern workplaces are seen as valuing collaboration, with workers often seeing higher evaluations when they collaborate effectively. Indeed, P1 (46, Man, White, Conflict Arbiter) and P2 (29, Woman, White, AI R&D) noted that working with a teammate, even an AI teammate, provides an opportunity to showcase one’s collaborative abilities, which in turn improves the way other people evaluate them and their work. As such, rather than detracting from human evaluation, AI teammates may actually provide an opportunity for humans to showcase their technical and collaborative capabilities within the workplace simultaneously.

Moreover, if these collaborations are able to yield a high-quality and effective product, then it would only serve to bolster these evaluations. Based on this, P2 went as far as to highlight the importance of women workers learning how to collaborate with AI in the coming years, as it will likely be a key factor in workplace evaluations and should be leveraged by women workers, who are otherwise disproportionately impacted by workplace gender dynamics. More specifically, the following quote from P2 illustrates this concept:

“Work with an AI teammate for your effectiveness, as demonstrated by the results [of Study 2]. I think you (women workers) should absolutely find every way to get an advantage, because there are plenty of forces, plenty of perceptions [about women] that are preventing you from being effective, that are out of your control, and just because of how humans interact, like how humans view each other.”

However, these criteria for evaluation may not always be consistent from work-

place to workplace. As noted by P4 (59, Woman, Asian, CEO Sci. Tech. Accelerator), contexts that place a greater emphasis on innovation in technology may be more likely to stress AI usage as a way to be more effective, *“Because, well, we’re adaptable. We’re more adaptive to new technology. And we see the opportunity.”* Even from an AI tooling and teaming perspective, P1, P4, and P5 (32, Man, White, Dir. of Business Intelligence) had noted that they see a weak distinction between these two concepts in their workplace, and that they are much more focused on the specific application of the AI system than on how that system is labeled.

However, despite AI usage’s impact on evaluation potentially being workplace dependent, it may still provide more universal impacts on effectiveness across workplaces. As noted by P1, their own personal position does not see any evaluation due to AI usage; however, their own personal effectiveness can benefit from their use of AI. In this sense, even if AI use were to be consistent across many workplaces, there may still be a strong inconsistency in how its use is evaluated depending upon how the workers themselves are evaluated. The following quote by P1 provides a summary of this idea:

“We are not really evaluated on the things that we would use AI for. So that’s like, we have a very loose evaluation criteria. Ours is mostly based on the feedback, or, I guess, if people want to utilize our services, right? So the demand for us to exist, I guess, is really our evaluation criteria. So if people stopped wanting to use the Ombuds office or feeling the need to use us to conduct training or for conflict management, then I think that would reflect on our performance. But you know, the training that we do and the things that we write [using AI] aren’t necessarily evaluated.”

Summarizing this theme, experts were clear that AI usage is likely going to

contribute to evaluations of human workers, but how it contributes is going to be dynamic. In some workplaces, its contribution will be explicit, as some groups may dictate how AI should be used and will evaluate humans on that usage. On the other hand, workplaces that value collaboration might leverage human-AI collaboration as an additional means to observe and evaluate human worker effectiveness. Lastly, even workplaces that do not explicitly evaluate AI may still see the downstream impact of AI utilization, as workers who work with or use AI to complete final products will ultimately be evaluated on the quality of those products. Overall, the experts interviewed provided an interesting perspective as to why and how the results of this work may serve to inform evaluation practices in AI-enabled workplaces.

5.5.2 Theme 2: Both Benevolent and Hostile Sexist Beliefs can Alter the Expectations of Women as Collaborators and AI Experts

The second Study 2 theme identified in the expert interviews centers around the role that sexism may play within AI-enabled workplaces. Interestingly, participants saw both benevolent aspects of sexism and hostile applications of sexism contributing to women's evaluations in AI-enabled workplaces. From a benevolent sexism perspective, expert interviews actually built on the findings of Theme 1.

Specifically, P1 and P3 had noted that women are often viewed as more collaborative in the workplace, which could explain why women workers were evaluated as more effective when working with AI and female teammates. In other words, women workers may actually have an advantage in AI Teammate-enabled workplaces because they are doubly rewarded for acting collaboratively. Experts 2 and 3 went on to discuss how this benevolent sexism bias may also interplay with how women are

expected to conduct themselves in the workplace, such as using more compassionate and less direct, or “bossy” language. As such, they felt that the interactions between the woman worker and AI teammate were more emblematic of this style of communication, which resulted in better perceptions of women workers. Interactions with the AI tool, however, engendered more direct or “bossy” language, which resulted in worse perceptions of women workers because she was acting counterstereotypically. Overall, then, it is clear that benevolent sexist traits within evaluators could play a role in evaluation for women and men workers in AI-enabled workplaces, with the following quote from P3 providing an example of how:

“First, the communication style, women communicating to a tool, more commanding. And second, it’s, again, also communication style. And it’s allocating work to others, so the woman will be perceived as receiving less credit. People have stereotypical expectations about women, how they should communicate and at work, how they should behave. If a woman is to receive credit at work, they’ll need to demonstrate a higher level of expertise and effort than a man, so a tiny little cue in their daily work behavior that suggests any type of less effort, for example, giving the AI [Tool] stuff to do, will be magnified in people’s perceptions, I think.”

P1 expanded upon this by bringing up how workplace dynamics interact with team conflict. As an expert in dealing with conflict, P1 explains that, while human-human conflict can stem from many factors, including from workplace gender dynamics, human-AI collaborations could provide an opportunity to reduce these conflicts. More specifically, similar to how women are often perceived as and rewarded for being collaborative, so would AI usage be perceived as beneficial if designed with collaboration in mind. Even more critically, AI could uniquely benefit workplaces where

workplace dynamics might foster conflict because AI does not have conflict-related traits that are characteristic of humans. The following quote from P1 explores this idea:

“If I was working on some kind of research project, and I had an idea, and that idea wasn’t great, AI is not going to tell me that that’s not a great idea. AI is going to provide the information that I’m requesting of it and, you know, collaborate on that information, help me to flesh it out. And maybe that idea wasn’t great, but through the use of that [AI] tool, and the fact that that [AI] tool doesn’t have those opinion-based barriers that are getting in the way, it allows me to flesh that out and possibly come up with a great product. Where, if I have a human teammate who is against this idea for whatever reason, they either don’t agree with the direction of the project, they don’t like the idea, they’re not in the mood to collaborate like, you know, there’s a million factors that could get in the way. Those factors could inhibit that idea from moving along the way that you could independently move along a project with the use of a[n AI] tool.”

Looking more specifically at hostile sexism, the quantitative results of this work revealed that evaluator endorsement of hostile sexism had differing impacts on evaluation when the human worker was working with an AI Tool DP compared to any of the Teammate DPs. Experts were directly asked about this finding to better understand how hostile sexism may begin to impact evaluations in AI-enabled workplaces, as this result was a bit perplexing on its own.

These experts collectively provided a number of possible hypotheses. On the one hand, experts felt that the integration of AI technology into the workplace would

provide an additional opportunity for the technical knowledge of women to be tested. After diving deeper into the concept of AI Tools, P2 noted that the less collaborative nature of tools may require a greater level of orchestration from human workers, and that people with more hostile sexist beliefs might be more critical of the technical abilities of others. On the other hand, as collaborative AI teammates are more agentic, the skill level required to use them might be perceived as lower, and thus less likely to garner these same reactions from evaluators that endorse hostile sexist beliefs.

P6 somewhat countered this notion, however, as they felt their workplaces had such strict evaluation criteria that it made it difficult for these types of gender biases to slip in and affect evaluations. Merging these two ideas, one can see that, if unchecked, hostile attitudes toward women would likely continuously perpetuate the idea that women have less ability and literacy with AI, making them less effective with AI tools vs. AI teammates, but workplaces can still counteract this through purposeful crafting of evaluation criteria to reduce the possibility of these gender biases creeping in. P2's quote provides an illustrative example of what this bias has looked like in her experience:

“I was in a meeting where the [other] women in the room treated me as a subject matter expert. They deferred to me. They said, ‘[P2’s name], I want to have your opinion about this. Can you share more about this thing?’ And the man in the room started berating me. The man who also does AI stuff was feeling overlooked, essentially, like, ‘Why do you keep turning to her and not asking me these questions?’... It got to the end of that meeting, and he just started saying, ‘What do you know about Kubernetes? What do you know about OpenShift? What do you know about this specific large language model? How do you know any of these

things?’ And demeaning me. There’s this weird divide across gender in AI development that I think is really interesting... There are layers to it. Women tend to support me using it, or support me being a part of making it. I’ve seen plenty of men put me down, say I don’t have technical expertise.”

Summarizing this theme, sexism is likely to play a role in the evaluation of workers in AI-enabled workplaces, but that role is likely going to be dynamic based on the workplace and the type of evaluations in question. In the event that worker evaluations are strongly reliant on collaboration, benevolent sexism may actually serve to benefit the evaluation of human workers. Yet, workplaces that strictly prioritize technical knowledge and literacy may provide an opportunity for hostile sexism to flourish. Regardless of the type of sexism at play within a workplace’s gender dynamics, AI-enabled workplaces have an ethical responsibility to their human workforce to make sure that biases are reduced as much as possible in evaluations, as P6 points out. Indeed, as an AI Governance Expert/Advisor, she emphasizes to her clients that they have an ethical obligation to their employees when designing and evaluating AI workplace integration over the entire life-cycle of AI use in the workplace.

5.6 Discussion

In answering Study 2’s research questions, Tables 5.24, 5.25, and 5.26 summarizes the main findings regarding how evaluations and attributions are influenced by the gender identity of a human worker and by the AI or gender identity of the dyadic partner (Table 5.24), how the aforementioned influences interact with the type of task being performed by the dyad (Table 5.25), and how evaluators’ pre-existing discriminatory beliefs (i.e., sexism and negative attitudes towards AI) influence evaluations and attributions (Table 5.26).

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ1: How are these evaluations and attributions influenced by the gender identity of a human worker?	Overall, the gender identity of the HW (<i>Woman</i> v. <i>Man</i>) did not significantly, directly influence evaluations and attributions of the HW, the DP, or the dyad as a whole. However, the gender identity of the HW did have a significant, interactive influence when coupled with <i>DP Type</i>: • When working with the <i>AI Teammate DP</i> or the <i>Female Teammate DP</i>, the <i>Woman HW</i> was evaluated as significantly more effective than the <i>Man HW</i> was when he worked with the same DPs. • Conversely, the <i>Woman HW</i> was evaluated as significantly less effective when she worked with the <i>AI Tool DP</i> than when she worked with any of the <i>Teammate DPs</i>. • Evaluations of the <i>Man HW</i>'s effectiveness were not significantly influenced by the type of DP he worked with, which suggests that there are unique perceptual disadvantages for women workers working with an AI Tool that do not impact men workers.
S.2.RQ2: How are these evaluations and attributions influenced by the AI identity or gender identity of a dyadic partner?	Overall, the identity of the DP did significantly, directly influence many evaluations and attributions to the HW, the DP, and the dyad as a whole, the most interesting of which are summarized below: • The <i>AI Tool DP</i> was evaluated as being significantly less effective than the three <i>Teammate DPs</i>, and HWs that worked with it were evaluated as being significantly less effective than HWs in the other dyads. • HWs were evaluated as having significantly less job security when working with either of the <i>AI (Tool, Teammate) DPs</i> compared to when they worked with either of the <i>human (Female, Male) Teammate DPs</i>. • HWs were attributed significantly more comparative credit when working with the <i>AI Teammate DP</i> than when they worked with the <i>Female Teammate DP</i>. • HW/DP dyads that featured the <i>Male Teammate DP</i> were evaluated as being significantly more mutually dependent on each other for successful task completion compared to dyads that featured either <i>AI (Tool, Teammate) DPs</i>. Dyads with the <i>Female Teammate DPs</i> were only evaluated as significantly more mutually dependent than dyads with an <i>AI Tool DP</i>.

Table 5.24: Study 2 Summary of Key Significant Results Answering **S.2.RQ1** and **S.2.RQ2**.

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ3: How, if at all, do the influences of S.2.RQ1 (HW Gender) and S.2.RQ2 (DP Type) vary based on the type of task completed in an AI-enabled workplace?

- Regarding **S.2.RQ1's** (*HW Gender*) influence:

- Evaluations of the *Man HW's* perceived power compared to the DP was **significantly higher** when the dyad was working on the *Meeting Scheduling Task* versus when they were working on the *Investment Banking Task*.

- Regarding **RQ.1.2's** (*DP Type*) influence:

- HWs were evaluated as being **significantly more effective** when working with any of the *Teammate DPs* on a *Meeting Scheduling Task* compared to working with the same DPs on an *Investment Banking Task*.
 - When working on the *Meeting Scheduling Task*, HWs were evaluated as having **significantly less job security** when working with either of the *AI (Tool, Teammate) DPs* compared to when they worked with either of the *human (Female, Male) Teammate DPs*.
 - HWs were evaluated as having **significantly more job security** when working with either of the *human (Female, Male) Teammate DPs* on the *Meeting Scheduling Task* compared to when they worked with the same DPs on the *Investment Banking Task*.
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Table 5.25: Study 2 Summary of Key Significant Results Answering **S.2.RQ3**.

S.2. Overarching RQ: How are evaluations and attributions of credit to a human worker, their dyadic partner, and their collaboration as a whole influenced by the gender dynamics of an AI-enabled workplace?

S.2.RQ4: How, if at all, do the discriminatory beliefs (i.e., sexism and AI attitudes) of an evaluator interact with gender dynamics to influence these evaluations and attributions?	- As evaluator endorsement of hostile sexist beliefs <i>increased</i>... • evaluations of the <i>AI Tool DP</i>'s effectiveness <i>increased</i>. • comparative credit trended towards the Woman HW when the dyad worked on an <i>Investment Banking Task</i>, but trended towards the DP in the <i>Meeting Scheduling Task</i>. • comparative credit trended towards the Man HW when the dyad worked on an <i>Meeting Scheduling Task</i>, but trended towards the DP in the <i>Investment Banking Task</i>. • evaluations of partnership mutual dependence <i>increased</i> when working with the <i>AI Tool DP</i>. - As evaluator endorsement of benevolent sexist beliefs <i>increased</i>... • evaluations of the <i>AI Tool DP</i>'s effectiveness <i>increased</i>. • comparative credit trended towards the AI Tool DP and the AI Teammate DP compared to the HW in those dyads, while it trended towards the HW in dyads feature the <i>Male Teammate DP</i>. - As negative attitudes towards AI <i>increased</i>... • evaluations of the <i>AI Tool</i>, <i>AI Teammate</i>, and <i>Female Teammate DPs</i>' effectiveness <i>decreased</i>, and external factors were allocated more weight when determining credit attribution for these DPs. • comparative credit trended towards the HW when working with the <i>AI Teammate</i> and <i>Female Teammate DPs</i>, while it trended towards the DP when working with the <i>AI Tool DP</i>. • HW job security <i>decreased</i> in all DP conditions, especially when working with <i>AI (Tool, Teammate) DPs</i>.
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Table 5.26: Study 2 Summary of Key Significant Results Answering **S.2.RQ4**.

Going forward, this discussion will situate these results in the existing literature on human-AI interaction/teaming and workplace gender dynamics in group work to unpack how evaluations and attributions are fundamentally shaped in an AI-enabled workplace by the functional design of the AI that human workers work with, intermingling with workplace gender and task dynamics. It will especially highlight how the functional design of an AI as either a Tool or a Teammate plays critical roles in evaluations and attributions of workers generally and women workers specifically. In doing so, Study 2's results make significant contributions to ongoing academic and popular discussions about the integration of AI into the everyday workplace [489], as well as making significant contributions towards furthering our understanding of the AI Tool vs. AI Teammate divide [153, 156].

Finally, this section ends with recommendations for how individuals and organizations looking to integrate AI into their everyday workplace can best account for AI functional design and workplace dynamics to craft successful integration practices. These recommendations represent the intermingling of academic research and practitioner expertise, as they were borne from direct suggestions from and this author's interpretation of the expert interviews.

5.6.1 Accounting for AI Functional Design and Gender Dynamics in Workplace Teams

This study sets out to understand if and how the gender dynamics and functional design of AI systems would influence the evaluations of both workers and technologies. Based on the results of this work, a clear conclusion can be drawn: the functional design of an AI system and the workplace gender dynamics of an AI-enabled workplace will influence the evaluation of human workers and their teammates/tools.

On the one hand, results for functional design show that worker evaluations trended more positively when working with AI teammates over AI tools. On the other hand, interactions between a dyadic partner’s gender identity and the type of task completed by a team demonstrate the potential for workplace gender dynamics to influence evaluation. While these main effect results should inform future evaluation, the most interesting findings to come from this work come from the intersection of gender dynamics and functional design. Indeed, the effects of dyadic partner type, interactions between dyadic partner type and worker gender, and the interactions between dyadic partner type and various covariates present an interesting narrative surrounding the design and integration of AI technology. As such, the following discussion highlights these interesting outcomes, their relation to prior work, and their implications for the future design and integration of AI into the workplace.

First, focusing on the manipulation of the dyadic partner type, both AI functional design and workplace gender dynamics can be found within its conditions. In regard to AI functional design, significant differences were found multiple times between the AI Tool and AI Teammate DP conditions. Some of these results mirror that of prior work, which has shown that job security is often perceived as lower for those work with AI teammates [156], and that evaluations of human workers are worse when working with AI tools [380]. However, in this study, the evaluators still showed more positive effectiveness evaluations for human workers working with AI teammates over AI tools, which is a shift from concepts theorized in prior works that advocate the idea that AI systems can only ever be treated as tools [421]. These effects are most likely explained by the way this study’s manipulation of AI functional design extended beyond a purely nominal representation of the word “teammate” to instead focus on a functional operationalization via differing levels of agency and interdependence [347, 353], which is a departure from prior theoretical work that has

yet to evaluate these functional differences [322].

Second, turning towards workplace gender dynamics, this work also sees differences between the AI Teammate DP condition, the Male Teammate DP condition, and the Female Teammate DP condition. In particular, this work showed differences in credit attributed to human female workers and AI Teammates, such that human workers were attributed more credit than their dyadic partners when working with the AI Teammate DP, while the DP was attributed more credit when the DP was the Female Teammate DP. Moreover, partnerships that featured a human worker working with Male Teammates were seen as generally more mutually dependent than those working with AI tools. This is likely reflective of the way that gender dynamics function in groupwork, such that men in groupwork situations are seen as taking on more of the credit for the shared task [403, 404], which could influence perceptions of mutual dependence.

Finally, the implications of these workplace gender dynamics become more complex when exploring the sole interaction effect between dyadic partner type and human worker gender and, thus, the interaction between workplace gender dynamics and AI functional design. This work found that the man worker was evaluated as significantly less effective than the woman worker when both worked with the AI Teammate and Female Teammate dyadic partners. Yet, women human workers saw significantly lower effectiveness when they worked with AI Tools compared to any of the other teammate dyadic partners that they worked with. These interactions highlight a surprising finding: that, despite completing the same task, women may have an evaluation bias over men working with AI, which is counter to many other works exploring gender and technology dynamics in the workplace [104]. Yet, women workers are also more likely to be evaluated differently depending upon the type of dyadic partner they work with, while men workers are evaluated similarly no matter

who they work with, which follows what prior works on gender dynamics in groupwork have found [403, 404].

By blending these three insights, some insights into the design of AI technology can be derived. Based on these results, although AI technology may not always carry with it a perceived gender identity (see Study 1 [411], and [96]), this work is still able to see gendered differences in the perceptions between AI and human collaborators. Most notably, it is likely not enough to simply remove gender identity from the research and design of AI systems; rather, researchers will need to understand how the identity of AI systems as either a Tool or a Teammate lends itself toward different gendered perceptions seen in modern workplaces. Moreover, these gendered perceptions may differ based on the functional design of AI in terms of its designed agency and interdependence, as evidenced by the AI Teammate DP and human Female Teammate DP sharing significant effects in terms of credit and effectiveness.

Outside of these manipulation-only insights, the role of covariates in this work also demonstrates the likely impact that workplace gender dynamics and AI functional design will play in evaluations of human workers generally and women workers specifically in AI-enabled workplaces. Indeed, this work found that both types of ambivalent sexist beliefs, hostile and benevolent [27, 188, 189], and negative attitudes towards AI use in society can be significant predictors for evaluation outcomes of workers collaborating with different dyadic partners. When compared to prior work, it isn't entirely novel that sexism and AI attitudes influence perception [104, 352]. Yet, the novel aspect of this work comes in the different dyadic partner conditions and their interaction with these beliefs.

Most notably, the AI Tool DP condition was shown to have consistent and strong relationships between multiple covariates and multiple evaluation criteria. Yet, the AI Teammate DP condition was shown to have weaker and sometimes even op-

posite trends compared to the AI Tool DP condition, with many evaluations of partnerships containing an AI Teammate DP matching closer to those that contained a human Teammate DP, particularly the Female Teammate DP. This clustering and difference in trends among dyadic partners highlight that AI Teammate DPs may not only be slightly more resilient toward these pre-existing biases, but that they may also engender more human-like perceptions and evaluations than the AI Tool.

Overall, then, the integration of AI Tools and, to a lesser extent, AI Teammates, will likely intermingle with evaluator biases in regards to AI's use in society and workplace gender dynamics that could hinder evaluations of both human workers and AI in AI-enabled workplaces. Indeed, evaluator bias as a critical consideration has gained recent traction in the AI community for its potential to allow racism and sexism to seep into the design of AI systems [77, 454]. However, based on Study 2's results, simply accounting for this bias during the design of an AI system for workplace integration is likely not enough to actually address these concerns. Rather, these biases are going to need to be accounted for and mitigated within evaluators of AI *and* of humans to ensure AI workplace integration accounts for the pre-existing ways that evaluator biases already creep into evaluations in all-human workplaces.

As research and practice continue to progress with the design and integration of AI technology, these impacts and intersections of workplace gender dynamics and AI functional design should be addressed when designing and evaluating AI workplace integration. Turning towards design, designers will need to account for the implications of designing AI as either tools (i.e., lacking agency, interdependence [347, 353]) or teammates (i.e., agentic, interdependent [347, 353]), keeping in mind that most worker evaluations and attributions of credit increased when working with an AI teammate compared to working with an AI tool. Furthermore, integrators will need to account for the gender dynamics of a team that is the target of AI integration, as

the gender makeup of that team and whether the AI is acting in the capacity of a tool or a teammate could substantially impact the ultimate effectiveness of that worker and their partner.

Lastly, evaluators will need to be trained to overcome potential biases related to gender in the workplace and technology use in the workplace, as those pre-existing biases may yield inconsistent or overly harsh evaluations of human workers who are, in reality, leveraging AI tools and teammates to increase their effectiveness in the workplace. Moving forward, it will be critical for human-centered computing and human-centered AI research to aid these design, integration, and evaluation efforts to ensure workers and AI are given the best opportunity to succeed in the workplace, regardless of their gender or functional design.

5.6.2 Design Recommendations for Approaching Future AI Workplace Integration with Consideration of Evaluating Human Workers

Grounded in both the quantitative and qualitative results of Study 2, the remainder of this chapter details two fundamental design recommendations aimed towards encouraging organizations to approach AI workplace integration with consideration of its human workforce, especially their women workers. It does so by first highlighting the vital role that AI functional design plays in how workers are evaluated when working with AI in AI-enabled workplaces, and explicitly calling for organizations to prioritize teamwork-based AI functional design when crafting AI workplace integration to maintain positive evaluations of *all* human workers. It then explicitly calls for organizations to interrogate and tackle workplace gender and technological biases within their own workplace *prior to* AI workplace integration, thus pushing for

organizations to be considerate of their specific workers at the ground level of future AI-enabled workplaces.

5.6.2.1 Recommendation 1: To Maintain Positive Evaluations of Human Workers, Prioritize Considering AI Teammates & Their Applicability in the Workplace Context

The first design recommendation of Study 2 stems from the direct comparison of AI tools and AI teammates. One of the most novel findings of this work highlights that AI tools, compared to AI teammates, led to significantly lower perceptions of performance for both the AI *and* the human worker. These differences highlight the impact that shifting from viewing AI tools and teammates as simply nominal differences to viewing AI tools and teammates as *functionally different* is essential in research and in industry practice. Moreover, the positive perceptions that come from working with AI teammates highlight the likely positive impacts that would come from designing AI as teammates rather than as tools when designing for AI workplace integration. As such, the first recommendation of this work is that the research and practice of AI design in collaborative settings should always consider the usage of AI teammate functional design principles of agency and interdependence to promote the positive perception of AI and human workers.

However, it is also critical to note that this recommendation is for the *consideration* of designing AI as teammates when designing for AI workplace integration. Indeed, in the event that an AI teammate’s functional design is *not* a good fit for a specific task, then their integration may actually be harmful to the processes that it intends to help [368, 388]. While prior works in HCC and HCI domains have often defaulted to the ‘tool’ perspective when designing and evaluating AI integration [421], the findings of this work suggest that viewing AI *teammates* as a default design

pattern may provide significant benefits to the evaluation of human workers and even AI systems. As such, future research and practice should first consider whether designing an AI as a teammate fits with its target task and workplace context, and only shift to a tool-oriented design perspective when it is determined that the teammate design would be inappropriate or would adversely affect task success.

5.6.2.2 Recommendation 2: Prior to Integrating AI Teammates, the Gender and Technologies Biases of Evaluators Need to Be Corrected

The second design recommendation of this work is more broadly concerned with workplace gender and technological dynamics as larger cultural concerns to address in the design and evaluation of AI workplace integration. Indeed, the results of this work highlight a clear interaction between different AI systems and the biases evaluators hold toward both AI and women. While prior work has continued to stress the importance of addressing evaluator bias during the design of AI systems [77], often through bias training and correction [39], the results of this work underscore that *solely* focusing on mitigating evaluation biases at the design stage of AI systems without addressing the larger workplace culture will be detrimental to human workers and the successful integration of AI technology into the workplace. As such, this study posits that anyone performing evaluations of human workers working with AI should undergo explicit bias training for both gender and AI biases, as well as other potential biases identified by future works.

In following this recommendation, research and practice will need to divest current efforts in evaluation bias to highlight how biases can be mitigated *after* AI workplace integration in addition to the existing efforts in design pre-integration. Without this correction, a seemingly unbiased and safe AI system could cause undue harm by enabling the harmful biases of evaluators to manifest and perpetuate without

adequate checks. Interestingly, with recent work highlighting the potential design and implementation of AI-powered evaluation systems [74], this recommendation is not necessarily limited to human evaluators. Indeed, AI-based evaluators would also need to undergo substantial bias training and correction prior to evaluating an AI-enabled workplace, as the perpetuation of these gender biases could be especially detrimental in these contexts.

Looking beyond these two recommendations, Study 2’s collective results highlighting how workplace gender dynamics and AI functional design intermingle to shape evaluations of and attributions of credit to human workers lays the groundwork for **Study 3’s focus on charting the paths that lead from these perceptions, dynamics, and designs to perceptions of human worker job security in AI-enabled workplaces.** Indeed, the ANCOVAs run in the quantitative results do not account for how dependent variables (i.e., evaluations and attributions) impact *each other*, such as how evaluations of effectiveness might influence attributions of comparative credit. Moreover, while the Study 2’s quantitative results do present limited insights on human worker job security (see section 5.4.2), it is still unclear how evaluations and attributions of credit work to influence perceptions of job security in AI-enabled workplaces. As such, Study 3 utilizes specific statistical techniques to chart a path that connects all of these factors, as well as supplemental expert interviews to contextualize the results of this path construction in real-world AI workplace integration perspectives.